

HISTORICAL MATERIALS FOR REVISION R45

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1. PRESERVATION MAP

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We study the minimum implementation cost of revising an installed policy while protecting operating performance after every state and date. One randomized Markov continuation serves all restarts. Backward feasibility repair yields complete finite-state search, including a finite-horizon extension without discounting. For controlled finite models with strictly separated operating actions, disadvantage and savings coordinates contract the bilinear relaxation: a repaired root-node certificate has a quadratic small-allowance error bound at fixed primitives. An observable-covariate disintegration further gives convergent certificates for a class of controlled atomic continuous-state models without smoothing their transitions. Exact exogenous aggregation and a verified density-transfer exercise provide complementary routes. The computations retain all registered solver stops, same-object relaxation comparisons, and independent rational endpoint checks. In the original maintenance cohort, 27 lotteries beat deterministic lower bounds, while all 30 positive-cost randomized intervals remain open. The results distinguish structural rates, complete search, and actual numerical closure without changing the common-policy objective.

KEYWORDS: Dynamic programming, Policy revision, Constrained Markov control, Global optimization, Verified computation.

1. INTRODUCTION

An organization may wish to improve an installed rule without replacing every decision it makes. Implementation cost then depends both on the decisions changed today and on the states reached under the revised rule tomorrow. A feasible revision must preserve the operating objective, whereas a cost-effective revision must economize on intervention along its own induced state distribution. These two requirements create a constrained dynamic problem. Pointwise selection of inexpensive changes need not solve it.

The relevant continuation must also be specified. Two histories that meet at the same date and date must use the same continuation under a Markov rule. Allowing a different continuation for each initial condition or each remembered randomization can reduce costs, but changes the economic object. Conversely, requiring a deterministic rule can increase implementation cost even when randomization is admissible. We therefore take a common randomized Markov rule as the natural choice variable. Performance is protected after every date and state; implementation cost is integrated under a stated initial distribution. A performance guarantee at every restart is not a guarantee of cost optimality at every restart.

The main contribution is a constructive connection between these requirements and a complete search. A policy that violates every operating constraint by at most δ can be repaired backward without adding a persistent random seed or a history variable. The repair improves operating value at every state. Its additional implementation cost is bounded explicitly by a modulus that vanishes with δ . Strict discounting provides a constant slack, while a finite-horizon allowance ladder also permits undiscounted decisions. At zero tolerance an exact restriction on operating-maximizing actions replaces the repair. No global duality assumption is used. The repair gives an error bound for rational policy lattices. For binary actions it also gives a stopping theorem for the interval Bellman branch-and-bound procedure used in the experiments.

tion constrained optimum.

controlled finite models, the operating constraint itself supplies a further numerical bound. Relative to a unique operating-optimal reference, the expected probability of a non-optimal action is bounded by the operating allowance divided by its Bellman disadvantage. Maintenance savings satisfy a cone relative to operating regret. Both factors of a McCormick approximation therefore shrink with the allowance. A repair directed toward the reference action targets only the rare nonreference mass. These observations yield an explicit $O(\varepsilon^2)$ root-width at fixed positive operating-action separation and strict discounting. The claim is a certified interval for the original constrained optimum, not only approximation of an unconstrained operating value. Constants and action ties remain visible.

Continuous-state analysis is also extended beyond exogenous transitions. When an observable follows an invertible autonomous evolution, it labels finite controlled problems. The optimum can be integrated exactly in principle and enclosed by verified covariate partitions. Actions may alter all regime transition probabilities. The policy uses only the current state and regime, so no hidden memory is introduced. Uniform finite-matrix perturbation and complete fiber solves provide convergence at a fixed positive allowance. This method is distinct from replacing atomic successors by smooth reset densities.

Witness-driven extension separates operating approximation error from policy-search error, one computable global bound. It supplies a positive-margin check and a constructive repair without passing the exact optimum to the repair routine. A further perturbation result bounds model error into operating and cost allowances; strict randomized separations also yield uniform administration charges. These statements extend the numerical method rather than approximate its economic objective.

The structural result makes a different part of the general problem computationally tractable. Exogenous transitions are exogenous, pointwise aggregation and a compatible family of within-cell measures together preserve the optimum over unrestricted measurable common policies. Identities can hold for nonlinear atomic transitions even when total-variation reset approximation fails. The resulting all-restart regret program is linear. Twelve designed service policies, spanning three transition structures, up to 64 regimes and 64 periods, and three or more periods, produce independently verified intervals below 9×10^{-8} . An unreduced Bellman interval is equally entitled classical comparator; we do not conceal its advantage on some small problems.

The second result transfers finite-state certificates to a continuous state space. Under a certified total-variation approximation of the transition kernel by a within-cell reset kernel, cell averaging transfers an arbitrary measurable Markov policy to a finite-state policy. Lifting reverses this operation. Uniform primitive errors give an explicit operating-tolerance adjustment and a repair modulus controls the remaining difference between the relaxed and original finite-state optima. The theorem makes the approximation assumption operational. It does not treat weak convergence of an atomic transition law as total-variation convergence, and it does not silently replace discrete shocks by continuous noise.

Numerical analysis separates three questions. First, does complete search return sound bounds for the common Markov problem, and how do those intervals compare with a direct mathematical-programming formulation? Second, does randomization change the conclusions in the original continuous-state maintenance configurations? Third, can a nontrivial lower bound survive nonlinear, two-state dynamics and nonzero operating-witness? Each experiment retains the same objective on the two sides of its interval. The inherited many-action finite-state environments and their two solver implementations were fixed. The environments were generated. The additional controlled, three- and four-action

described as an untouched code-frozen holdout. The original 42 configurations are explicitly a development cohort, not a prospective test. The resulting distinctions are economically consequential. In the development cohort, 27 in-domain randomized deployments cost strictly less than a certified lower bound for every admissible deterministic policy under the same uniform initial law. This is a statement about the value of admissible randomization, not about a neural label attached to the installed rule. The linear experiment changes the zero lower bound into a positive bound; its finest horizon-interval is $[8.8718, 11.8492]$. That interval is informative but not tight. In the prospective in-state comparison, both methods face the same node allowance and wall-clock limit. Stronger linear relaxation often improves the lower endpoint, but longer horizons retain substantial residual intervals. The theory supplies completeness, not a claim that complete search is expensive.

1.1. *Relation to the literature*

Constrained Markov decision processes provide the natural starting point (Altman, 1999). Occupation measures are particularly effective when the constraints are expectations under a designated initial law. Our formulation additionally requires every restarted problem to meet a separating guarantee using one shared continuation. A separate occupation measure for each restart would need compatibility restrictions: at a shared date and state, its action probabilities must not depend on which restart generated the occupancy. Dropping that restriction produces a different feasible class. Our finite formulation imposes it directly through a single policy variable for each date and state. This observation does not preclude occupation-measure methods; it clarifies the coupling they must preserve in this problem.

Finite approximations to constrained control are not new. Saldi (2018) proves value convergence for discounted and average-cost constrained problems, and develops approximately optimal policies and rates under additional continuity conditions. Saldi, Linder, and Yüksel (2018) provides a systematic treatment of quantized stochastic control. Our transfer result combines finite-horizon, all-restart constraints and combines the within-cell reset identity with an implicit Markov feasibility repair. It states the total-variation condition needed to average unrelated measurable policies rather than assuming that a convenient policy sieve is automatically present in the constrained optimum. The theorem is not a replacement for those approximation theories; its additional object is the common-continuation restart constraint and its computable distance-to-cost modulus.

Primal-dual policy methods also solve constrained expected-return problems. Ding, Zhang, and Jovanovic (2020) establishes global convergence guarantees for natural policy-gradient methods in discounted constrained MDPs. Ni and Kamgarpour (2025) studies safe exploration under cumulative constraints. Those learning problems have information and feasibility requirements different from our known-model, all-restart certification problem. Safe online improvement (Laroche, Trichelair, and Tachet des Combes, 2019) and approximation-error analysis for fitted dynamic programming (Munos and Szepesvári, 2008, Scherrer, 2013) never distinguish model uncertainty, policy approximation, and optimization error. We claim a sample-efficiency comparison against methods given less information.

The global optimization ingredients are classical. Convex envelopes of factorable products dominate in McCormick (1976); the numerical LP proposals here use a modern simplex implementation associated with Huangfu and Hall (2018). We do not attribute novelty to bilinear envelopes, best-bound search, or weak duality. The contribution is a common-policy economic formulation whose local approximation errors can be repaired into an admissible deployment,

ject comparator, not as a deliberately weak price portfolio. A rational residual correction proposed LP multipliers into lower bounds even when floating-point dual feasibility is not exact. This separation of numerical proposal and proof follows the broader program of exact computation (Rump, 2010). PRESERVED ARTICLE AND TECHNICAL HISTORY

infinite optimization emphasizes the difference between checking finitely many constraints and controlling them on a region (Hettich and Kortanek, 1993). Our continuous-state models therefore include whole-cell inequalities and isolated switching points. Information-theoretic bounds supply another route to dynamic-programming lower and upper bounds (Smith, and Sun, 2010, Brown and Smith, 2014); they are related to, but not identical to, the restart support functions used below. The support functions remain useful accelerators though no finite price set is asserted to exhaust a global dual.

Economic motivation is adjustment rather than unconstrained operating optimization. Real-world economics has long emphasized model structure, approximation error, and state-space complexity (Judd, 1998, Brumm and Scheidegger, 2017). Empirical replacement models (Rust (1987) discipline operating primitives with data. Our revision cost is instead the cost of departing from an installed decision rule, not a cost paid whenever the previous physical action changes. Adding physical switching costs would require including the previous action in the state. The present maintenance environments are designed, known-model economies. Their operating and implementation units are not estimated monetary welfare, and their costs are not estimates of empirical prevalence. The convergence and exact-reduction theorems rather than a claimed calibration, carry the methodological argument. The density gradient descent now executes the reset error chain; the original endogenous-transition models remain as evidence rather than being relabeled as instances of the exogenous theorem.

2. THE ECONOMIC PROBLEM

Let X be a standard Borel space, $A = \{1, \dots, m\}$ a finite action set, and $t = 0, \dots, T$. Rewards $r_t(x, a)$, implementation costs $k_t(x, a)$, and terminal operating payoff $g(x)$ are bounded Borel functions. The transition kernels P_t^a are known Borel probability kernels. Write $|r_t| \leq R$, $|k_t| \leq K$, and $0 \leq k_t \leq K$. Unless otherwise indicated, $0 < \beta < 1$ and $\varepsilon > 0$. Define

$$T_t^a v(x) = r_t(x, a) + \beta P_t^a v(x), \quad V_T = g, \quad V_t = \max_{a \in A} T_t^a V_{t+1}. \quad (1)$$

Since A is finite, V_t is Borel and permits a Borel maximizing selector, for example the lowest a maximizing action.

A randomized Markov policy is a sequence of Borel kernels $p_t(\cdot | x)$ on A . Its values satisfy

$$J_T^p = g, \quad J_t^p = \sum_a p_t(a | x) T_t^a J_{t+1}^p, \quad (2)$$

$$C_T^p = 0, \quad C_t^p = \sum_a p_t(a | x) \{k_t(x, a) + \beta P_t^a C_{t+1}^p(x)\}. \quad (3)$$

The policy p_t is used by all histories reaching (t, x) . Randomization is newly drawn conditional on current date and state; a persistent hidden coin indexing different future policies is not used.

Given an initial probability law ν , the central object is

$$K_R(\nu, \varepsilon) = \inf_{p \in \mathcal{F}_R(\varepsilon)} \int C_0^p d\nu, \quad \mathcal{F}_R(\varepsilon) = \{p : J_t^p(x) \geq V_t(x) - \varepsilon \forall t, x\}. \quad (4)$$

nal policy is feasible, so both classes are nonempty. On a finite state space, the product policy simplex is compact, J^p and C^p are continuous, and the feasible set is closed. The num in (4) is then attained. We use infimum, not an unproved function-space minimum, in Borel-state formulation.

or an installed rule $a_t^0(x)$, the principal cost specification is

$$k_t(x, a) = w_t(x) \mathbf{1}\{a \neq a_t^0(x)\}, \quad w_t \geq 0. \quad (5)$$

cost is charged for the action actually drawn. Its expectation under a lottery is not an indicator the lottery's most probable action differs from the installed action. The unchanged installed policy has zero revision cost even when it is operating-infeasible. Consequently, dividing an optimality interval by its revision cost is undefined, not an informative relative-gap statistic. We report $(U - L)/U$ when $U > 0$, and report zero-cost exactness separately.

The uniform law in the development economy weights all normalized initial conditions equally. We use a numerical reference distribution, not an estimated stationary law. We also integrate exactly outwardly under densities $2x$ and $2(1 - x)$. These calculations change the initial objective by leaving the all-state operating constraints and installed rule unchanged. A pointwise certificate would cover arbitrary initial laws, including atoms; an almost-everywhere cost certificate need not.

3. FEASIBILITY REPAIR AND COMPLETE POLICY SEARCH

et

$$D_C = K \sum_{s=0}^{T-1} (s+1) \beta^s, \quad (6)$$

$$D_J = \max_{1 \leq h \leq T} \left\{ 2R \sum_{s=0}^{h-1} (s+1) \beta^s + 2Gh\beta^h \right\}. \quad (7)$$

maximum over remaining horizons in D_J matters: the terminal term $h\beta^h$ need not increase with h . Total variation between action distributions is one half of their ℓ_1 distance.

LEMMA 1—Policy perturbations: *If $\sup_{t,x} \text{TV}(p_t(\cdot | x), q_t(\cdot | x)) \leq \theta$, then*

$$\sup_{t,x} |J_t^p - J_t^q| \leq D_J \theta, \quad \sup_{t,x} |C_t^p - C_t^q| \leq D_C \theta.$$

PROOF: For costs, an action's remaining cost lies in $[0, K \sum_{s=0}^{h-1} \beta^s]$ at a state with h periods remaining. Changing its distribution changes the conditional expectation by at most the length of this interval times θ . Adding β times the next-date value difference and iterating gives $\sum_{s=0}^{h-1} (s+1) \beta^s \theta$. For operating rewards use the interval length $2R$ for each stage and $2G$ for the terminal reward. A reward at distance s can be affected by $s+1$ action changes, and the terminal reward by h changes. Taking the maximum over h gives (7). The same conclusions follow by a finite-horizon coupling, but do not require one. *Q.E.D.*

THEOREM 1—Common-continuation repair: *Suppose $p \in \mathcal{F}_R(\varepsilon + \delta)$, $\delta \geq 0$. There is a Borel normalized Markov policy $\hat{p} \in \mathcal{F}_R(\varepsilon)$ with $J_t^{\hat{p}} \geq J_t^p$ for every (t, x) and*

$$\sup_{t,x} \text{TV}(\hat{p}_t, p_t) \leq \rho_\varepsilon(\delta) := \frac{\delta}{(1 - \beta)\varepsilon + \delta}, \quad \int C_0^{\hat{p}} d\nu \leq \int C_0^p d\nu + D_C \rho_\varepsilon(\delta). \quad (8)$$

$$0 \leq K_R(\nu, \varepsilon) - K_R(\nu, \varepsilon + \delta) \leq D_C \rho_\varepsilon(\delta). \quad (9)$$

F: Proceed backward, with $J_T^{\hat{p}} = g$ given repaired continuation Π^a and HISTORY_1 , $p_t(a | x)q_a$, $q_* = \max_a q_a$, and $b = V_t - \varepsilon$. By the induction hypothesis, $q_p \geq J_t^p \geq b - \delta$. Also $J_{t+1}^{\hat{p}} \geq V_{t+1} - \varepsilon$ implies $q_* \geq V_t - \beta\varepsilon$. If $q_p \geq b$, retain p_t . Otherwise choose a reasonable maximizing action a_* and set

$$\hat{p}_t = (1 - \alpha)p_t + \alpha e_{a_*}, \quad \alpha = \frac{b - q_p}{q_* - q_p}.$$

If $d = b - q_p \in (0, \delta]$, the denominator is at least $(1 - \beta)\varepsilon + d$. Thus $\alpha \leq \rho_\varepsilon(\delta)$ and $\hat{p}_t \geq q_p \geq J_t^p$. All operations are Borel; the division is used only where the denominator is positive. The construction uses only t and x and the already fixed next-date continuation. ρ_ε bounds cost. Apply the construction to arbitrarily near-minimizers at tolerance $\varepsilon + \delta$ and their infimum to obtain (9). *Q.E.D.*

Repair is not a convex combination of whole policies. Such a combination, if selected and remembered, would enlarge the information state. Here the mixture is selected anew at each state using one repaired continuation. The slack $(1 - \beta)\varepsilon$ explains why the theorem requires strict discounting and a positive tolerance. No vanishing modulus at $\beta = 1$ or $\varepsilon = 0$ is possible by this argument.

3.1. Certified search without an exact operating solve

An exact value in the preceding statements is not an oracle that the method must possess. A repaired upper witness supplies the slack needed by the repair. This matters when operating on a policy, rather than policy search alone, is expensive. Let $L_t \leq V_t \leq U_t$ be bounded witnesses with $U_T = g$. Define checked scalar bounds

$$\begin{aligned} e &\geq \max_{t < T} \|U_t - L_t\|_\infty, & \sigma &= (1 - \beta)\varepsilon - d, \\ d &\geq \max \left\{ 0, \sup_{t < T, x} \left[U_t(x) - \max_a T_t^a U_{t+1}(x) \right] \right\}. \end{aligned} \quad (10)$$

In the finite-state statement, let $m = |A|$, let $M \geq 1$ be an integer, and set $\theta_M = \min\{1, (m - 1)/M\}$. Lattice probabilities are multiples of $1/M$. The same primitive transition and reward functions occur in the witness check and the repair. Error in a different operator must first be bounded in its enclosure. A small residual is not, by itself, a certificate that $U \geq V$; the witness conditions and terminal condition are also required.

THEOREM 2—Witness-driven repair and global error: *Suppose $\sigma > 0$ and $\delta \geq 0$. If a common repaired Markov policy satisfies $J_t^p \geq U_t - \varepsilon - \delta$ for all $t < T$, the backward construction starting from $U_T = g$ produces a common policy $\hat{p}$ such that*

$$J_t^{\hat{p}} \geq U_t - \varepsilon \geq V_t - \varepsilon, \quad J_t^{\hat{p}} \geq J_t^p, \quad \sup_{t, x} \text{TV}(\hat{p}_t, p_t) \leq \frac{\delta}{\sigma + \delta}. \quad (11)$$

be the smallest cost among rational-lattice policies satisfying $J_t^p \geq L_t - \varepsilon - D_J \theta_M$.
 Find a minimizer using U , and let its cost be $U_{M,L,U}$. Then

$$8 \quad \max\{0, v_{M,L} - D_C \theta_M\} \leq K_R(\nu, \varepsilon) \leq U_{M,L,U}, \quad (12)$$

$$U_{M,L,U} - \max\{0, v_{M,L} - D_C \theta_M\} \leq D_C \left\{ \theta_M + \frac{e + D_J \theta_M}{\sigma + e + D_J \theta_M} \right\}. \quad (13)$$

As $M \rightarrow \infty$, $e \rightarrow 0$, and $d \rightarrow 0$ give convergence to the actual constrained optimum using
 rounded witnesses rather than an exact operating solve. All displayed errors and the stopping
 are computable for rational represented finite-state primitives.

PROOF: Fix a repaired continuation and put $q_a = T_t^a J_{t+1}^{\hat{p}}$, $q_p = \sum_a p_t(a | x) q_a$, and $b =$
 $V - \varepsilon$. At the final decision date $J_T^{\hat{p}} = U_T$; at every earlier date the induction hypothesis gives
 $J_t^{\hat{p}} \geq U_{t+1} - \varepsilon$. Positivity and (10) imply

$$q_* := \max_a q_a \geq \max_a T_t^a U_{t+1} - \beta \varepsilon \geq U_t - d - \beta \varepsilon = b + \sigma.$$

Inductive improvement gives $q_p \geq J_t^p \geq b - \delta$. Where $q_p < b$, set $\alpha = (b - q_p)/(q_* - q_p)$
 mix p_t with the least-index maximizing action. For $z = b - q_p \in (0, \delta]$, one has $\alpha \leq$
 $(\sigma + z) \leq \delta/(\sigma + \delta)$. Otherwise leave p_t unchanged. This proves (11); the cost bound follows
 from Lemma 1. Finite action maximization and division on a strictly positive-denominator Borel
 preserve measurability.

For the lattice statement, round any genuinely feasible policy as in Theorem 3. Its rounded
 cost exceeds $V - \varepsilon - D_J \theta_M \geq L - \varepsilon - D_J \theta_M$, and its cost increases by at most $D_C \theta_M$. This
 gives the lower endpoint. A lattice minimizer is at worst $e + D_J \theta_M$ below the upper-witness
 cost. Apply the first assertion with that value of δ to obtain the upper endpoint and (13).
 Non-negative costs justify clipping the lower endpoint at zero. *Q.E.D.*

Failure of $\sigma > 0$ means that this particular witness pair has not certified the repair margin;
 it does not establish infeasibility of the original problem. Witness refinement and complete
 search have different roles in (13). Refining the witnesses alone does not close the remaining
 policy-search gap. Conversely, an increasingly fine policy lattice does not erase a fixed operating-
 cost error. The exact-value theorem is recovered by $L = U = V$ and $d = e = 0$.

3.2. A complete finite-state construction

Let X have n states. For an integer $M \geq 1$, let $\mathcal{P}_M$ be the policies whose action probabil-
 ities are multiples of $1/M$. Put $\theta_M = \min\{1, (m-1)/M\}$ and $\delta_M = D_J \theta_M$. Evaluate every
 candidate by the common Bellman equations and let

$$v_M = \min \left\{ \int C_0^p d\nu : p \in \mathcal{P}_M, J_t^p \geq V_t - \varepsilon - \delta_M \ \forall t \right\}.$$

For rational primitives these comparisons, values, and policy probabilities use rational arithmetic.

THEOREM 3—Convergence to the constrained optimum: *Repair a minimizer defining v_M
 and denote its feasible cost by U_M . Then*

$$U_M := \max\{0, v_M - D_C \theta_M\} \leq K_R(\nu, \varepsilon) \leq U_M, \quad U_M - L_M \leq D_C \{\theta_M + \rho_\varepsilon(\delta_M)\}. \quad (14)$$

F: Round the first $m - 1$ probabilities of any feasible policy down to multiples of $\frac{1}{M}$ and assign the remainder to the last action. The total-variation error is at most $(m - 1)/M$, by one. Lemma 1 shows that the rounded policy belongs to the relaxed finite set and has at most the original cost plus $D_C \theta_M$. Hence $v_M - D_C \theta_M \leq K_R$. Theorem 1 applied to the grid policy gives $U_M \leq v_M + D_C \rho_\varepsilon(\delta_M)$. The weak-composition count gives the number of simplex points at each date and state. The displayed error tends to zero. *Q.E.D.*

Theorem controls distance to (4), not distance to a one-step relaxation. Its count is initial in $nT(m - 1)$; it is a completeness construction, not a polynomial-time claim. The deterministic finite-state problem can instead be solved by evaluating its m^{nT} vertices with minimal tolerance. The randomized repair is neither required nor valid as a deterministic operation.

3.3. The executed spatial search

For binary actions, write p_{ti} for the probability of action 1. A policy box $Q = \prod_{ti} [\ell_{ti}, u_{ti}]$ has the same coordinates across all restarts. Backward interval Bellman evaluation gives upper values $\bar{J}$ and unconstrained implementation lower values $\underline{C}$ for that box. Every operating tests can contract its coordinate intervals: if q_0^+, q_1^+ are next-date operating values, an admissible p must satisfy

$$(1 - p)q_0^+ + pq_1^+ \geq V_t(i) - \varepsilon. \quad (15)$$

Remove this half-interval with $[\ell_{ti}, u_{ti}]$. Empty intersections or inconsistent directed operating values remove the box. Every removal is justified by a necessary inequality. Feasible policies are removed because a sufficient candidate check fails.

The remaining cost lower bound is $\int \underline{C}_0 d\nu$. Repairing the midpoint of the contracted box gives a feasible upper policy. The algorithm selects a live box with the smallest lower bound and contracts a coordinate maximizing $\beta^t(u_{ti} - \ell_{ti})$. It preserves both children and every live leaf. The lower endpoint is the minimum of all live or cost-pruned leaf lower bounds and the current upper cost. It stops when their difference is no greater than the requested tolerance, or when the remaining interval when the preassigned budget is exhausted.

PROPOSITION 1—Stopping bound for interval Bellman search: *Let $w = \max_{ti}(u_{ti} - \ell_{ti})$ for a surviving policy box. Its midpoint-repair cost minus its interval Bellman lower bound is at*

$$H(w) = D_C \{w + \rho_\varepsilon(D_J w)\}. \quad (16)$$

For a node or time cap, the best-bound bisection procedure reaches every prescribed gap η in finitely many steps. In particular, choose $w_ > 0$ with $H(w_*) \leq \eta$. A sufficient bisection depth is*

$$d \left\lceil \log_2 \frac{1}{\beta^{T-1} w_*} \right\rceil, \quad d = nT, \quad (17)$$

where the ceiling is replaced by zero when its argument is at most one. The associated full binary-tree bound is exponential in this depth. Adding valid McCormick lower bounds cannot weaken the stopping guarantee.

reduces the maximum operating value simultaneously at every state. It is a common Markov policy, not a family of incompatible histories. A surviving contracted box has its operating value at least $V_t - \varepsilon$ at every date and state. Each such maximizing policy differs from midpoint policy by total variation at most w . Lemma 1 therefore makes the midpoint $D_J w$ -approximately feasible. The same rectangular Bellman argument for minimum cost gives a lower bound no smaller than midpoint cost minus $D_C w$. Repair and (8) imply (16). A box satisfying $v \leq \eta$ cannot need refinement while the global gap exceeds η : its repaired cost is already an incumbent candidate, so its lower bound is within η of the global incumbent or higher. On a still-refined path, weighted longest-side bisection makes every physical width at most w_* by depth in (17). There are finitely many nodes up to that depth. Valid stronger lower bounds remove or improve boxes. The supplement details the effect of the necessary contractions; shrink the rectangles and preserve every feasible point. *Q.E.D.*

This is a convergence theorem for the implemented search rule, not an inference from favorable outcomes. It also explains why a 60-second run can end with a large gap: a finite worst-case bound can be prohibitively large. The numerical results report that distinction directly.

3.4. Finite-horizon repair without strict discounting

Strict discounting supplies the constant margin in Theorem 1, but it is not necessary for finite-horizon completeness. A date-dependent allowance supplies a different margin. This distinction is relevant for organizations that assess a finite service commitment without discounting its constituent dates. In this subsection only, allow $0 < \beta \leq 1$; the constants D_J, D_C retain their β -sum definitions.

THEOREM 4—A finite-horizon slack ladder: *Let $\varepsilon > 0$, $0 < \delta < \varepsilon$, and let p satisfy $J_t^p \geq \varepsilon - \delta$ at every state and date. Choose $z \in (0, 1)$ with $\delta \leq \varepsilon z^T$ and put*

$$\gamma_t = \varepsilon z^{T-t}, \quad \tau_t = \varepsilon - \gamma_t, \quad 0 \leq t \leq T. \quad (18)$$

toward mixing toward an action maximizing the repaired operating continuation, with current value $V_t - \tau_t$, constructs a common Borel Markov policy q satisfying

$$J_t^q \geq V_t - \tau_t \geq V_t - \varepsilon, \quad J_t^q \geq J_t^p, \quad \sup_{t,x} \text{TV}(q_t, p_t) \leq \frac{2z}{1+z}.$$

Consequently $C_0^q \leq C_0^p + D_C 2z/(1+z)$ pointwise. At $\delta = 0$ retain p ; at $\delta \geq \varepsilon$ an operating-optimal policy gives the bound D_C . With $z = (\delta/\varepsilon)^{1/T}$ the resulting modulus vanishes as $\delta \downarrow 0$, tending at $\beta = 1$.

PROOF: The terminal allowance is zero. Given an already repaired continuation, write $V_t - \tau_t$, $q_a = T_t^a J_{t+1}^q$, $q_p = \sum_a p_t(a | x) q_a$, and $q_* = \max_a q_a$. Inductive improvement is $q_p \geq J_t^p$, while

$$q_* - b \geq s_t := \tau_t - \beta \tau_{t+1} \geq \gamma_{t+1} - \gamma_t > 0.$$

If $b \geq q_*$, retain the current probabilities. Otherwise the mixture share $\alpha = (b - q_p)/(q_* - q_p)$ is defined. Its deficit is at most $\delta + \gamma_t$, so

$$\alpha \leq \frac{\delta + \gamma_t}{s_t + \delta + \gamma_t} \leq \frac{\delta + z\gamma_{t+1}}{\delta + \gamma_{t+1}} \leq \frac{2z}{1+z}.$$

improves operating value; the first maximizing action in a fixed order is Borel. Lemma 1
 the cost inequality. No action-independent transition assumption is used. Q.E.D.

ing the repair modulus in Theorem 3 by this modulus extends its finite state horizon
 ment to $\beta = 1$. At fixed T its worst-case rate is $O(M^{-1/T})$, rather than the discounted
 te. It is therefore an extension of the guarantee, not an assertion of efficient generic
 tion. A rational implementation need not evaluate an irrational root: a dyadic z with
 gives a completely rational construction. For $\beta < 1$, either valid repair can be used; their
 erified costs, rather than a presumed ordering of their worst-case constants, determine
 and date is retained.

ame ladder can use operating witnesses. For checked $L_t \leq V_t \leq U_t$, $U_T = g$, let d_t
 $\tau_t - \max_a T_t^a U_{t+1}$ from above, and choose $0 \leq \tau_t \leq \varepsilon$, $\tau_T = 0$, with

$$\sigma_t = \tau_t - \beta \tau_{t+1} - d_t > 0.$$

$U_t - \varepsilon - \delta$, the same proof with target $U_t - \tau_t$ bounds the current mixture by $(\delta + \varepsilon -$
 $+ \delta + \varepsilon - \tau_t)$. This is a checked, date-specific condition at $\beta = 1$, not a claim that every
 mate witness has adequate slack. In a lattice bracket the deficit is $\delta = e + D_J \theta_M$. The
 width, residual, and policy-rounding terms remain separate.

POSITION 2—Zero tolerance: At $\varepsilon = 0$, define $A_t^*(x) = \arg \max_a T_t^a V_{t+1}(x)$. The min-
 implementation cost is the ordinary cost Bellman recursion restricted to $A_t^*(x)$. A de-
 tic Borel Markov selector attains its minimum at every state and date; randomization
 improve it. This holds for finite T and $0 < \beta \leq 1$.

F: A feasible policy has $J_t^p = V_t$ everywhere, since no policy exceeds V_t . Thus
 ort is contained in A_t^* at each date, by the operating Bellman equality. Conversely
 policy supported on these correspondences is operating-optimal by backward induction.
 ing implementation cost on each nonempty finite correspondence is an unconstrained
 tion Bellman problem with Borel selectors. Q.E.D.

proposition assumes the optimal-action ties are known exactly; a small numerical Bellman
 does not establish an exact tie. It addresses zero tolerance without taking a singular
 a positive-tolerance repair bound.

4. EXACT CERTIFICATES FOR STRUCTURED ATOMIC STATE DYNAMICS

finite search is complete but potentially expensive. A distinct question is when the
 n-continuation problem can be reduced exactly, without searching over probability boxes
 ing atomic successors by densities. The following conditions answer that question for
 us state dynamics. They are restrictive and testable: neither action independence alone
 all spatial mesh suffices.

PTION 1—Exogenous aggregation identities: At each date let $\{E_{ti}\}_{i=1}^{n_t}$ be a finite
 rtition and let μ_{ti} be a probability measure supported in E_{ti} . The rewards and imple-
 on costs are constant on each cell for each action; the terminal payoff is constant on
 l cells. The transition P_t does not depend on the action. For a stochastic matrix $\bar{P}_t$,

$$P_t(x, E_{t+1,j}) = \bar{P}_{tij} \quad (x \in E_{ti}), \quad (19)$$

$$12 \quad \int P_t(x, B) \mu_{ti}(dx) = \sum_j \bar{P}_{tij} \mu_{t+1,j}(B) \quad (B \text{ Borel}). \quad (20)$$

initial law is $\nu = \sum_i \nu_i \mu_{0i}$.

Equation (19) is a pointwise identity, including cell boundaries. Equation (20) is an identity of measures, not equality only on partition cells. It allows each $P_t(x, \cdot)$ to be finitely atomic. In particular it need not be a reset kernel.

THEOREM 5—Exact common-policy aggregation: *Under Assumption 1, for finite T , $0 < \beta < 1$, and $\varepsilon \geq 0$, the optimum over all Borel common randomized Markov policies equals the finite-state optimum with primitives $(\bar{r}, \bar{k}, \bar{g}, \bar{P}, \nu)$. A finite optimal policy lifted constantly on cells is feasible at every original state and date and attains that continuum optimum. No cellularity or cellwise-constant restriction is imposed on the original competitor policies.*

PROOF: The operating optimum is cellwise constant by (19) and backward induction. For an arbitrary Borel policy p , define $\bar{p}_t(a | i) = \int p_t(a | x) \mu_{ti}(dx)$. Action independence is essential: the operating continuation is $P_t J_{t+1}^p$, because $\sum_a p_t(a | x) = 1$. Integrating the Bellman equation for μ_{ti} and using (20) shows that the averages of J^p obey the finite Bellman equations for the policy $\bar{p}$. The same argument applies to C^p . Every pointwise operating constraint therefore has its finite counterpart after averaging, while the initial cost is unchanged by the mixture for ν . The finite infimum is no larger than the original infimum. Conversely lift any finite policy constantly on cells. Equation (19) makes its original values equal to the finite values at every point, proving feasibility and the reverse inequality. The finite optimum is attained by compactness. *Q.E.D.*

A deterministic original policy can average to a nondegenerate lottery. The theorem therefore concerns K_R , not equality between finite and continuum deterministic optima. It cannot by itself certify a deterministic lower bound for a continuum model. The deterministic separations in the final maintenance cohort continue to use their own whole-domain deterministic certificates.

4.1. An exact linear program, rather than a bilinear search

Write $\ell_{tia} = \max_b \bar{r}_{tib} - \bar{r}_{tia} \geq 0$ and let $w_0 = \nu$, $w_{t+1} = w_t \bar{P}_t$ be the exogenous state occupancies. Because the continuation is action-independent, operating regret $D_t = V_t - J_t^p$ satisfies a linear recursion. The continuum optimum in Theorem 5 is exactly the program

$$\min_{p, D} \sum_{t,i,a} \beta^t w_{ti} \bar{k}_{tia} p_{tia}, \quad (21)$$

$$D_{ti} = \sum_a \ell_{tia} p_{tia} + \beta \sum_j \bar{P}_{tij} D_{t+1,j},$$

$$\sum_a p_{tia} = 1, \quad p_{tia} \geq 0, \quad 0 \leq D_{ti} \leq \varepsilon, \quad D_T = 0.$$

with at most s nonzeros per row gives $O(nT(m+s))$ constraint nonzeros. Rational programming admits polynomial bit-complexity algorithms; no such complexity claim is for the particular simplex implementation used here. The unrestricted continuum reduction and finite linear algebra are separate claims. The latter is ordinary linear programming over a goal optimizer.

Similarly, the corresponding occupation-measure representation of the general, action-dependent problem has one flow y_{tia}^s for each restart $s = (u, j)$. With $z_{ti}^s = \sum_a y_{tia}^s$, it imposes mass one at s , the usual transition flow, and

$$y_{tia}^s = z_{ti}^s p_{tia} \quad \text{for every restart } s \text{ and } t \geq u. \quad (22)$$

ating constraint is $\sum_{t \geq u, i, a} \beta^{t-u} r_{tia} y_{tia}^s + \beta^{T-u} \sum_i g_i z_{Ti}^s \geq V_u(j) - \varepsilon$. A further flow at ν evaluates implementation cost. Zero-occupancy states cause no division: (22) is as a product equality. With controlled transitions these are genuine common-policy restrictions; independent restart occupation measures omit them. With exogenous z_{ti}^s all z_{ti}^s are fixed, so compatibility becomes linear. Equation (21) eliminates the restart flows by retaining their regret recursion. This is the structural reason the trial search disappears in this class.

LP approximations and their convergence rates are established tools in constrained Dufour and Prieto-Rumeau, 2013, Saldi, 2018). The claim here is the exact preservation of restart common-policy constraint under the two identities above, including atomic z_{ti}^s and its use with a repair-based certification framework. It is not priority for occupation z_{ti}^s , state aggregation, or linear programming.

4.2. Arithmetic certificates and economic charges

equality-constrained bounded LP $\min\{c'z : Az = b, l \leq z \leq u\}$ has, for every rational b , the lower certificate

$$L(y) = y'b + \sum_j \min\{(c - A'y)_j l_j, (c - A'y)_j u_j\}. \quad (23)$$

$y'(Az - b)$ from the objective and minimize each remaining term over its interval to the inequality. No exact dual feasibility or optimizer status is required. In (21), probabilities p_{tia} and regrets in $[0, \varepsilon]$. Rationally rounded LP probabilities are evaluated and, where repaired backward; their actual cost is an upper certificate. An independent checker checks both endpoints from primitives, probabilities, and multipliers. It imports neither the solver nor the constructor.

Exogenous state dynamics make the implementation economics particularly transparent. A positive state- and action-dependent administration charge that leaves operating payoffs and costs unchanged simply adds to $\bar{k}_{tia}$. Program (21) remains exact, rather than requiring a break-even approximation. An overhead that changes rewards must also alter ℓ and the global optimum; an overhead that changes the transition law can invalidate exogeneity and is not treated by the general common-policy program. Implementation units in the following economies are designed adjustment units, not estimated monetary welfare.

5. REGRET-SCALED GLOBAL BOUNDS WITH ENDOGENOUS TRANSITIONS

The linear reduction above requires exogenous transitions. A different structure is available when transitions change future state probabilities. Tight operating allowances limit how often

ive to that action then contracts both factors in the remaining bilinear products. This yields an additive bound for a classical McCormick relaxation, rather than only a finite-termination argument.

Throughout this section the state space is finite, $0 < \beta < 1$, and $\varepsilon > 0$. Let a_{ti}^* be the unique operating-maximizing action at (t, i) , and let H be its implementation cost. Define the operating advantage and the implementation advantage

$$d_{tia} = V_{ti} - r_{tia} - \beta P_t^a V_{t+1}(i), \quad (24)$$

$$g_{tia} = H_{ti} - k_{tia} - \beta P_t^a H_{t+1}(i). \quad (25)$$

vanish at a_{ti}^* , while $d_{tia} > 0$ for the other actions. Put $\underline{d} = \min_{t,i,a \neq a_{ti}^*} d_{tia}$. With a fixed model, $\underline{d} > 0$. This is a checkable separation condition, not a uniform assertion over a family in which operating actions become tied.

For a common policy p , write $D^p = V - J^p$ and $S^p = H - C^p$. Their exact recursions are

$$D_{ti}^p = \sum_a p_{tia} (d_{tia} + \beta P_t^a D_{t+1}^p(i)), \quad S_{ti}^p = \sum_a p_{tia} (g_{tia} + \beta P_t^a S_{t+1}^p(i)), \quad (26)$$

with zero terminal values. Let

$$\lambda_- = \min \left\{ 0, \min_{t,i,a \neq a_{ti}^*} \frac{g_{tia}}{d_{tia}} \right\}, \quad \lambda_+ = \max \left\{ 0, \max_{t,i,a \neq a_{ti}^*} \frac{g_{tia}}{d_{tia}} \right\}.$$

LEMMA 2—Disadvantage and savings cones: *Every feasible common policy satisfies*

$$\sum_{a \neq a_{ti}^*} d_{tia} p_{tia} \leq \varepsilon, \quad \sum_{a \neq a_{ti}^*} p_{tia} \leq \min\{1, \varepsilon / \underline{d}\}, \quad \lambda_- D_{ti}^p \leq S_{ti}^p \leq \lambda_+ D_{ti}^p. \quad (27)$$

In particular, $0 \leq D^p \leq \varepsilon$ and $\lambda_- \varepsilon \leq S^p \leq \lambda_+ \varepsilon$.

PROOF: Operating optimality gives $D^p \geq 0$. The first recursion in (26) and feasibility give the first inequality; strict action separation gives the second. Iterating the two recursions expresses D^p and S^p as discounted expected sums of d and g under the same policy and state. The stage inequalities $\lambda_- d \leq g \leq \lambda_+ d$ also hold at the maximizing action, where both d and g are zero. Taking expectations proves the cone inequalities. The last bounds use the signs λ_- and λ_+ . *Q.E.D.*

To eliminate $p_{ti,a^*} = 1 - \sum_{a \neq a^*} p_{tia}$ and write $\Delta P_t^a = P_t^a - P_t^{a^*}$. The two recursions become

$$D_{ti} = \beta P_t^{a^*} D_{t+1}(i) + \sum_{a \neq a^*} p_{tia} \{d_{tia} + \beta \Delta P_t^a D_{t+1}(i)\}, \quad (28)$$

$$S_{ti} = \beta P_t^{a^*} S_{t+1}(i) + \sum_{a \neq a^*} p_{tia} \{g_{tia} + \beta \Delta P_t^a S_{t+1}(i)\}. \quad (29)$$

The new objective is $\nu' H_0 - \nu' S_0$. Use the probability simplexes, the first inequality in (27), the savings cones, and ordinary nonnegative-cost bounds. Replace each remaining product $p_{tia} D_{t+1,j}$ by its four McCormick inequalities over its certified rectangle. All unrestricted policies remain in this linear relaxation. The construction preserves one probability simplex per date, state, and action; it does not relax common continuations into restart-specific policies.

$$\sup_{t,i} \text{TV}(\hat{p}_{ti}, p_{ti}) \leq \chi \frac{\delta}{(1-\beta)\varepsilon + \delta}, \quad \nu' C_0^{\hat{p}} \leq \nu' C_0^p + D_C \chi \frac{\delta}{(1-\beta)\varepsilon + \delta}. \quad (30)$$

F: A repaired continuation obeys $J_{t+1}^{\hat{p}} \geq V_{t+1} - \varepsilon$. Since a_{ti}^* maximizes the original
 g Bellman equation, its value with that continuation is at least $V_{ti} - \beta\varepsilon$. It therefore lies
 above the target whenever a repair is needed. The proof of Theorem 1 applies with this
 ion and bounds the mixing fraction by $\rho_\varepsilon(\delta)$. The total-variation change is exactly that
 times $1 - p_{ti,a^*}$, at most $\chi\rho_\varepsilon(\delta)$. Operating value still improves at every backward step.
 1 gives the cost bound. Q.E.D.

LEM 6—Quadratic small-allowance relaxation width: Let $L_\varepsilon^{\text{LP}}$ be the exact optimum
 gret-scaled relaxation. Set $e_{D,T-1} = 0$ and, for $t < T - 1$, set

$$e_{D,t} = \frac{\beta\varepsilon^2}{4} \max_i \sum_{a \neq a_{ti}^*} \frac{\|\Delta P_t^a(i, \cdot)\|_1}{d_{tia}}, \quad e_{S,t} = (\lambda_+ - \lambda_-)e_{D,t}, \quad e_{S,T-1} = 0.$$

$E_{D,T} = E_{S,T} = 0$ and $E_{D,t} = e_{D,t} + \beta E_{D,t+1}$, $E_{S,t} = e_{S,t} + \beta E_{S,t+1}$. Write $\bar{E}_D =$
 $E_{D,t}$ and $\chi_\varepsilon = \min\{1, \varepsilon/\underline{d}\}$. An exact minimizing LP policy can be repaired to a feasible
 f cost $U_\varepsilon^{\text{rep}}$ such that

$$0 \leq U_\varepsilon^{\text{rep}} - L_\varepsilon^{\text{LP}} \leq E_{S,0} + D_C \chi_\varepsilon \frac{\bar{E}_D}{(1-\beta)\varepsilon + \bar{E}_D}. \quad (31)$$

ed finite model with a strictly positive operating-action gap and fixed $\beta < 1$, the right
 $O(\varepsilon^2)$ as $\varepsilon \downarrow 0$.

F: For any point in a McCormick rectangle, the discrepancy between its product
 and the actual product of its two factor coordinates is at most one quarter of the
 of their interval lengths. Here a nonreference probability has length at most ε/d_{tia} .
 coordinate has length ε , and a savings coordinate has length at most $(\lambda_+ - \lambda_-)\varepsilon$.
 g the triangle inequality to (28)–(29) gives one-step errors e_D, e_S . Recombining the
 e and nonreference terms gives the stochastic kernel P^p ; its sup-norm is one. Backward
 son therefore bounds actual regret by $\varepsilon + \bar{E}_D$ and the actual policy cost by $L_\varepsilon^{\text{LP}} + E_{S,0}$.
 s local disadvantage inequality bounds its nonreference mass by χ_ε . Lemma 3 proves
 r fixed primitives, $E_D, E_S = O(\varepsilon^2)$ and $\chi_\varepsilon = O(\varepsilon)$; the repair ratio is $O(\varepsilon)$. Q.E.D.

theorem does not bound the time required to solve an arbitrary LP, nor the size of
 l global search. Its gain is quantitative: shrinking the operating allowance contracts
 linear factors, and rare-departure repair prevents a first-order feasibility penalty from
 ng the quadratic bound. An exact operating tie, a vanishing action gap, or a discount
 nding to one can invalidate the uniform rate. Such a case retains the ordinary finite-state
 tes and complete search; it is not discarded. For a nonoptimal feasible LP proposal,
 bjective gap must be added to (31). In computation we instead report independently

rounding losses.

The use of McCormick envelopes and variable-bound tightening is classical (McCormick, 1975). The additional result is the disadvantage/savings reduction and its tolerance-dependent error rate for the same all-restart constrained problem. The direct unscaled formulation remains a legitimate competitor. The experiment below tests the incremental contribution of these bounds without claiming dominance over every implementation of global optimization.

6. A SINGLE ERROR BUDGET FOR CONTINUOUS-STATE APPROXIMATION

A finite-state theorem alone does not justify replacing a continuous model by a grid. The following sufficient conditions permit such a transfer without restricting the original policy to be piecewise constant.

ASSUMPTION 2—Certified reset approximation: *For a finite Borel partition $(E_i)_{i=1}^{n_h}$, let μ_i be probability measures supported on E_i and write $\nu = \sum_i \nu_i \mu_i$. There are cellwise rewards $r_{h,t}$, costs $k_{h,t} \in [0, K]$, and terminal values g_h , and a kernel*

$$Q_{h,t}^a(x, dy) = \sum_j P_{h,t}^a(i, j) \mu_j(dy), \quad x \in E_i,$$

certified uniform errors d_r, d_k, d_g and α_h satisfying

$$|r - r_h| \leq d_r, \quad |k - k_h| \leq d_k, \quad |g - g_h| \leq d_g, \quad |(P_t^a - Q_{h,t}^a)f(x)| \leq \alpha_h \|f\|_\infty$$

every bounded Borel f . The approximation rewards obey the same bounds R, G as the original rewards, or R, G are enlarged to cover both models. All finite transition entries and error bounds are computable with verified directions.

The last inequality uses the operator convention for total variation; it is not the half- ℓ_1 convention for action probabilities above. Define $M_{J,t} = R \sum_{s=0}^{T-t-1} \beta^s + G \beta^{T-t}$ and $M_{C,t} = R \sum_{s=0}^{T-t-1} \beta^s$. Set

$$a_T = d_g, \quad a_t = d_r + \beta a_{t+1} + \beta \alpha_h M_{J,t+1}, \quad (32)$$

$$b_T = 0, \quad b_t = d_k + \beta b_{t+1} + \beta \alpha_h M_{C,t+1}, \quad (33)$$

$$a_h = \max_t a_t, \quad b_h = b_0.$$

Let $v_h(s)$ be the optimum of the finite common-Markov model at operating allowance s and initial masses (ν_i) .

THEOREM 7—End-to-end transfer: *Under Assumption 2, if $\varepsilon > 2a_h$, then*

$$v_h(\varepsilon + 2a_h) - b_h \leq K_R(\nu, \varepsilon) \leq v_h(\varepsilon - 2a_h) + b_h. \quad (34)$$

Choose finite computation returns intervals $[L_+, U_+]$ and $[L_-, U_-]$ for these relaxed and tightened optima, of widths at most η_+ and η_- . A valid global interval is

$$[\max\{0, L_+ - b_h\}, U_- + b_h], \quad (35)$$

$$\eta_+ + \eta_- + 2b_h + D_C \frac{4a_h}{(1 - \beta)(\varepsilon - 2a_h) + 4a_h}. \quad (36)$$

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ified primitive errors tend to zero and both finite-solver widths tend to zero, this interval es to the original measurable-policy optimum. Its measured width supplies a termination he actual economic objective.

F: Fix any measurable Markov policy p , not necessarily cellwise constant. Backward ion of its original and reset-model Bellman equations gives (32)–(33). Bellman com- gives the same operating error for optimal values. In the reset model, averaging the within E_i , $\bar{p}_t(a | i) = \int p_t(a | x) \mu_i(dx)$, averages its value exactly: after every transition in-cell law is again μ_j , and stage rewards and costs are cellwise constant. This identity backward, including the common continuation. An original feasible policy has reset g value at least the reset optimum minus $\varepsilon + 2a_h$ pointwise, hence also after cell averag- finite-model cost differs from the original initial cost by at most b_h . Taking an infimum e lower inequality. Conversely, lift a finite policy feasible at $\varepsilon - 2a_h$ by using its action ities throughout each cell. The reset value is constant in each cell; the two operating take the lifted original policy feasible, and its cost differs by at most b_h . This proves the equality. Finally apply Theorem 1 in the finite model with target allowance $\varepsilon - 2a_h$ and $4a_h$, then add the two finite-computation widths and the cost errors. *Q.E.D.*

6.1. Where the transfer theorem applies

compact domain, a transition density with computable uniform continuity in the source and an L_1 -approximable density in the destination variable can furnish the reset nation. The cell reference measures must also represent the initial law as in Assumption 2. mple, with Lebesgue-uniform cell measures and a uniform initial law, a uniformly z transition density gives a computable $\alpha_h = O(h)$ on a shape-regular mesh. Cellwise g and cost approximations give $d_r, d_k, d_g = O(h)$ when their moduli are Lipschitz. discontinuities in an installed rule can be retained as exact partition boundaries. In e (36) is $O(h + \eta_+ + \eta_-)$ at fixed positive ε and $\beta < 1$, with explicitly displayed s. The number of cells still scales as $h^{-\dim X}$; the theorem does not remove the curse of onality.

primary affine-shock model and the nonlinear two-state model below have atomic n laws. Smearing their successor atoms into cells generally leaves the total-variation unded away from zero. We therefore do *not* apply Theorem 7 to those models by asserting mesh. Their reported intervals are direct whole-domain bounds for the unchanged models. They remain valid without a convergence claim for that particular spatial itation. This distinction prevents the approximation theorem from being used as an orted explanation for closing an atomic-model gap.

7. CERTIFICATES THAT ACCELERATE THE GLOBAL PROBLEM

omplete search, exact structural reduction, and reset transfer provide different routes to certificate, each under its stated assumptions. The inherited one-sided certificates are ccelerators and diagnostic tools, but are not substituted for that argument.

Let $L_t \leq V_t \leq U_t$ be bounded Borel operating witnesses, including the terminal date. Every restart feasible deterministic policy must select an action in

$$18 \quad \mathcal{N}_t^{L,U}(x) = \{a : L_t(x) - T_t^a U_{t+1}(x) \leq \varepsilon\}. \quad (37)$$

ed, its operating continuation cannot exceed $V_{t+1} \leq U_{t+1}$. The set is nonempty because operating-optimal action satisfies the test. Its graph is Borel, since it is a finite union of sublevel sets. Backward minimization of $k_t + \beta P_t^a B_{t+1}$ over this set yields a Borel lower bound; the lowest-index minimizer is a Borel selector. Thus the general-state recursion does rely on an implicit measurable-selection assertion.

his is a necessary-action outer class, not the original constrained problem. A selected continuation can accumulate losses that violate later operating requirements. Equality between integrated cost lower bound and a separately feasible policy's cost certifies $K_D(\nu, \varepsilon)$. Equality of the full functions certifies more. Tightening L, U alone does not force equality; complete which is what controls the remaining structural gap.

7.2. Randomized support floors and a connected local program

for $\lambda \geq 0$, a verified supersolution W_t^λ of

$$W_T^\lambda \geq \lambda g, \quad W_t^\lambda \geq \max_a \{\lambda r_t - k_t + \beta P_t^a W_{t+1}^\lambda\}$$

ies $C_t^p \geq \lambda(V_t - \varepsilon) - W_t^\lambda$ for every feasible policy, including randomized Markov policies. ne a floor F_t by the maximum of zero and finitely many such terms, using L_t in place of V_t n necessary. Given a valid next-date lower function B_{t+1} , define

$$\mathcal{D}_t(x) = \left\{ p \in \Delta(A) : \sum_a p_a T_t^a U_{t+1}(x) \geq L_t(x) - \varepsilon \right\}.$$

$$B_t(x) = \max \left\{ F_t(x), \min_{p \in \mathcal{D}_t(x)} \sum_a p_a \{k_t(x, a) + \beta P_t^a B_{t+1}(x)\} \right\}, \quad B_T = 0. \quad (38)$$

feasible action distribution of any admissible policy belongs to the local constraint, and its inuation cost dominates B_{t+1} . Induction proves that (38) is a global randomized cost lower and. It does not prove that the local optimizer can be combined into a feasible policy or that e support prices close a global duality gap.

or three actions and one local operating inequality, the local polytope has only pure feasible ces and two-action boundary mixtures. Our new primary calculation evaluates these vertices exact rational arithmetic on directed whole-cell bounds. Unlike the earlier diagnostic with ro global floor, it uses the nonzero F_t at every date and is paired with a feasible policy for ame uniform-initial randomized objective. It improves that lower endpoint in 23 of the 42 igitations. The prior adaptive local-LP theorem and all of its experimental records remain e retained supplement; their target is still the local relaxation, not (4).

7.3. Whole-domain randomized deployments

simple backward construction supplies feasible upper policies without changing the atomic el. Given a feasible next-date value J_{t+1} , write $q_0 = T_t^{a_0} J_{t+1}$, $q_* = \max_a T_t^a J_{t+1}$, and

$$p_t^{\text{change}}(x) = \frac{b(x) - q_0(x)}{q_s(x) - q_0(x)}. \quad (39)$$

ominator is positive wherever this expression is used. The resulting operating value $\max\{q_0, b\}$ and is feasible at every state. In the piecewise-affine model, J_t remains affine while the mixing probability can be a ratio of affine functions. Cost is therefore not falsely represented as an exact piecewise-affine function.

verifier includes isolated switching points in every operating identity. Directed cost boxes are against the three absolutely continuous initial laws. All affine transition slopes are positive, so a finite set of switching points and its finite-horizon preimages have zero measure under these laws. That fact justifies almost-everywhere cost enclosures; it does not ignore knots in a restart operating constraint or using the same cost enclosure for a different initial law.

7.4. A verified direct mathematical-programming comparator

Direct finite-state formulation minimizes $\sum_i \nu_i C_{0i}$ subject to (2)–(3), $J_{ti} \geq V_{ti} - \varepsilon$, over common policy simplexes. The constraints are bilinear in policy probabilities and transition values. A McCormick relaxation on each probability box provides a classical lower program. The baseline uses the same incumbent repair, contraction rule, box refinement, node allowance, initial law, and all-restart constraints as interval Bellman search. The final work is the stronger LP bound and an LP-proposed feasible candidate after repair. A single-point LP status is not a certificate. For a minimization relaxation $c'z$ subject to $Az \leq b$ and $\ell \leq z \leq u$, every rational vector $\lambda \geq 0$ gives

$$\mathcal{L}(\lambda) = -\lambda'b + \sum_j \min\{(c + A'\lambda)_j \ell_j, (c + A'\lambda)_j u_j\} \leq \min c'z. \quad (40)$$

follows by adding $\lambda'(Az - b) \leq 0$ and minimizing the resulting affine function over the bounded nonnegative solver marginals are evaluated in (40) using exact rational arithmetic. If the LP falls back to the interval Bellman lower bound; a floating infeasibility declaration moves the policy box. A separate verifier rebuilds every relaxation and every search-tree node from the primitive model.

8. CONTROLLED ATOMIC MODELS WITH OBSERVABLE COVARIATES

Atomic transitions need not preclude a complete continuum certificate. They do preclude the displacement of an atom as a small total-variation error. This section uses a diffusion: an observed covariate labels finite controlled problems, and its label remains constant from the current state. The regime transition is allowed to depend on the action.

ASSUMPTION 3—Observable finite fibers: *The state is $(i, z) \in \{1, \dots, n\} \times Z$, where Z is compact and Borel. At date t , the next state is $(j, \Phi_t(z))$ with probability $P_t^a(i, j; z)$. Each $\Phi_t: Z \rightarrow Z$ is a Borel bijection with Borel inverse, independent of i, j, a . Primitive rewards, terminal payoffs, and regime probabilities are bounded Borel functions satisfying the Lipschitz bounds. Initially, z has law ξ and the regime has a fixed conditional law q , independent of t .*

ate t . For fixed ζ , evaluate the primitives at $(i, F_t(\zeta))$ to obtain a finite controlled model. Note its common-policy constrained optimum by $v(\zeta, \varepsilon)$ under initial regime law q .

THEOREM 20—Disintegration without hidden memory: *Under Assumption 3, for $0 < \beta < 1$ and $\varepsilon > 0$,*

$$K_R(\nu, \varepsilon) = \int_Z v(\zeta, \varepsilon) \xi(d\zeta). \quad (41)$$

the infimum on the left ranges over all Borel common randomized Markov policies, not only those that are constant on covariate cells. No exogeneity of the regime transition is required.

PROOF: A Borel policy induces the fiber policy $p_t^\zeta(a \mid i) = p_t(a \mid i, F_t(\zeta))$. The Bellman equations agree at every date and regime. An all-state feasible policy consequently has fiber cost at least $v(\zeta, \varepsilon)$ for every ζ , proving the lower inequality after integration.

For the reverse inequality, apply the rational-lattice construction of Theorem 3 separately in each fiber. At any fixed denominator, there are finitely many candidate probability arrays. Their common values, relaxed-feasibility tests, lowest-index minimizers, and repaired policies are uniform in ζ . Uniform primitive bounds make the lattice and repair cost errors uniform in ζ . Thus the finite construction supplies a Borel family of exactly feasible policies whose costs approach $v(\zeta, \varepsilon)$ uniformly. It also establishes measurability of v as a limit of Borel lower endpoints. Approximate this family by using $\zeta = F_t^{-1}(z)$ at the current date. Its continuation is common to every policy reaching (t, i, z) ; it remembers no unobserved initial condition. Integrate its uniform approximation error and let the denominator tend to infinity. *Q.E.D.*

PROPOSITION 3—Computable covariate-cell enclosures: *Partition Z into finitely many cells B_c and choose a representative finite fiber for each cell. Suppose primitive errors throughout cell c are bounded by $d_{r,c}, d_{k,c}, d_{g,c}$ and the finite regime matrices satisfy*

$$\sup_{t,i,a,\zeta \in B_c} \sum_j |P_t^a(i, j; F_t(\zeta)) - P_{c,t}^a(i, j)| \leq \alpha_c.$$

Compute a_c, b_c from (32)–(33), using these bounds. If $\varepsilon > 2a_c$ for every cell, and finite computations supply lower endpoints L_c^+ at allowance $\varepsilon + 2a_c$ and feasible-policy upper endpoints U_c^- at allowance $\varepsilon - 2a_c$, then

$$\sum_c \xi(B_c) \max\{0, L_c^+ - b_c\} \leq K_R(\nu, \varepsilon) \leq \sum_c \xi(B_c) (U_c^- + b_c). \quad (42)$$

As the primitive bounds and finite-solver widths tend uniformly to zero under refinement, these bounds converge to the original Borel-policy optimum at the fixed allowance ε .

PROOF: Each pair of fibers has the same finite regime coordinates. Their matrix ℓ_1 bound controls expectations of arbitrary bounded functions on these coordinates. Standard Bellman contraction gives the operating and cost perturbation bounds, for every policy, and gives the same bounding bound for optimal values. The representative model at the relaxed allowance supplies a lower bound for each true fiber; its tightened feasible policy lifts to a feasible policy in every cell of that cell. Integrate these inequalities, using Theorem 8. The tolerance-to-cost modulus from Theorem 1, together with the finite-solver widths, proves convergence. Nonnegativity of v fixes truncating each lower endpoint at zero. *Q.E.D.*

at $\delta_{\Phi_t(z)}$ and $\delta_{\Phi_t(z')}$ are close in total variation. The covariate is never smeared. A fixed firm type is the special case $\Phi_t(z) = z$. Nonlinear invertible covariate dynamics are omitted. By contrast, the original maintenance transitions move the continuous condition in a condition-dependent manner and do not satisfy this assumption. The fiber theorem extends the rolled atomic model class with a genuine convergence mechanism; it does not relabel maintenance intervals as converged.

9. NUMERICAL EVIDENCE

Experiments keep model class, information, initial distribution, and operating allowance. The supplement reports every configuration; no failed or budget-stopped case is removed. The computation report records proof-object sizes, memory, wall time, the exact environment, the boundary between independent arithmetic checks and shared-code validation.

9.1. *Controlled transitions and regret-scaled relaxation*

A controlled audit uses a condition regime $i = 0, \dots, n - 1$, three or four actions, and reward $r_i - c_a$ with a small rational price perturbation. More intensive actions shift transition probability from deterioration toward improved condition and replacement. The installed rule, relative revision weights, uniform initial regime law, and terminal reward are fixed by a rational primitive contract. Thus the transition law is genuinely endogenous, unlike exogenous service economies below. The primitives are designed, not estimated.

These cases were declared before evaluation, together with solver and verifier source hashes. At the initial execution, accumulating rational expressions with unrelated denominators make certificate arithmetic unnecessarily expensive; an integer-string limit also interrupted a replacement accumulation outward dyadic enclosures of individual summands. Original results, the amendment, and both remote hash registrations remain in the computation. No case, operating allowance, action set, transition generator, or LP proposal rule was altered. This is a predeclared-case experiment with a disclosed arithmetic amendment, not an altered code-frozen holdout.

These methods use the reference-eliminated Bellman formulation, identical operating-regret and probability caps, the same LP engine and tolerances, and the same exact candidate. Method U uses ordinary implementation-cost bounds. Method S additionally uses the cones and contracted savings bounds. This is a same-object root-relaxation ablation; Method S branches in this experiment. The 30-second cap applies to the LP call, not to the construction or proof checking. All those costs are recorded separately. A stopped computation contributes a sound fallback bound and the operating-optimal feasible reference, not an altered primal or dual estimate.

Savings bounds materially reduce the certified interval in 5 of the eight pairs (Table I); a width below 10^{-8} in absolute width is not counted. The eight-regime, 32-period, four-action case with allowance 10^{-3} gives $[16.213382, 16.258594]$, a relative width of 0.279%. Other cases receive little benefit, and 4 LP calls reach their cap. In particular, a stronger mathematical proof need not give a stronger *computed* certificate within a fixed time limit. Intersecting independently valid intervals is permitted, but the table retains the individual methods rather than merging an unfavorable run behind a portfolio.

With allowance 10^{-4} the scaled positive-cost interval has width at most 0.000008657, compared to 0.09521686 for the unscaled relaxation. This calculation illustrates why tight allowances are not a separated-action formulation even though they worsen the generic repair denominator.

Case	n	T	m	Method	Lower	Upper	Gap/upper (%)	LP status
00	23	4	3	U	2.628082	2.718904	3.341	Solved
00		4	3	S	2.652991	2.712135	2.181	Solved
01	8	8	3	U	4.948069	5.284000	6.358	Solved
01	8	8	3	S	5.167607	5.235970	1.306	Solved
02	16	12	3	U	4.859588	5.388508	9.816	Solved
02	16	12	3	S	4.859588	5.388508	9.816	Solved
03	16	24	4	U	17.921643	19.730330	9.168	Solved
03	16	24	4	S	18.516903	19.416854	4.635	Solved
04	32	16	3	U	5.364739	5.495882	2.387	Solved
04	32	16	3	S	5.384431	5.485796	1.848	Solved
05	8	32	4	U	15.974376	16.278277	1.867	Solved
05	8	32	4	S	16.213382	16.258594	0.279	Solved
06	16	32	4	U	0.000000	28.613836	100.000	Cap
06	16	32	4	S	0.000000	28.613836	100.000	Cap
07	32	32	3	U	0.000000	15.797290	100.000	Cap
07	32	32	3	S	0.000000	15.797290	100.000	Cap

unscaled. S: regret-scaled. Endpoints are outward decimal displays of independently checked rational endpoints. "U" retains the complete sound fallback interval. These are root bounds for the same constrained common-policy minimum, not floating-point LP objective intervals. The supplement reports all allowances, dimensions, timings, memory, proof sizes, and worst-case constants.

TABLE II

THREE ORDERS OF ALLOWANCE REFINEMENT IN ONE FIXED EIGHT-PERIOD MODEL

ε	U width	S width	S width/ ε^2	S total (s)	S check (s)
0.1000	2.181082557	2.181330051	218.13	0.049	0.035
0.0100	0.194637170	0.174925073	1749.3	0.045	0.035
0.0010	0.030918109	0.004397164	4397.2	0.044	0.035
0.0001	0.009521686	0.000008657	865.63	0.043	0.035

The model has three regimes and three actions. U and S are defined in Table I. Total time includes primitive instruction, LP proposal, and constructor arithmetic; the independent check is additional. Widths are displayed outward. The ratio is descriptive: the theorem is a fixed-model upper rate, not an assertion that every pair of observed widths has slope two.

does not imply that the action-separation constants are uniformly favorable. The supplement reports $\underline{d}$, the savings-cone slopes, and each term of (31); these sufficient bounds can be much larger than the measured interval. None of the eight larger-state cases is relabeled as having met absolute target merely because its relative gap is small.

9.2. An executed controlled atomic continuum certificate

Consider a continuum of observable establishment sizes $z \in [0, 1]$, fixed over the decision horizon. Condition evolves according to the four-regime, three-action controlled maintenance specified in the replication contract; the horizon is eight and $\beta = 0.95$. Operating rewards and the terminal reward are multiplied by $s(z) = 1 + z$, while revision costs are multiplied by $c(z) = 1 + z/2$. Size and the initially uniform regime are independent, and size is uniform. The

cells	Lower	Upper	Gap/upper (%)	Finite widths	Partition remainder
2	9.926685	10.269496	3.339	0.056918	0.285893
4	10.015768	10.223818	2.035	0.066491	0.141558
8	10.055476	10.198274	1.401	0.072194	0.070603
16	10.074215	10.184476	1.083	0.074982	0.035279

te endpoint problems and all four weighted sums are independently checked. “Finite widths” is the sum of endpoint fiber widths with exact cell weights. The remainder is the displayed total interval width minus that it includes value change and differences between adjacent lower endpoints, not an independently estimated quadrature error.

g allowance is 0.01 for every size, condition, and restart. Every transition is atomic in
 inuous size coordinate. Actions nonetheless change the condition transition probabilities.
 ng v_0 for the base finite-model optimum, Theorem 8 gives

$$K_R = \int_0^1 (1 + z/2) v_0 \left(\frac{0.01}{1 + z} \right) dz. \quad (43)$$

an optimization identity, not an assumed restriction to a parametric policy. On a cell
 onotonicity of v_0 in its allowance supplies a lower endpoint from the finite solve at
 + a) and a feasible upper policy from the tighter solve at $0.01/(1 + b)$. Multiply these
 ts by the exact positive weight $(b - a) + (b^2 - a^2)/4$ and sum. The upper policy uses
 rved size cell at every date and retains common continuations.

6-cell interval is [10.074215, 10.184476], with relative width 1.083%. It is a paired
 ertificate for an uncountable-state controlled atomic model, not a density approximation
 ection of pointwise policy tests. Partition refinement reduces the displayed interval, but
 e root widths remain material. Arbitrarily refining size cells alone does not force this
 n to reach zero width: the finite fibers must also be solved more accurately, as required
 osition 3. The original action-dependent continuous-condition maintenance models are
 t models and retain their existing status.

9.3. Closed atomic continuous-state service economies

der an organization selecting one of m service intensities $a/(m - 1)$ in demand regime i .
 ized demand is $q_i = (2i + 1)/(2n)$, operating reward is $-(q_i - a/(m - 1))^2$, and terminal
 s zero. The installed rule is one intensity below the lowest-index reward-maximizing
 runcated at zero. Keeping that rule has zero implementation cost; a realized departure
 $-q_i + |a - a_i^0|/(m - 1)$. Demand is exogenous to service intensity. Initial regime masses
 $-1)/(n(n + 1))$. Banded, cyclic, and dense transition matrices are specified exactly in
 lement. These are designed known-model economies, not calibrated staffing data.
 ate is not merely a finite regime. It is $(i, u, v) \in \{0, \dots, n - 1\} \times [0, 1]^2$. Conditional
 ime transition $i \rightarrow j$, the continuous coordinates move deterministically according to

$$u' = (u + v^2) \bmod 1, \quad v' = (v + (u')^2) \bmod 1. \quad (44)$$

ear is a translation on one circle conditional on the other coordinate and hence preserves
 measure; their composition does too. Taking uniform measures on the two-dimensional
 ifies both aggregation identities exactly. Every transition law is finitely atomic, including

n	T	m	Law	β	ε	Lower	Upper	Gap
8	8	3	B	0.95	0.01	13.120621355	13.120621379	2.291×10^{-8}
16	16	3	B	0.95	0.01	22.366537651	22.366537682	2.938×10^{-8}
32	32	3	B	0.95	0.01	32.576896364	32.576896401	3.589×10^{-8}
64	64	3	B	0.95	0.01	39.128135567	39.128135608	4.013×10^{-8}
8	8	3	C	0.99	0.05	12.174612806	12.174612832	2.535×10^{-8}
16	16	3	C	0.99	0.05	23.440340924	23.440340965	4.034×10^{-8}
32	32	3	C	0.99	0.05	43.707336818	43.707336878	5.966×10^{-8}
16	16	3	D	0.95	0.01	18.290633115	18.290633156	3.952×10^{-8}
8	8	3	B	1	0.01	15.605690050	15.605690075	2.431×10^{-8}
16	16	3	B	1	0.01	32.013733752	32.013733787	3.352×10^{-8}
32	32	3	C	1	0.05	50.441701001	50.441701067	6.497×10^{-8}
16	16	5	B	0.95	0.01	20.009468627	20.009468717	8.820×10^{-8}

points are rounded outward to nine decimal places; the last column bounds the unrounded arithmetic gap. B, C, and D denote banded, cyclic, and dense regime laws. Both continuous coordinates remain atomic under (44). Independent checks concern finite arithmetic; measure preservation and the aggregation theorem are analytic proofs.

discontinuities of the modulo representation. The theorem applies to arbitrary policies that observe and use both continuous coordinates; it does not assume them away by restricting policies in advance. Payoffs depend on the regime, so this is a structured continuum application rather than evidence of general high-dimensional numerical performance.

The twelve cases were registered on the new branch before execution, including the 60-second limit and absolute certificate target 10^{-4} . Source code was not frozen by that protocol commit; the experiment is consequently distinguished from the earlier fully code-frozen prospective one. No case was removed or replaced. Table IV reports all twelve. Every pair of endpoints independently reconstructed in rational arithmetic. All costs are positive, and every interval meets the registered target.

The largest program has 16,384 variables and 8,192 equalities, corresponding to 4,096 n -date policy simplexes. The supplement separates model assembly, LP proposal, exact certificate construction, independent checking, and process memory. It also reports the same-object reduced Bellman LP with ordinary presolve. Neither formulation branches on probabilities. Structure elimination is faster on several larger rows, whereas the unreduced LP is faster on some small rows. These observations establish tractable closure for this class; they do not demonstrate superiority over the best classical LP formulation or over the earlier McCormick code on its concurrent, action-dependent problems.

9.4. An executed nonzero total-variation transfer budget

For an additional continuous-state model keep the service economy but replace (44) by the in-cell density

$$P(i, u; \{j\} \times dv) = \bar{P}_{ij}[1 + \zeta f(u)f(v)] dv, \quad f(u) = 2u_1 - 1, \quad 0 \leq \zeta \leq 1.$$

Let u, v each denote a point of $[0, 1]^2$. This density is nonnegative and integrates to one. Integrating over a uniform source cell gives a uniform destination cell, so the exact aggregation theorem supplies an independent structural reference optimum. The reset approximation sets $\zeta = 0$. Its operator error, in the convention of Theorem 7, is $\alpha_h = \zeta/2$; its probability total

ζ	Operator α_h	a_h	$2b_h$	Transferred width	Status
0.1	0.05	0.955941	5.846110	—	condition fails
0.001	0.0005	0.009559	0.058461	0.929921	paired
1e-06	5e-07	0.000010	0.000058	0.000934	paired

Successful rows use independently checked finite lower and feasible upper policies at allowances $\varepsilon + 2a_h$ and Failure of the sufficient reset condition is not economic infeasibility: the exact structural reference remains available.

α is $\zeta/4$. Reward, implementation, and terminal approximation errors are all zero. The row uses exactly the same uniform reference measures.

$T = 8$, $m = 3$, $\beta = 0.95$, and $\varepsilon = 0.05$, we construct every primitive and propagated and solve both adjusted finite problems whenever $\varepsilon > 2a_h$. Table V retains the failed reset condition as well as the successful transfers. The supplement gives the two finite widths, $2b_h$, and the repair-cost bound separately. In particular a small measured transfer must not be confused with a small worst-case repair bound.

Exercise instantiates the reset theorem rather than substituting a mesh trend for its assumptions. It also exhibits its conditioning: the largest density perturbation defeats the prescribed range, and the next one gives a wide interval. The exact aggregation result does not incur transfer error. Neither result is applied to the original action-dependent maintenance kernels, so not satisfy the new exogeneity assumption.

9.5. Prospectively fixed common-Markov problems

In generating eight new environments, we committed their seeds, joint parameter changes, parameter hashes, node cap 4,096, wall-clock cap 60 seconds per solver and case, and absolute allowance 10^{-3} . The freeze is commit 13fd3ee; it follows development seed 390001 precedes all eight evaluated seeds. Both solvers see the same known model. The two-state models represent low and high condition; four-state models use a pair of condition indicators in their transition probabilities. Horizons are 3, 4, 8, and 12; discount rates are 0.8 or 0.95; allowances are 0.01 or 0.05; operating prices, transition persistence, transition effects, and initial masses vary jointly. The exact formulas and resulting rational values are in the supplement and proof objects.

Table VI contains all eight pairs, including unfinished searches. A positive absolute-gap certificate is not necessarily a small relative-gap certificate when the optimum is small. We report both. The linear relaxation often improves the lower endpoint substantially, but more per node. The four-state longer-horizon cases expose the combinatorial burden: the range interval is not hidden by a success count. The comparison does not establish that Bellman search dominates direct global programming. Instead it identifies the value and the stronger relaxation under matched information and caps.

9.6. The unchanged continuous-state development cohort

The original maintenance model has $x \in [0, 1]$, three actions, two shocks with probabilities $2/5$, $\beta = 19/20$, stage reward $x - c_a$ with $c = (0, 3/20, 9/20)$, terminal payoff $x/2$, and discount weight $1 + x$. Its two affine successors for actions 0, 1, and 2 respectively are

$$(3x/4 + 1/20), \quad (17x/20 + 1/10, 17x/20 + 7/50), \quad (x/10 + 4/5, x/10 + 17/20).$$

	States	T	Method	Lower	Upper	Gap/upper (%)	Seconds
26	2	3	I	2.8414	2.8425	0.03	0.99
	2	3	M	2.8414	2.8425	0.03	1.77
	2	4	I	0.0000	0.0000	0.00	0.00
	2	4	M	0.0000	0.0000	0.00	0.38
	2	8	I	7.2534	9.0046	19.45	17.19
	2	8	M	8.5290	9.0046	5.28	60.41
	2	12	I	7.1419	7.5553	5.47	16.32
	2	12	M	7.4446	7.5553	1.46	60.40
	4	3	I	2.6157	2.6217	0.23	10.82
	4	3	M	2.6206	2.6217	0.04	35.99
	4	4	I	0.0233	0.0244	4.00	0.47
	4	4	M	0.0235	0.0244	3.13	2.60
	4	8	I	4.6631	7.8481	40.58	60.04
	4	8	M	6.1990	7.8481	21.01	60.41
	4	12	I	4.8936	6.6439	26.34	60.04
	4	12	M	5.8606	6.6439	11.79	60.43

Interval Bellman search. M: verified McCormick search. Bounds concern the identical common randomized Markov certificate for each pair. Reported times include model construction and the operating solve; the search’s 60-second cap is checked between iterations, not by interrupting a proof-producing node. A zero upper cost is an exact zero-cost case. Full stop reasons, node counts, and memory appear in the supplement.

TABLE VII

COMMON RANDOMIZED MARKOV CERTIFICATES ON THE DEVELOPMENT COHORT

Cases	Zero-cost exact	Strict gain over K_D	Improved lower	Maximum gap/upper (%)
14	4	10	8	14.61
14	4	8	7	32.29
14	4	9	8	34.23

gain means that a verified randomized upper cost is below the deterministic lower bound for the same initial law constraints. It does not mean that the randomized optimum is attained by that candidate. Improved lower compares connected local-support bound with the prior randomized support lower bound. All 42 uniform-initial rows and both nonuniform sensitivity tables are retained.

seven installed rules, horizons 4, 8, and 12, and allowances 0.01, 0.05 are unchanged. Neural spline rules are fixed computational incumbents, not estimated economic policies. Exact rating values are inexpensive in this model and their computation is charged. We do not use cohort to claim performance when operating approximation itself is difficult.

The inherited deterministic comparison has 12 zero-cost and 12 positive-cost integrated equalities. Ten of the positive-cost equalities occur at horizon 4, two at horizon 8, and none at horizon 12. The maximum deterministic relative gaps at each horizon are reported alongside absolute gaps in the supplement. In the horizon-8 neural31003 case at $\varepsilon = 0.05$, the two initial functions are identical pointwise, but the arrays differ at later restart dates. Thus initial equality does not imply equality at every restart. The new diagnostic records the full rational agreement locus.

The lotteries in (39) certify strict improvement over *every* feasible deterministic policy in 27 of the 42 uniform-initial configurations (Table VII). The horizon-12 defer case at allowance

T	Positive	Median (%)	Max. (%)	Strict gains	Min. gain	Max. gain
4	10	5.24	14.61	10	0.021898	1.041948
8	10	14.33	32.29	8	0.058701	0.896320
12	10	15.75	34.23	9	0.044193	0.747177

$(U_R - L_R)/U_R$ among positive-cost intervals. Savings are $L_D - U_R$, not a difference between lower and upper bounds. The last two columns condition on strictly positive savings. All 42 signed savings are in the supplement. These are exact re-tabulations of retained certificates, not new optimization runs.

TABLE IX
SELECTED NONLINEAR COMMON-MARKOV INTERVALS

T	Mesh per axis	Lower	Upper	Gap/upper (%)	Independent check
8	64	0.9438	3.9348	76.01	full
16	128	4.1908	7.6644	45.32	not independent
32	64	6.5389	16.7884	61.05	full
32	128	8.1264	12.5226	35.11	full R40
32	256	8.8718	11.8492	25.13	full R41

on initial law on the square and operating allowance 1/2. Inherited full independent scalar checks cover (8, 64) and (32, 64). R40 additionally replays the complete (32, 128) object, including the candidate upper bound. R41 subsequently independently rebuilt the complete (32, 256) object; this revision incorporates that successful arithmetic audit with its source and proof hashes.

For example, has a randomized interval [15.3172, 15.4670] and a positive deterministic lower bound. This addresses the economically relevant policy-class difference directly, without requiring a fixed-restart history-conditioned frontier. The 12 zero-cost cases are already covered by feasibility and nonnegativity and are not counted as nontrivial search successes. All five-cost randomized intervals retain a reported nonzero width; the narrowest and widest intervals are distinguishable in the full table.

Increasing the initial density to $2x$ yields 28 strict deterministic separations; changing it to x yields 27. These are sensitivity results on the same developed environments, not independent observations. The connected local program improves 23 uniform lower endpoints, including the longer-horizon cases; it thus affects the primary global interval rather than a secondary local diagnostic. It still does not establish global strong duality or close every

9.7. Nonlinear two-state bounds

The retained nonlinear model has $(x, y) \in [0, 1]^2$, four actions, bilinear transitions, two shocks, a terminal payoff $(x + y)/2 - c_a$, terminal payoff $(x + y)/4$, and revision weight $1 + (x + y)/2$. Its exact primitive specification is reproduced in the supplement. Whole-box operating witnesses support supersolutions use outward arithmetic. The new lower construction uses 11 global prices and five nonnegative local multipliers, including zero, at every date. It is paired exactly with the retained feasible upper policies. We run no new upper-portfolio competition to compare unequal proposal success fractions as evidence of optimality.

The old zero lower endpoint was not forced by the definition of revision cost: the operating bounds can force departures from the installed rule. Nonzero support supersolutions now

h 25.13%. This is a genuine improvement over $[0, 11.8492]$, but it is not a near-exact
 mum or a high-dimensional scalability result. Both endpoints and the all-restart operating
 uality are independently rebuilt at $T = 32, N = 64$, the R40 audit extends this to the
 plete $T = 32, N = 128$ object, and the R41 audit extends it to the finest $T = 32, N = 256$
 ct. The latter independently checks 92,274,688 support-action inequalities over 2,097,152
 -date cells; the original interval is unchanged. Memory for the retained support arrays grows
 $N^2(T + 1)(11 + 4)$ bytes before temporary arrays and interpreter overhead; the machine
 rd reports the actual payload and process high-water mark.
 he allowance $1/2$ belongs to the inherited exploratory nonlinear design. It is not made
 comparable to 0.01 simply by normalizing state variables: the operating scale and remaining
 zon matter. We additionally compute lower-bound diagnostics at $\varepsilon = 0.05$ on four stated
 igitations. None is paired with a policy certified only at $1/2$. Without a newly feasible
 er policy those tighter-tolerance rows remain lower-bound-only results. A failed sufficient
 certificate is labeled uncertified, not infeasible. A box crossing an installed-rule threshold
 charge the upper revision weight conservatively; it does not assert that every realized state
 at box changes action.

10. VERIFICATION, ECONOMIC INTERPRETATION, AND LIMITS

certificate has three distinct burdens. Its mathematical reduction must preserve the target
 ey class. Its finite arithmetic must check the asserted inequalities with the correct direction.
 ublication must correspond to the checked objects. Hashes address the third burden, not
 first two. Independent arithmetic addresses part of the second burden and can expose a
 el-contract error, but is not a proof that every economic assumption is appropriate.
 he finite-state verifier uses SymPy rationals and imports neither search implementation
 he optimizer. It rebuilds the common Markov equations, primitive contracts, every policy-
 split, every necessary contraction, the LP residual bounds, and the global leaf minimum.
 continuous-state verifier independently checks the new lotteries, whole-cell costs, support
 rsolutions, and local primal vertices, using the prior independent primitive parser rather
 the constructor's piecewise-affine operations. The nonlinear verifier uses symbolic rational
 er geometry and arbitrary-precision scalar integer induction instead of the constructor's
 orized box operations. Its recorded coverage now includes the complete finest $(32, 256)$
 ct, not merely selected rows. The R42 structural checker separately reconstructs every
 finite primal and dual certificate without importing its constructor. The verification report
 ifies precisely which objects passed; it never substitutes cross-run equality or a mutation
 for an inequality proof.

he synthesis preserves all prior scientific material. Deterministic bounds, history-conditioned
 tiers, the two-period analytic randomization example, adaptive local-LP analysis, 66 eco-
 ic sensitivities, failed runs, and the stopped-control derivations remain available in full. The
 rate historical volume preserves the prior article and technical supplement verbatim; the
 ent supplement supplies a direct map of the new results. The older 349-page historical annex
 ains unchanged as an archival object rather than a prerequisite for reading this article. The
 ped consumption–portfolio problem is not relabeled solved: its inherited full-domain bound
 pproximately 7.1818 does not meet its 0.01 target. A killed Bellman comparison supports
 ess propagation but supplies neither missing action coverage nor a diffusion-discretization
 r bound (Kushner and Dupuis, 2001).

he economic conclusion established by the new primary calculation is precise. When random-
 action assignment is admissible and costs attach to realized departures, common Markov

rule. This conclusion holds on the certified configurations and initial laws. It does not how frequently organizations permit such lotteries, how costly they are to administer, monetary welfare effect of a real deployment. Fixed randomization overhead, parameter nty, and empirical calibration would require additional primitives and certificates. The perturbation holdout study tests a fixed algorithm beyond its development seed, but is not a generalization theorem.

10.1. Model uncertainty and the economic margin for randomization

certificates can be translated into a robustness statement without treating designed al primitives as estimated data. Fix a nominal model, initial law, and policy class. a nonempty set of alternative models has verified uniform bounds

$$|J_t^{p,M} - J_t^{p,0}| \leq a, \quad |V_t^M - V_t^0| \leq a, \quad |C_0^{p,M} - C_0^{p,0}| \leq b \quad (45)$$

y common policy p , state, and date, and every M in the set. Superscript 0 denotes the model. The error recursion in Section 6 supplies such bounds from uniform reward, minimal, and kernel enclosures on a common state space; no reset aggregation is needed for d-space perturbation statement. Uniformity over policies is essential. A fitted residual only under the proposed policy does not establish (45).

POSITION 4—Portable operating and cost intervals: *Let $\varepsilon > 2a$. A nominal policy at allowance $\varepsilon - 2a$ is feasible at allowance ε in every alternative model. If L_+ is al lower bound at allowance $\varepsilon + 2a$ and U_- is the nominal cost of such a tightened policy, then every model-specific common-policy optimum belongs to*

$$[\max\{0, L_+ - b\}, U_- + b].$$

the interval encloses the optimum that minimizes worst-case implementation cost over feasible in all the alternative models. In the latter assertion the upper endpoint uses a policy, not a different policy for each model.

F: For the proposed policy, $J^{p,M} \geq J^{p,0} - a \geq V^0 - \varepsilon + a \geq V^M - \varepsilon$. Conversely feasible in a model M satisfies $J^{p,0} \geq J^{p,M} - a \geq V^M - \varepsilon - a \geq V^0 - \varepsilon - 2a$. Its cost is at least L_+ and its cost in M at least $L_+ - b$. Take infima for the model-specific nt. For the robust problem, the first implication supplies one policy for all models and nd applies to every robustly feasible policy in each model. Taking the worst-case cost s the bounds. *Q.E.D.*

ct randomized separation also has an economically interpretable administration-cost ce. Let U_R be the certified cost of a feasible common lottery and L_D a lower bound for feasible deterministic policy, for the same operating constraint and initial law. Suppose g payoffs and dynamics are unchanged by administering the lottery. A nonnegative extra ration charge at date t bounded by h_t increases its expected discounted cost by at most $\sum_{t < T} \beta^t h_t$. If

$$U_R + H < L_D, \quad (46)$$

lottery remains cheaper than every deterministic feasible policy even after this additional This conclusion does not require administration at every visited state; charging the

T	Installed rule	ε	Det. lower	Lottery upper	h_* lower bound
4	defer	0.01	4.93668	4.73011	0.055684
4	defer	0.05	4.93206	3.89012	0.280857
4	neural31001	0.01	0.11117	0.08928	0.005902
8	defer	0.01	10.81072	10.63794	0.025667
8	defer	0.05	10.75071	9.85440	0.133151
8	neural31001	0.01	0.46864	0.40232	0.009853
12	defer	0.01	15.61281	15.46691	0.015872
12	defer	0.05	15.47129	14.72413	0.081278
12	neural31001	0.01	3.49393	3.36015	0.014553

form initial law, unchanged development economy. Endpoints and the threshold are rounded outward. A positive threshold means that every smaller constant per-date overhead preserves a strict cost advantage over every feasible deterministic policy. All 126 rule/horizon/tolerance/initial-law margins are retained in the machine-readable results, including zeros.

and everywhere is conservative. Deterministic rules are assumed not to pay the randomization-specific charge.

for a common per-date ceiling h , the certified break-even threshold is

$$h_* = (L_D - U_R)_+ \bigg/ \sum_{t=0}^{T-1} \beta^t. \quad (47)$$

ry $h < h_*$ preserves the strict ranking. Equality at h_* yields no strict conclusion. For model uncertainty, use a deterministic nominal lower bound at $\varepsilon + 2a$ and a nominal lottery feasible $-2a$; then replace the numerator by $(L_{D,+} - U_{R,-} - 2b)_+$. The current development reports $a = b = 0$: they calculate economic margins under known designed primitives, not irrational uncertainty radii.

11. CONCLUSION

local operating requirement becomes a global revision certificate only when continuation compatibility and the cost of restoring feasibility are controlled together. Backward repair supports that connection. A date-dependent allowance ladder covers finite undiscounted problems, the exact operating-action restrictions handle zero tolerance. These extensions retain the same common randomized Markov objective rather than replacing it by a necessary-action relaxation. For endogenous finite-state dynamics, the operating-action disadvantage provides a probability bound, and implementation savings obey a regret cone. Their combination with reference-acted repair yields a quadratic small-allowance bound for a root relaxation at fixed action variation. This is a structural rate, not a polynomial guarantee for the unrestricted global problem. In the controlled audit, tighter relaxation geometry can materially improve a certificate, difficult high-discount cases and solver stops remain in the evidence.

observable-covariate disintegration supplies a separate controlled atomic continuum route. It serves every Borel common policy by using a label recoverable from the current state, and its action bounds converge when the finite fiber solves and primitive errors are refined. Exact endogenous aggregation remains a tractable complementary class; its twelve executed service economies retain tightly closed positive-cost intervals. The density exercise retains all primitive, repair, and repair terms, including a failed sufficient tightening condition.

ons remain valid, all 30 positive-cost randomized intervals remain open, and the non-interval remains wide despite complete independent arithmetic verification. Generic eness, a small-allowance rate, observable-fiber convergence, exact structural solvability, signed policy-class comparison and preference questions. Stating that a suspension their unsuccessful executions permits the method to advance without altering its objective or converting a certificate into an unsupported empirical claim.

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This supplement gives exact primitive contracts, complete computational accounting, the transfer budget, a discounted infinite-horizon tail result, and a theorem-to-checker map. The prior article and technical supplement are retained without modification in a separate historical volume. Numerical proposal, independent arithmetic verification, structural mathematical assumptions, and publication provenance are distinguished throughout.

1. CONTROLLED-MODEL PRIMITIVE CONTRACT AND COMPLETE ACCOUNTING

The controlled cases use the exact generator in `regret.py`, independently reconstructed by `lit.py`. Regimes are $i = 0, \dots, n - 1$. For each regime and action, four integer weights are given, in that order, from

$$\max\{1, 38 - 10a\} + u_1, 36 + u_2, 12 + 7a + u_3, 2 + 11a + u_4), \quad u_j \in \{0, 1, 2, 3\}.$$

Normalize these weights and assign them to successors $\max(0, i - 1)$, i , $\min(n - 1, i + 1)$, and n , combining coincident successors. The seeded standard-library pseudorandom generator is advanced in regime/action order. The stage reward is

$$r_{ia} = \frac{i}{n - 1} - c_a - \frac{av}{100000}, \quad c = (0, 9/100, 8/25, 11/20), \quad v \in \{1, \dots, 19\},$$

where v is drawn immediately after the four transition weights. The installed action is zero for $i \leq \lfloor n/2 \rfloor$ and is the seed modulo two otherwise. The implementation cost is $(1 + i/(n - 1))$ and i is the indicator of a realized departure. The terminal operating value is $i/[2(n - 1)]$, and the initial law is uniform. Transitions, rewards, and costs are constant over dates; Bellman advantages generally are not. All primitives and every declared seed are retained in the protocol and compressed proof objects.

The original registration is `commit 18cbcf4c1a4c35660be824f6984f68a1d38288a9`. The closed arithmetic amendment is `950b1f1ac2ebaf974d4ef528add8b2c2e4ce72b`. The amendment retains all cases and changes only certificate accumulation and its theoretical diagnostics. Its numerical constructor, independent reader, and runner hashes are checked against the final sources. This sequence is not called a fully code-frozen holdout. The old source partial execution record remain in the R43 history directory.

TABLE I: Controlled audit: complete proposal and verification work

Case	Method	Coordinates	Variables	Nonzeros	Build	LP	Arithmetic	Check	Bytes
00	U	24	144	1102	0.005	0.006	0.011	0.015	1380
00	S	24	144	1150	0.005	0.004	0.010	0.017	1290
01	U	128	1040	8611	0.039	0.050	0.081	0.123	6804
01	S	128	1040	8867	0.040	0.072	0.084	0.130	7484
02	U	384	3408	29018	0.176	0.340	0.290	0.456	19833

	Nodes	Coordinates	Variables	Nodes	Bound	LP	Primal	Gap	Bytes
	384	3408	29786	0.159	0.367	0.297	0.481	19963	
J	1152	10200	89840	0.570	9.077	0.940	1.574	56498	
S	1152	10200	91376	0.590	27.514	0.969	1.580	56699	
J	1024	9488	81406	0.612	2.182	0.915	1.573	52718	
S	1024	9488	83454	0.634	2.032	0.938	1.599	52390	
J	768	6488	57088	0.307	2.328	0.576	0.895	37114	
S	768	6488	58112	0.315	1.935	0.632	0.955	40608	
J	1536	13720	121698	0.852	30.052	1.000	1.951	2491	
S	1536	13720	123746	0.870	30.052	1.002	1.955	2495	
J	2048	19472	169793	1.635	30.078	1.608	3.468	3575	
S	2048	19472	173889	1.641	30.077	1.673	3.541	3583	

are seconds. Coordinates are $nT(m-1)$, not just nT . Each row uses one root relaxation, tire certified bound improvement is a one-node result; this is not a node-count scaling ent. The LP has a 30-second cap. Model construction, exact constructor arithmetic, and dent verification are additional. Process high-water memory is recorded in the raw rows. e runner is sequential, that memory includes earlier allocations and cannot identify a er-method memory advantage. Comparing isolated peak memory would require separate ocesses. No such comparison is asserted.

2. DIRECTED CERTIFICATE ARITHMETIC AND INDEPENDENT RECONSTRUCTION

the root relaxation as $\min c'z$ subject to $Ez = e$, $Az \leq b$, and $\ell \leq z \leq u$. Any rational and any rational $\mu \geq 0$ give the lower bound

$$y'e - \mu'b + \sum_j \min\{h_j\ell_j, h_ju_j\}, \quad h = c - E'y + A'\mu.$$

exact reference cost $\nu'H_0$ and truncate at zero. A floating-point LP success flag is neither y nor sufficient for this inequality. Numerical equality marginals and sign-corrected ty marginals are merely proposals for y, μ .

culating all distinct rational denominators before rounding is unnecessarily expensive. lemented reader and constructor instead use denominator 2^{40} . They round each constant tion downward. For each residual coefficient, they accumulate dyadic lower and upper es $h_j^- \leq h_j \leq h_j^+$. The safe contribution is the downward-rounded minimum of

$$h_j^-\ell_j, h_j^-u_j, h_j^+\ell_j, h_j^+u_j.$$

ll four products is necessary when a savings coordinate can be negative. This procedure or arbitrary proposed multipliers with the required sign, and keeps denominator growth ontrol. The complete lower endpoint, including reference cost, is retained as a rational

ne upper endpoint, the constructor repairs backward toward the reference action. It a local inward margin before rounding nonreference probabilities down to denominator unused probability is assigned to the reference action. At a fixed repaired continuation, dding-induced regret change is at most $(m-1)2^{-40} \max_a |q_a - q_{a^*}|$. If the reserved aves insufficient reference slack, the constructor uses the feasible reference action.

Case	ε	$\underline{d}$	$\lambda_+ - \lambda_-$	$\overline{E}_D$	$E_{S,0}$	Width bound	
34	00	1/100	0.00371	603	0.0027	1.63	16.9
	01	1/100	0.00061	1.2e+03	0.0167	19.8	75.1
	02	1/100	0.000116	1.24e+04	0.126	1.53e+03	1.64e+03
	03	1/100	0.000331	5.05e+03	0.0563	285	799
	04	1/1000	2.2e-05	8.49e+04	0.00709	602	767
	05	1/1000	0.000156	1.03e+04	0.000715	4.99	376
	06	1/20	6.2e-05	2.99e+04	4.73	1.27e+05	1.28e+05
	07	1/100	7.44e-05	2.53e+04	0.129	3.18e+03	4.04e+03

s are descriptive decimal approximations of retained rational quantities. Error recursions are rounded upward to c numbers. The sufficient bound concerns an exact minimizing LP policy and exact repair; a numerical proposal and endpoint arithmetic losses are not included in this theoretical column. Actual certified widths in the main table de those losses and can be larger when a solver stops. Constants are not suppressed when they are uninformative.

se are proposal choices. The independent reader ignores the repair routine and checks serialized probabilities by the original operating and implementation Bellman equations. Every probability is nonnegative and sums to one, and every date/regime satisfies the original, modified allowance.

he independent endpoint reader imports neither the constructor nor NumPy, SciPy, or any mizer. It reconstructs the operating optimum, reference policy, disadvantage and savings coefficients, all variable bounds, all product envelopes, and every cone and Bellman constraint. Further independent audit reconstructs the declared random primitive sequence and checks a complete proof against its registered parameters. It also independently recomputes the minimum weighted sums. These checks do not mechanically prove the analytic repair, rate, or integration theorems. Four retained mutations test rejection of a forged endpoint, negative on probability, negative inequality multiplier, and invalid transition mass.

3. ALL CONTROLLED CONTINUUM ENDPOINTS

he model has a persistent observed size $z \in [0, 1]$. Regime transitions are controlled but dependent of size. Operating primitives are multiplied by $1 + z$ and implementation costs by $z/2$. Thus the finite allowance is $0.01/(1 + z)$; it is not 0.01 after rescaling. The following e includes all 17 endpoints used by every displayed partition.

TABLE III: Every finite endpoint used in the atomic continuum certificate

Endpoint	Allowance	Lower	Upper	Width	Check (s)
0	0.01000000	7.765997	7.883084	0.117085733	0.054
1	0.00941176	7.822697	7.929826	0.107128306	0.054
2	0.00888888	7.872910	7.971275	0.098364582	0.053
3	0.00842105	7.918003	8.007721	0.089717171	0.053
4	0.00800000	7.958346	8.040570	0.082223429	0.054
5	0.00761904	7.994660	8.070364	0.075703281	0.053
6	0.00727272	8.027518	8.097495	0.069976171	0.054

Endpoint	Allowance	Lower	Upper	Width	Check (s)	
7	0.00695652	8.057475	8.122294	0.064817828	0.054	
8	0.00666666	8.084816	8.145021	0.060203701	0.054	
9	0.00640000	8.109872	8.165848	0.055975509	0.054	
10	0.00615384	8.132909	8.185072	0.052162255	0.054	PRESERVED ARTICLE AND TECHNICAL HISTORY
11	0.00592592	8.154153	8.202063	0.047908675	0.054	
12	0.00571428	8.173805	8.218688	0.044882388	0.054	
13	0.00551724	8.192044	8.234190	0.042144623	0.054	
14	0.00533333	8.209073	8.248540	0.039466645	0.054	
15	0.00516129	8.224969	8.260548	0.035578105	0.054	
16	0.00500000	8.239809	8.273341	0.033531010	0.054	

point j corresponds to $z = j/16$. Allowances are printed downward for display only; the solver and checker use the exact rational allowance. Lower endpoints are rounded down, upper endpoints and widths up. The 2-, 4-, 8-, and 16-cell calculations reuse subsets of these rounded endpoints. No new model or initial law is substituted when the partition changes. The general observable-fiber theorem, the bijectivity of the covariate evolution has an interesting content: the label that selects a continuation is recoverable from the current observed noninvertible maps can merge different labels. Selecting a different continuation for a merged label would add memory and would not prove the stated common-Markov result. Hence, an action-dependent continuous-condition transition is not covered merely because it has many successors. The original affine and nonlinear condition models are therefore supported as separate, open-width evidence.

4. EXACT SERVICE-ECONOMY CONTRACTS

States are indexed by $i = 0, \dots, n - 1$. Demand, rewards, installed actions, costs, and initial laws are defined in the main article. Ties in reward maximization use the lowest action index. Laws are rational. The three regime laws are as follows. In the banded law, $P_{i,i+1} = P_{i+1,i} = (\text{mod } 3)/16$ and P_{ii} is the remaining row mass; other entries are zero. The cyclic law has $P_{i,i+1} = 1/4$ and $P_{i,(i+1) \bmod n} = 1/4$. The dense law has $P_{ij} = 1/(2n) + (1/2)\mathbf{1}\{i = j\}$. Each law is stochastic and independent of the action. These definitions, including the installed rule and transition distribution, are reconstructed independently by the checker.

The continuous state space consists of n disjoint copies of $[0, 1]^2$ equipped with their Borel σ -algebras; the half-open convention assigns each modulo boundary a unique image. If $(u, v) = (u + v^2 \bmod 1, v)$, then for every bounded Borel h , Fubini's theorem and translation invariance on the circle give $\int h(S_1(u, v)) du dv = \int h(u, v) du dv$. The analogous identity for $S_2(u, v) = (u, v + u^2 \bmod 1)$. Their composition preserves uniform measure. Conditioning on each possible regime successor, this is precisely the pushforward identity required by the aggregation theorem. Its proof concerns arbitrary bounded Borel test functions, not transition indicators. Pointwise regime transition probabilities give the other aggregation theorem. The transition from an individual state is nonetheless a finite sum of Dirac masses. The within-cell density alternative is $1 + \zeta f(u) f(v)$, where $f(u) = 2u_1 - 1$ on $[0, 1]^2$. Since $\int |f| = 1/2$, it is a probability density for $|\zeta| \leq 1$, and its source average is the destination law. For a fixed source, its L_1 difference from the reset density is $|\zeta f(u)|/2$. The supremum over sources is $|\zeta|/2$ (a supremum need not be attained on the half-open cell). The operator norm over test functions bounded by one. Probability total variation is

TABLE IV: All structured cases: work and proof-object size

Case	nT	Variables	Nonzeros	LP (s)	Full LP (s)	Exact (s)	Check (s)	Bytes
1	64	256	538	0.0032	0.0036	0.0057	0.0065	2233
2	256	1024	2226	0.0063	0.0140	0.0266	0.0355	5648
3	1024	4096	9058	0.0191	0.1127	0.1546	0.2239	17114
4	4096	16384	36546	0.2618	1.2414	0.9772	1.6081	56086
5	64	256	496	0.0032	0.0036	0.0051	0.0063	1976
6	256	1024	2016	0.0079	0.0119	0.0248	0.0333	5123
7	1024	4096	8128	0.0326	0.0753	0.1440	0.2201	15921
8	256	1024	5376	0.0074	0.0227	0.0391	0.0394	5135
9	64	256	538	0.0032	0.0038	0.0057	0.0064	2188
10	256	1024	2226	0.0064	0.0140	0.0267	0.0348	5616
11	1024	4096	8128	0.0325	0.0781	0.1515	0.2075	15251
12	256	1536	3250	0.0079	0.0154	0.0321	0.0416	6577

LP” is the eliminated regret program; “Full LP” retains operating and implementation man values with the same common policy. Both use HiGHS with presolve and the same erical feasibility tolerances and 60-second solver limit. They solve the same rational mathe- cal problem, but their floating-point objective estimates are not the certificates. The exact points use a separately evaluated rational policy and rational equality-multiplier residual ection. A 10^{-9} inward allowance is a proposal device only. Policy and dual multipliers are ded on denominator 10^{12} ; the checker always uses the original declared allowance. Every s exact repair fraction was zero in the recorded execution; the feasibility assertion follows n exact evaluation, not from that observation or the floating-point solver status.

he raw result file additionally reports model preparation time, solver iterations, original mulation dimensions, and process high-water resident memory. Memory is measured per tructor subprocess and includes Python, NumPy, SciPy, and the optimizer; it is not isolated f-array memory. The independent checker runs in a separate optimizer-free process. Table es are single-run descriptive timings, not statistical speed estimates. For the largest sparse ram the exact checker is more expensive than the LP proposal. This cost is included rather being hidden in publication overhead.

or the regret program, order the equality multipliers as s_{ti} for simplex equations and d_{ti} for et equations. With $\ell_{ia} = \max_b r_{ib} - r_{ia}$, the reduced costs are

$$h_{tia} = \beta^t w_{ti} k_{ia} - s_{ti} + \ell_{ia} d_{ti}, \quad h_{ti}^D = -d_{ti} + \beta \sum_j P_{ji} d_{t-1,j},$$

re the preceding-date term is zero at $t = 0$. The lower certificate is

$$\max \left\{ 0, \sum_{ti} s_{ti} + \sum_{tia} \min(0, h_{tia}) + \varepsilon \sum_{ti} \min(0, h_{ti}^D) \right\}.$$

6. ALL TERMS OF THE CONTINUOUS TRANSFER BOUND

The density experiment uses $R = \max_{i,a} |r_{ia}|$, $K = \max_{i,a} k_{ia}$, and $G = 0$. Starting at the initial date with zero error and zero payoff envelope, backward induction computes

$$b_{t+1} + \beta \alpha_h B_{t+1}^J, \quad b_t = \beta b_{t+1} + \beta \alpha_h B_{t+1}^C, \quad B_t^J = R + \beta B_{t+1}^J, \quad B_t^C = K + \beta B_{t+1}^C.$$

where a_h and b_h are the maxima over dates. The two finite problems have allowances ε and $\varepsilon - 2a_h$. The first supplies a lower endpoint; the second supplies a feasible lifted policy. Their independently checked arithmetic widths are η_+ and η_- . No finite solve is required when the tightening is nonpositive. The repair term is $D_C 4a_h / [(1 - \beta)(\varepsilon - 2a_h) +$

TABLE V

COMPUTED TERMS IN THE SUFFICIENT TRANSFER-WIDTH BOUND

ζ	η_+	η_-	$2b_h$	Repair bound	Total bound
0.001	2.232×10^{-8}	2.291×10^{-8}	0.058461	73.581005	73.639466
1e-06	2.291×10^{-8}	2.291×10^{-8}	0.000058	1.153667	1.153725

The displayed total is computed from rational values, with decimal displays only for readability. The measured interval width in the main article is appreciably smaller than this worst-case bound. For $\zeta = 1/10$, a_h exceeds the permissible tightening threshold; that is, it is not sustained with its actual error budget and a failed sufficient condition. The exact structural analysis is still available because its two aggregation identities hold at that parameter as well. The failure of a particular error bound from failure of the economic model or of the computational problem.

7. AN EXACT TAIL BRIDGE FOR DISCOUNTED INFINITE-HORIZON REVISION

The main computational experiments are finite horizon. There is also a simple infinite-horizon extension with an explicit terminal trust boundary. Consider time-homogeneous bounded payoffs, finite actions, and $0 < \beta < 1$. Let V^∞ be the exact operating fixed point and let a^* be the stationary maximizing selector. Allow common Markov policies to be time dependent, and suppose $J_t^p(x) \geq V^\infty(x) - \varepsilon$ for every date and state. Denote this infinite-horizon minimum by v_T . Let v_T be the finite-horizon revision minimum with terminal operating payoff V^∞ and terminal implementation cost.

PROPOSITION 1—Tail enclosure: *For any initial law and $\varepsilon \geq 0$,*

$$v_T \leq K_R^\infty \leq v_T + \frac{K\beta^T}{1 - \beta}.$$

A finite certificate $[L_T, U_T]$ gives the infinite-horizon certificate $[L_T, U_T + K\beta^T / (1 - \beta)]$.

T by the operating-optimal selector weakly improves operating values at earlier dates. Its changed prefix is therefore feasible for the finite problem with terminal payoff V^∞ , and its stated implementation cost does not exceed its full nonnegative cost. Taking infima proves lower bound. Conversely, append a^* to any finite feasible prefix. This supplies an operating-feasible common Markov policy at all dates and adds at most $K\beta^T/(1-\beta)$ implementation cost. Approximate minimizers prove the upper bound without assuming a Borel function-space minimizer exists. *Q.E.D.*

This proposition does not give a stationary-policy optimum: the admissible prefix is time dependent. It also does not treat an approximate terminal value as exact, or an average-cost problem as discounted. A numerical use requires a certified terminal fixed point and tail policy, additional witness errors. No infinite-horizon execution is claimed in the tables.

8. DATA-DEPENDENT REPAIR ACCOUNTING

For any two finite-horizon common policies p, q , define the implementation action values $C_t^p(x, a) = k_t(x, a) + \beta P_t^a C_{t+1}^p(x)$. Their exact performance difference under the state law induced by q is

$$\int (C_0^q - C_0^p) d\nu = \sum_{t=0}^{T-1} \beta^t \mathbb{E}_q \left[\sum_a (q_t(a | X_t) - p_t(a | X_t)) Q_{k,t}^p(X_t, a) \right].$$

Subtract the two cost Bellman equations and iterate the resulting q -transition recursion to prove identity. If $q_t = (1 - \alpha_t)p_t + \alpha_t \delta_{a_t}$, its additional cost is at most the discounted q -occupancy measure of α_t times the action span of $Q_{k,t}^p$. For a given finite repaired policy this can be much sharper than $D_C \sup \alpha_t$. The generic stopping theorem still needs a uniform bound over explored policies; a favorable realized repair cost does not replace that requirement. The new structural cases use exact primal evaluation, and do not charge an unnecessary worst-case repair cost to their actual upper endpoints.

9. ALL SIGNED SAVINGS IN THE RETAINED PRIMARY COHORT

The next table is an exact re-tabulation of R40 rational endpoints. It does not change any policy or rerun the primary optimizer. A positive signed saving $L_D - U_R$ proves a gain over any feasible deterministic policy; a nonpositive value establishes no such separation. None of the signs establishes equality of L_R and U_R .

TABLE VI: All 42 uniform-initial primary certificates and signed deterministic savings

T	Installed rule	ε	L_R	U_R	$L_D - U_R$	Width (%)
4	defer	0.01	4.617419	4.730105	0.206584	2.38
4	defer	0.05	3.664042	3.890120	1.041948	5.81
4	neural31001	0.01	0.085102	0.089277	0.021898	4.68
4	neural31001	0.05	0.026223	0.030155	0.044177	13.04
4	neural31002	0.01	0.120480	0.125168	0.024289	3.75

instances	rule	C	E_R	C_R	E_D	C_R	$Width(\%)$
4	neural31002	0.05	0.047964	0.054026	0.058588		11.22
4	neural31003	0.01	0.094786	0.099085	0.022055		4.34
4	neural31003	0.05	0.031787	0.037227	0.047069		14.61
4	occupancy_stress	0.01	3.906911	4.020718	0.186083		2.83
4	occupancy_stress	0.05	3.020221	3.248042	0.941433		7.01
4	spline129	0.01	0.000000	0.000000	0.000000		0.00
4	spline129	0.05	0.000000	0.000000	0.000000		0.00
4	spline33	0.01	0.000000	0.000000	0.000000		0.00
4	spline33	0.05	0.000000	0.000000	0.000000		0.00
8	defer	0.01	10.503551	10.637940	0.172786		1.26
8	defer	0.05	9.449120	9.854394	0.896320		4.11
8	neural31001	0.01	0.343445	0.402312	0.066330		14.63
8	neural31001	0.05	0.104088	0.140458	-0.015946		25.89
8	neural31002	0.01	0.225489	0.277716	0.058701		18.81
8	neural31002	0.05	0.040296	0.059511	-0.015198		32.29
8	neural31003	0.01	0.552828	0.653902	0.211403		15.46
8	neural31003	0.05	0.086348	0.100439	0.244576		14.03
8	occupancy_stress	0.01	9.173627	9.365726	0.125235		2.05
8	occupancy_stress	0.05	7.983805	8.565982	0.682582		6.80
8	spline129	0.01	0.000000	0.000000	0.000000		0.00
8	spline129	0.05	0.000000	0.000000	0.000000		0.00
8	spline33	0.01	0.000000	0.000000	0.000000		0.00
8	spline33	0.05	0.000000	0.000000	0.000000		0.00
12	defer	0.01	15.317224	15.466905	0.145915		0.97
12	defer	0.05	14.142037	14.724122	0.747177		3.95
12	neural31001	0.01	2.770956	3.360150	0.133784		17.53
12	neural31001	0.05	1.584778	2.023532	0.379048		21.68
12	neural31002	0.01	2.051734	2.385046	0.151782		13.98
12	neural31002	0.05	1.195731	1.455145	0.309646		17.83
12	neural31003	0.01	0.772482	1.010276	0.217318		23.54
12	neural31003	0.05	0.093464	0.142107	-0.001624		34.23
12	occupancy_stress	0.01	13.420033	13.655059	0.044193		1.72
12	occupancy_stress	0.05	12.158413	12.877517	0.365025		5.58
12	spline129	0.01	0.000000	0.000000	0.000000		0.00
12	spline129	0.05	0.000000	0.000000	0.000000		0.00
12	spline33	0.01	0.000000	0.000000	0.000000		0.00
12	spline33	0.05	0.000000	0.000000	0.000000		0.00

10. THEOREM-TO-CHECKER MAP AND EVIDENCE CLASSIFICATION

black ladder	Analytic Borel proof; 48 exact tests with action-dependent three-state, three-action kernels, including $\beta = 1$	Finite tests are not a machine proof; the rate deteriorates with horizon.
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Atomic aggregation	Two explicit kernel identities and a proof over arbitrary Borel policies; exact shear pushforward identity	Not a convergence theorem for arbitrary controlled atomic kernels or approximate lumpability.
Structured LP intervals	Independent rational probabilities, all-restart regrets, actual cost, and residual-corrected lower bounds for every new row	No reliance on solver success as a proof; no assertion about the continuum deterministic optimum.
Density transfer	Nonzero operator error and full propagated budget, plus independent finite certificates at both adjusted allowances	No assertion that every error budget is useful; the failed tightening row is retained.
R41 finest nonlinear audit	Complete independent scalar rebuild of the retained object (32, 256, 1/2) in the previously successful recorded run	Not a new upper policy, tighter operating tolerance, narrower optimum interval, or local re-run in R42.
Protocol and publication	Pre-execution case registration; content hashes and additive preservation checks	The source was not frozen in the case-registration commit; hashes do not verify a mathematical inequality.
Economic interpretation	Known-model adjustment costs and all-restart operating protection	No estimated primitives, empirical confidence sets, or monetary welfare estimates.

he inherited R41 audit reports 92,274,688 support-action inequalities and 2,097,152 state-cells. It independently reproduces upper cost and lower bound

$$U = \frac{1667622841980339}{140737488355328}, \quad L = \frac{2497217525995195}{281474976710656},$$

regret at most $2147483439/4294967296 < 1/2$. Its arithmetic replay took approximately 6 seconds. The manifest retains the source commit, verifier hash, proof hash, successful kflow identifier, and retrieved result. R42 does not misclassify this recorded prior run as a local re-execution.

11. UNCHANGED EARLIER TECHNICAL MATERIAL

ll earlier repository files remain intact. The separate file HISTORY_R42 .pdf combines complete R40 article and technical supplement without editing their pages; the latter already udes the R38 article and supplement. Revision-relative labels in that archive are historical. current article and this supplement state R42's additions directly. The original primary and inear experiments, failed sufficient certificates, adaptive local-LP work, model sensitivities,

We study the minimum implementation cost of revising an installed policy while preserving operating performance after every state and date. A common randomized Markov continuation must serve all restarts. We construct a backward feasibility repair with an explicit cost modulus. A witness-driven form avoids an exact operating solve. It yields convergent global policy search for finite-state problems and an end-to-end approximation bound for continuous-state problems admitting certified total-variation aggregation. Exact rational spatial search and a verified McCormick comparator target the same constrained objective. A prospectively fixed eight-environment experiment reports all budget stops. On the original 42 continuous-state development configurations, whole-domain lotteries certify strict improvement over every deterministic policy in 27 cases. Nonlinear two-state lower bounds become positive, but longer-horizon gaps remain material. Independent checks distinguish arithmetic validity from approximation assumptions. The results identify when local operating guarantees can be converted into global revision certificates without changing the policy class.

KEYWORDS: Dynamic programming, Policy revision, Constrained Markov control, Global optimization, Verified computation.

1. INTRODUCTION

An organization may wish to improve an installed rule without replacing every decision it makes. Implementation cost then depends both on the decisions changed today and on the states reached under the revised rule tomorrow. A feasible revision must preserve the operating objective, whereas a cost-effective revision must economize on intervention along its own induced state distribution. These two requirements create a constrained dynamic problem. Pointwise selection of inexpensive changes need not solve it.

The relevant continuation must also be specified. Two histories that meet at the same date and state must use the same continuation under a Markov rule. Allowing a different continuation for each initial condition or each remembered randomization can reduce costs, but changes the economic prospect. Conversely, requiring a deterministic rule can increase implementation cost even when randomization is admissible. We therefore take a common randomized Markov rule as the central choice variable. Performance is protected after every date and state; implementation cost is integrated under a stated initial distribution. A performance guarantee at every restart is not a guarantee of cost optimality at every restart.

The main contribution is a constructive connection between these requirements and a complete rational search. A policy that violates every operating constraint by at most δ can be repaired backward without adding a persistent random seed or a history variable. The repair improves operating value at every state. Its additional implementation cost is bounded explicitly by a modulus that vanishes with δ . Strict discounting and a positive operating allowance provide slack; no global duality assumption is used. This repair gives an error bound for rational policy lattices. For binary actions it also gives a stopping theorem for the interval Bellman reach-and-bound procedure used in the experiments. Unlike convergence to a necessary-action maximization, the limiting object is the original common-continuation constrained optimum.

The witness-driven extension separates operating approximation error from policy-search error in one computable global bound. It supplies a positive-margin check and a constructive

model error into operating and cost allowances; strict randomized separations also yield even administration charges. These statements extend the numerical method rather than its economic objective.

and result transfers finite-state certificates to continuous state space. Under discretization approximation of the transition kernel by a within-cell reset kernel, cell averaging an arbitrary measurable Markov policy to a finite-state policy. Lifting reverses this action. Uniform primitive errors give an explicit operating-tolerance adjustment and a costment. The repair modulus controls the remaining difference between the relaxed and finite-state optima. The theorem makes the approximation assumption operational. It does not treat weak convergence of an atomic transition law as total-variation convergence, and it does not silently replace discrete shocks by continuous noise.

Numerical analysis separates three questions. First, does complete search return sound results for the common Markov problem, and how do those intervals compare with a direct mathematical-programming formulation? Second, does randomization change the conclusions of the original continuous-state maintenance configurations? Third, can a nontrivial revision bound survive nonlinear, two-state dynamics and nonzero operating-witness error? Each experiment retains the same objective on the two sides of its interval. The eight new finite-state experiments and both solver implementations were fixed before those environments were created. The original 42 configurations are explicitly a development cohort, not a prospective

resulting distinctions are economically consequential. In the development cohort, 27 remain randomized deployments cost strictly less than a certified lower bound for every deterministic policy under the same uniform initial law. This is a statement about the admissible randomization, not about a neural label attached to the installed rule. The repair experiment changes the zero lower bound into a positive bound; its finest horizontal interval is $[8.8718, 11.8492]$. That interval is informative but not tight. In the prospective state comparison, both methods face the same node allowance and wall-clock limit. Longer linear relaxation often improves the lower endpoint, but longer horizons retain initial residual intervals. The theory supplies completeness, not a claim that complete search is comprehensive.

1.1. Relation to the literature

Constrained Markov decision processes provide the natural starting point (Altman, 1999). Occupation measures are particularly effective when the constraints are expectations under a fixed initial law. Our formulation additionally requires every restarted problem to meet a waiting guarantee using one shared continuation. A separate occupation measure for each problem would need compatibility restrictions: at a shared date and state, its action probabilities depend on which restart generated the occupancy. Dropping that restriction produces a feasible class. Our finite formulation imposes it directly through a single policy variable at date and state. This observation does not preclude occupation-measure methods; it is the coupling they must preserve in this problem.

Approximations to constrained control are not new. Saldi (2018) proves value convergence for discounted and average-cost constrained problems, and develops approximately optimal policies and rates under additional continuity conditions. Saldi, Linder, and Yüksel provides a systematic treatment of quantized stochastic control. Our transfer result is finite-horizon, all-restart constraints and combines the within-cell reset identity with an Markov feasibility repair. It states the total-variation condition needed to average unrepresentable policies rather than assuming that a convenient policy sieve is automatically

ries; its additional object is the common-continuation restart constraint and its computable
 ance-to-cost modulus.

Primal–dual policy methods also solve constrained expected-return problems. Ding, Zhang,
 ar, and Jovanovic (2020) establishes global convergence guarantees for natural policy-
 cient methods in discounted constrained MDPs. Ni and Kamgarpour (2025) studies safe
 oration under cumulative constraints. Those learning problems have information and fea-
 ility requirements different from our known-model, all-restart certification problem. Safe
 line improvement (Laroche, Trichelair, and Tachet des Combes, 2019) and approximation-
 r analysis for fitted dynamic programming (Munos and Szepesvári, 2008, Scherrer, 2013)
 er distinguish model uncertainty, policy approximation, and optimization error. We claim
 ample-efficiency comparison against methods given less information.

The global optimization ingredients are classical. Convex envelopes of factorable products
 inate in McCormick (1976); the numerical LP proposals here use a modern simplex im-
 mentation associated with Huangfu and Hall (2018). We do not attribute novelty to bilinear
 opes, best-bound search, or weak duality. The contribution is a common-policy economic
 mulation whose local approximation errors can be repaired into an admissible deployment,
 an explicit global stopping bound. We use the direct McCormick formulation as a serious
 e-object comparator, not as a deliberately weak price portfolio. A rational residual correction
 s proposed LP multipliers into lower bounds even when floating-point dual feasibility is
 perfect. This separation of numerical proposal and proof follows the broader program of
 ied computation (Rump, 2010).

Semi-infinite optimization emphasizes the difference between checking finitely many con-
 nts and controlling them on a region (Hettich and Kortanek, 1993). Our continuous-state
 ficates therefore include whole-cell inequalities and isolated switching points. Information-
 ation bounds supply another route to dynamic-programming lower and upper bounds
 wn, Smith, and Sun, 2010, Brown and Smith, 2014); they are related to, but not identical
 , the restart support functions used below. The support functions remain useful accelerators
 though no finite price set is asserted to exhaust a global dual.

The economic motivation is adjustment rather than unconstrained operating optimization.
 merical economics has long emphasized model structure, approximation error, and state-
 e complexity (Judd, 1998, Brumm and Scheidegger, 2017). Empirical replacement models
 as Rust (1987) discipline operating primitives with data. Our revision cost is instead the
 of departing from an installed decision rule, not a cost paid whenever the previous physical
 on changes. Adding physical switching costs would require including the previous action in
 state. The present maintenance environments are designed, known-model economies. Their
 alized operating and implementation units are not estimated monetary welfare, and their
 ficates are not estimates of empirical prevalence. The general convergence results, rather
 a claimed calibration, carry the methodological argument.

2. THE ECONOMIC PROBLEM

Let X be a standard Borel space, $A = \{1, \dots, m\}$ a finite action set, and $t = 0, \dots, T$. Rewards
 $r_t(x, a)$, implementation costs $k_t(x, a)$, and terminal operating payoff $g(x)$ are bounded Borel
 functions. The transition kernels P_t^a are known Borel probability kernels. Write $|r_t| \leq R$,
 $|k_t| \leq G$, and $0 \leq k_t \leq K$. Unless otherwise indicated, $0 < \beta < 1$ and $\varepsilon > 0$. Define

$$T_t^a v(x) = r_t(x, a) + \beta P_t^a v(x), \quad V_T = g, \quad V_t = \max_{a \in A} T_t^a V_{t+1}. \quad (1)$$

maximizing action.

domized Markov policy is a sequence of Borel kernels $p_t(\cdot | x)$ on A . Its values satisfy

$$J_T^p = g, \quad J_t^p = \sum_a p_t(a | x) \{k_t(x, a) + \beta P_t^a J_{t+1}^p\} \quad (2)$$

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$$C_T^p = 0, \quad C_t^p = \sum_a p_t(a | x) \{k_t(x, a) + \beta P_t^a C_{t+1}^p(x)\}. \quad (3)$$

the p_t is used by all histories reaching (t, x) . Randomization is newly drawn conditional on current date and state; a persistent hidden coin indexing different future policies is not used.

Given an initial probability law ν , the central object is

$$K_R(\nu, \varepsilon) = \inf_{p \in \mathcal{F}_R(\varepsilon)} \int C_0^p d\nu, \quad \mathcal{F}_R(\varepsilon) = \{p : J_t^p(x) \geq V_t(x) - \varepsilon \forall t, x\}. \quad (4)$$

A deterministic policy restricts p_t to vertices of the simplex and defines K_D . An operating-policy is feasible, so both classes are nonempty. On a finite state space, the product of many simplexes is compact, J^p and C^p are continuous, and the feasible set is closed. The infimum in (4) is then attained. We use infimum, not an unproved function-space minimum, in the all-state formulation.

Given an installed rule $a_t^0(x)$, the principal cost specification is

$$k_t(x, a) = w_t(x) \mathbf{1}\{a \neq a_t^0(x)\}, \quad w_t \geq 0. \quad (5)$$

The principal is charged for the action actually drawn. Its expectation under a lottery is not an indicator of the lottery's most probable action differs from the installed action. The unchanged installed rule has zero revision cost even when it is operating-infeasible. Consequently, dividing an interval by its revision cost is undefined, not an informative relative-gap statistic. We report $(J - L)/U$ when $U > 0$, and report zero-cost exactness separately.

We use a uniform law in the development economy weights all normalized initial conditions equally. We use a numerical reference distribution, not an estimated stationary law. We also integrate exactly over the action space under densities $2x$ and $2(1 - x)$. These calculations change the initial objective function by integrating the all-state operating constraints and installed rule unchanged. A pointwise approximation would cover arbitrary initial laws, including atoms; an almost-everywhere cost approximation would need not.

3. FEASIBILITY REPAIR AND COMPLETE POLICY SEARCH

$$D_C = K \sum_{s=0}^{T-1} (s+1) \beta^s, \quad (6)$$

$$D_J = \max_{1 \leq h \leq T} \left\{ 2R \sum_{s=0}^{h-1} (s+1) \beta^s + 2Gh\beta^h \right\}. \quad (7)$$

The maximum over remaining horizons in D_J matters: the terminal term $h\beta^h$ need not increase with h . Total variation between action distributions is one half of their ℓ_1 distance.

$$\sup_{t,x} |J_t^p - J_t^q| \leq D_J \theta, \quad \sup_{t,x} |C_t^p - C_t^q| \leq D_C \theta.$$

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PROOF: For costs, an action's remaining cost lies in $[0, K \sum_{s=0}^{h-1} \beta^s]$ at a state with h periods remaining. Changing its distribution changes the conditional expectation by at most the length of this interval times θ . Adding β times the next-date value difference and iterating gives $\sum_{s=0}^{h-1} (s+1)\beta^s \theta$. For operating rewards use the interval length $2R$ for each stage and $2G$ for the terminal reward. A reward at distance s can be affected by $s+1$ action changes, and the terminal reward by h changes. Taking the maximum over h gives (7). The same conclusions follow by a finite-horizon coupling, but do not require one. *Q.E.D.*

THEOREM 1—Common-continuation repair: *Suppose $p \in \mathcal{F}_R(\varepsilon + \delta)$, $\delta \geq 0$. There is a Borel randomized Markov policy $\hat{p} \in \mathcal{F}_R(\varepsilon)$ with $J_t^{\hat{p}} \geq J_t^p$ for every (t, x) and*

$$\sup_{t,x} \text{TV}(\hat{p}_t, p_t) \leq \rho_\varepsilon(\delta) := \frac{\delta}{(1-\beta)\varepsilon + \delta}, \quad \int C_0^{\hat{p}} d\nu \leq \int C_0^p d\nu + D_C \rho_\varepsilon(\delta). \quad (8)$$

particular,

$$0 \leq K_R(\nu, \varepsilon) - K_R(\nu, \varepsilon + \delta) \leq D_C \rho_\varepsilon(\delta). \quad (9)$$

PROOF: Proceed backward, with $J_T^{\hat{p}} = g$. Given a repaired continuation, put $q_a = T_t^a J_{t+1}^{\hat{p}}$, $\sum_a p_t(a|x)q_a$, $q_* = \max_a q_a$, and $b = V_t - \varepsilon$. By the induction hypothesis, $q_p \geq J_t^p \geq \varepsilon - \delta$. Also $J_{t+1}^{\hat{p}} \geq V_{t+1} - \varepsilon$ implies $q_* \geq V_t - \beta\varepsilon$. If $q_p \geq b$, retain p_t . Otherwise choose measurable maximizing action a_* and set

$$\hat{p}_t = (1 - \alpha)p_t + \alpha e_{a_*}, \quad \alpha = \frac{b - q_p}{q_* - q_p}.$$

Since $d = b - q_p \in (0, \delta]$, the denominator is at least $(1 - \beta)\varepsilon + d$. Thus $\alpha \leq \rho_\varepsilon(\delta)$ and $\hat{p}_t \geq q_p \geq J_t^p$. All operations are Borel; the division is used only where the denominator is strictly positive. The construction uses only t and x and the already fixed next-date continuation. Lemma 1 bounds cost. Apply the construction to arbitrarily near-minimizers at tolerance $\varepsilon + \delta$ and take their infimum to obtain (9). *Q.E.D.*

The repair is not a convex combination of whole policies. Such a combination, if selected and remembered, would enlarge the information state. Here the mixture is selected anew at each state using one repaired continuation. The slack $(1 - \beta)\varepsilon$ explains why the theorem needs strict discounting and a positive tolerance. No vanishing modulus at $\beta = 1$ or $\varepsilon = 0$ is warranted by this argument.

3.1. Certified search without an exact operating solve

The exact value in the preceding statements is not an oracle that the method must possess. Certified upper witness supplies the slack needed by the repair. This matters when operating approximation, rather than policy search alone, is expensive. Let $L_t \leq V_t \leq U_t$ be bounded

$$e \geq \max_{t < T} \|U_t - L_t\|_\infty, \quad \sigma = (1 - \beta)\varepsilon - d,$$

$$d \geq \max \left\{ 0, \sup_{t < T, x} \left[U_t(x) - \max_a T_t^a U_{t+1}(x) \right] \right\}. \quad (10)$$

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finite-state statement, let $m = |A|$, let $M \geq 1$ be an integer, and set $\theta_M = \min\{1, (m - 1)/M\}$. Lattice probabilities are multiples of $1/M$. The same primitive transition and reward functions occur in the witness check and the repair. Error in a different operator must first be corrected in its enclosure. A small residual is not, by itself, a certificate that $U \geq V$; the witness check and terminal condition are also required.

LEMMA 2—Witness-driven repair and global error: *Suppose $\sigma > 0$ and $\delta \geq 0$. If a common policy $\hat{p}$ satisfies $J_t^{\hat{p}} \geq U_t - \varepsilon - \delta$ for all $t < T$, the backward construction produces a common policy $\hat{p}$ such that*

$$J_t^{\hat{p}} \geq U_t - \varepsilon \geq V_t - \varepsilon, \quad J_t^{\hat{p}} \geq J_t^p, \quad \sup_{t, x} \text{TV}(\hat{p}_t, p_t) \leq \frac{\delta}{\sigma + \delta}. \quad (11)$$

On a finite state space, let D_C be the cost of a rational-lattice policy p using U , and let its cost be $U_{M, L, U}$. Then

$$\max\{0, v_{M, L} - D_C \theta_M\} \leq K_R(\nu, \varepsilon) \leq U_{M, L, U}, \quad (12)$$

$$U_{M, L, U} - \max\{0, v_{M, L} - D_C \theta_M\} \leq D_C \left\{ \theta_M + \frac{e + D_J \theta_M}{\sigma + e + D_J \theta_M} \right\}. \quad (13)$$

As $M \rightarrow \infty$, $e \rightarrow 0$, and $d \rightarrow 0$ give convergence to the actual constrained optimum using witnesses rather than an exact operating solve. All displayed errors and the stopping time are computable for rational represented finite-state primitives.

PROOF: Fix a repaired continuation and put $q_a = T_t^a J_t^{\hat{p}}$, $q_p = \sum_a p_t(a | x) q_a$, and $b = J_t^{\hat{p}} - \varepsilon$. At the final decision date $J_T^{\hat{p}} = U_T$; at every earlier date the induction hypothesis gives $J_t^{\hat{p}} \geq U_t - \varepsilon$. Positivity and (10) imply

$$q_* := \max_a q_a \geq \max_a T_t^a U_{t+1} - \beta \varepsilon \geq U_t - d - \beta \varepsilon = b + \sigma.$$

Inductive improvement gives $q_p \geq J_t^p \geq b - \delta$. Where $q_p < b$, set $\alpha = (b - q_p)/(q_* - q_p)$ and clip p_t with the least-index maximizing action. For $z = b - q_p \in (0, \delta]$, one has $\alpha \leq z/(q_* - q_p) \leq \delta/(\sigma + \delta)$. Otherwise leave p_t unchanged. This proves (11); the cost bound follows from Lemma 1. Finite action maximization and division on a strictly positive-denominator Borel measurable function preserve measurability.

In the lattice statement, round any genuinely feasible policy as in Theorem 3. Its rounded cost exceeds $V - \varepsilon - D_J \theta_M \geq L - \varepsilon - D_J \theta_M$, and its cost increases by at most $D_C \theta_M$. This is the lower endpoint. A lattice minimizer is at worst $e + D_J \theta_M$ below the upper-witness check. Apply the first assertion with that value of δ to obtain the upper endpoint and (13). The lattice costs justify clipping the lower endpoint at zero. Q.E.D.

does not establish infeasibility of the original problem. Witness refinement and complete search have different roles in (13). Refining the witnesses alone does not close the remaining search gap. Conversely, an increasingly fine policy lattice does not erase a fixed operating-cost error. The exact-value theorem is recovered by $L = U = V$ and $d = e = 0$.

3.2. A complete finite-state construction

Let X have n states. For an integer $M \geq 1$, let $\mathcal{P}_M$ be the policies whose action probabilities are multiples of $1/M$. Put $\theta_M = \min\{1, (m-1)/M\}$ and $\delta_M = D_J \theta_M$. Evaluate every candidate by the common Bellman equations and let

$$v_M = \min \left\{ \int C_0^p d\nu : p \in \mathcal{P}_M, J_t^p \geq V_t - \varepsilon - \delta_M \forall t \right\}.$$

For all rational primitives these comparisons, values, and policy probabilities use rational arithmetic.

THEOREM 3—Convergence to the constrained optimum: *Repair a minimizer defining v_M to denote its feasible cost by U_M . Then*

$$U_M := \max\{0, v_M - D_C \theta_M\} \leq K_R(\nu, \varepsilon) \leq U_M, \quad U_M - L_M \leq D_C \{\theta_M + \rho_\varepsilon(\delta_M)\}. \quad (14)$$

As a prescribed positive absolute accuracy is reached at a finite M . At most $\binom{M+m-1}{m-1}^{nT}$ policies need be evaluated at that level. Running maxima of lower bounds and running minima of upper bounds give nested certificates without changing their validity.

PROOF: Round the first $m-1$ probabilities of any feasible policy down to multiples of $1/M$ and assign the remainder to the last action. The total-variation error is at most $(m-1)/M$, bounded by one. Lemma 1 shows that the rounded policy belongs to the relaxed finite set and has at most the original cost plus $D_C \theta_M$. Hence $v_M - D_C \theta_M \leq K_R$. Theorem 1 applied to the minimizing grid policy gives $U_M \leq v_M + D_C \rho_\varepsilon(\delta_M)$. The weak-composition count gives the number of simplex points at each date and state. The displayed error tends to zero. *Q.E.D.*

The theorem controls distance to (4), not distance to a one-step relaxation. Its count is exponential in $nT(m-1)$; it is a completeness construction, not a polynomial-time claim. The deterministic finite-state problem can instead be solved by evaluating its m^{nT} vertices with original tolerance. The randomized repair is neither required nor valid as a deterministic-approximating operation.

3.3. The executed spatial search

For binary actions, write p_{ti} for the probability of action 1. A policy box $Q = \prod_{ti} [\ell_{ti}, u_{ti}]$ has the same coordinates across all restarts. Backward interval Bellman evaluation gives propagating upper values $\bar{J}$ and unconstrained implementation lower values $\underline{C}$ for that box. Necessary operating tests can contract its coordinate intervals: if q_0^+, q_1^+ are next-date operating upper values, an admissible p must satisfy

$$(1-p)q_0^+ + pq_1^+ \geq V_t(i) - \varepsilon. \quad (15)$$

es remove the box. Every removal is justified by a necessary inequality. Feasible policies
 r removed because a sufficient candidate check fails.
 remaining cost lower bound is $\int \underline{C}_0 d\nu$. Repairing the midpoint of the contracted box
 s a feasible upper policy. The algorithm preserves a live box with the minimal lower bound
 ts a coordinate maximizing $\beta^t(u_{ti} - \ell_{ti})$. It preserves both children and every live leaf.
 l lower endpoint is the minimum of all live or cost-pruned leaf lower bounds and the
 ent upper cost. It stops when their difference is no greater than the requested tolerance,
 ts the remaining interval when the preassigned budget is exhausted.

POSITION 1—Stopping bound for interval Bellman search: *Let $w = \max_{ti}(u_{ti} - \ell_{ti})$
 rving policy box. Its midpoint-repair cost minus its interval Bellman lower bound is at*

$$H(w) = D_C\{w + \rho_\varepsilon(D_J w)\}. \quad (16)$$

*a node or time cap, the best-bound bisection procedure reaches every prescribed
 gap η in finitely many steps. In particular, choose $w_* > 0$ with $H(w_*) \leq \eta$. A sufficient
 use bisection depth is*

$$d \left\lceil \log_2 \frac{1}{\beta^{T-1} w_*} \right\rceil, \quad d = nT, \quad (17)$$

*ceiling replaced by zero when its argument is at most one. The associated full binary-
 e bound is exponential in this depth. Adding valid McCormick lower bounds cannot
 te the stopping guarantee.*

F: With only rectangular probability restrictions, ordinary Bellman maximization
 s the maximum operating value simultaneously at every state. It is a common Markov
 not a family of incompatible histories. A surviving contracted box has its operating
 alue at least $V_t - \varepsilon$ at every date and state. Each such maximizing policy differs from
 point policy by total variation at most w . Lemma 1 therefore makes the midpoint $D_J w$ -
 ally feasible. The same rectangular Bellman argument for minimum cost gives a lower
 o smaller than midpoint cost minus $D_C w$. Repair and (8) imply (16). A box satisfying
 η cannot need refinement while the global gap exceeds η : its repaired cost is already
 mbent candidate, so its lower bound is within η of the global incumbent or higher. On
 -refined path, weighted longest-side bisection makes every physical width at most w_* by
 h in (17). There are finitely many nodes up to that depth. Valid stronger lower bounds
 ove or improve boxes. The supplement details the effect of the necessary contractions;
 ink the rectangles and preserve every feasible point. *Q.E.D.*

s a convergence theorem for the implemented search rule, not an inference from favorable
 es. It also explains why a 60-second run can end with a large gap: a finite worst-case
 an be prohibitively large. The numerical results report that distinction directly.

4. A SINGLE ERROR BUDGET FOR CONTINUOUS-STATE APPROXIMATION

te-state theorem alone does not justify replacing a continuous model by a grid. The
 g sufficient conditions permit such a transfer without restricting the original policy to
 wise constant.

probability measures supported on E_i and write $\nu = \sum_i \nu_i \mu_i$. There are cellwise rewards $r_{h,t} \in [0, K]$, costs $k_{h,t} \in [0, K]$, and terminal values g_h , and a kernel

$$50 \quad Q_{h,t}^a(x, dy) = \sum_j P_{h,t}^a(i, j) \mu_j(dy), \quad x \in E_i,$$

certified uniform errors d_r, d_k, d_g and α_h satisfying

$$|r - r_h| \leq d_r, \quad |k - k_h| \leq d_k, \quad |g - g_h| \leq d_g, \quad |(P_t^a - Q_{h,t}^a)f(x)| \leq \alpha_h \|f\|_\infty$$

every bounded Borel f . The approximation rewards obey the same bounds R, G as the original rewards, or R, G are enlarged to cover both models. All finite transition entries and their bounds are computable with verified directions.

The last inequality uses the operator convention for total variation; it is not the half- ℓ_1 convention for action probabilities above. Define $M_{J,t} = R \sum_{s=0}^{T-t-1} \beta^s + G \beta^{T-t}$ and $M_{C,t} = R \sum_{s=0}^{T-t-1} \beta^s$. Set

$$a_T = d_g, \quad a_t = d_r + \beta a_{t+1} + \beta \alpha_h M_{J,t+1}, \quad (18)$$

$$b_T = 0, \quad b_t = d_k + \beta b_{t+1} + \beta \alpha_h M_{C,t+1}, \quad (19)$$

$$a_h = \max_t a_t, \quad b_h = b_0.$$

$v_h(s)$ be the optimum of the finite common-Markov model at operating allowance s and initial masses (ν_i) .

THEOREM 4—End-to-end transfer: *Under Assumption 1, if $\varepsilon > 2a_h$, then*

$$v_h(\varepsilon + 2a_h) - b_h \leq K_R(\nu, \varepsilon) \leq v_h(\varepsilon - 2a_h) + b_h. \quad (20)$$

Suppose finite computation returns intervals $[L_+, U_+]$ and $[L_-, U_-]$ for these relaxed and tightened optima, of widths at most η_+ and η_- . A valid global interval is

$$[\max\{0, L_+ - b_h\}, U_- + b_h], \quad (21)$$

its width is at most

$$\eta_+ + \eta_- + 2b_h + D_C \frac{4a_h}{(1 - \beta)(\varepsilon - 2a_h) + 4a_h}. \quad (22)$$

If certified primitive errors tend to zero and both finite-solver widths tend to zero, this interval converges to the original measurable-policy optimum. Its measured width supplies a termination bound for the actual economic objective.

PROOF: Fix any measurable Markov policy p , not necessarily cellwise constant. Backward substitution of its original and reset-model Bellman equations gives (18)–(19). Bellman comparison gives the same operating error for optimal values. In the reset model, averaging the policy within E_i , $\bar{p}_t(a | i) = \int p_t(a | x) \mu_i(dx)$, averages its value exactly: after every transition within-cell law is again μ_j , and stage rewards and costs are cellwise constant. This identity flows backward, including the common continuation. An original feasible policy has reset

finite-model cost differs from the original initial cost by at most b_h . Taking an infimum over all such policies gives the lower inequality. Conversely, lift a finite policy feasible at $\varepsilon - 2a_h$ by using its action in every cell throughout each cell. The reset value is constant in each cell; the two operating policies make the lifted original policy feasible and its cost differs by at most b_h . This proves the equality. Finally apply Theorem 1 in the finite model with target allowance $\varepsilon - 2a_h$ and $4a_h$, then add the two finite-computation widths and the cost errors. *Q.E.D.*

4.1. Where the transfer theorem applies

On a compact domain, a transition density with computable uniform continuity in the source variable and an L_1 -approximable density in the destination variable can furnish the reset operation. The cell reference measures must also represent the initial law as in Assumption 1. For example, with Lebesgue-uniform cell measures and a uniform initial law, a uniformly continuous transition density gives a computable $\alpha_h = O(h)$ on a shape-regular mesh. Cellwise density and cost approximations give $d_r, d_k, d_g = O(h)$ when their moduli are Lipschitz. Discontinuities in an installed rule can be retained as exact partition boundaries. In the theorem (22) is $O(h + \eta_+ + \eta_-)$ at fixed positive ε and $\beta < 1$, with explicitly displayed constants. The number of cells still scales as $h^{-\dim X}$; the theorem does not remove the curse of dimensionality.

The primary affine-shock model and the nonlinear two-state model below have atomic transition laws. Smearing their successor atoms into cells generally leaves the total-variation error bounded away from zero. We therefore do *not* apply Theorem 4 to those models by asserting cellwise approximation. Their reported intervals are direct whole-domain bounds for the unchanged models. They remain valid without a convergence claim for that particular spatial discretization. This distinction prevents the approximation theorem from being used as an ad hoc explanation for closing an atomic-model gap.

5. CERTIFICATES THAT ACCELERATE THE GLOBAL PROBLEM

Complete search and the transfer theorem provide the convergence argument. The inherited certificates are useful accelerators and diagnostic tools, but are not substituted for that argument.

5.1. Deterministic outer restrictions and measurability

Let $k_t \leq V_t \leq U_t$ be bounded Borel operating witnesses, including the terminal date. Every admissible deterministic policy must select an action in

$$\mathcal{N}_t^{L,U}(x) = \{a : L_t(x) - T_t^a U_{t+1}(x) \leq \varepsilon\}. \quad (23)$$

The admissible operating continuation cannot exceed $V_{t+1} \leq U_{t+1}$. The set is nonempty because the operating-optimal action satisfies the test. Its graph is Borel, since it is a finite union of Borel level sets. Backward minimization of $k_t + \beta P_t^a B_{t+1}$ over this set yields a Borel lower bound; the lowest-index minimizer is a Borel selector. Thus the general-state recursion does not require an implicit measurable-selection assertion.

The set is a necessary-action outer class, not the original constrained problem. A selected action can accumulate losses that violate later operating requirements. Equality between the certified cost lower bound and a separately feasible policy's cost certifies $K_D(\nu, \varepsilon)$. Equality of all functions certifies more. Tightening L, U alone does not force equality; complete certification is what controls the remaining structural gap.

or $\lambda \geq 0$, a verified supersolution W_t^λ of

$$52 \quad W_T^\lambda \geq \lambda g, \quad W_t^\lambda \geq \max_a \{\lambda r_t - k_t + \beta P_t^a W_{t+1}^\lambda\}$$

ies $C_t^p \geq \lambda(V_t - \varepsilon) - W_t^\lambda$ for every feasible policy, including randomized Markov policies. ne a floor F_t by the maximum of zero and finitely many such terms, using L_t in place of V_t n necessary. Given a valid next-date lower function B_{t+1} , define

$$\mathcal{D}_t(x) = \left\{ p \in \Delta(A) : \sum_a p_a T_t^a U_{t+1}(x) \geq L_t(x) - \varepsilon \right\}.$$

$$B_t(x) = \max \left\{ F_t(x), \min_{p \in \mathcal{D}_t(x)} \sum_a p_a \{k_t(x, a) + \beta P_t^a B_{t+1}(x)\} \right\}, \quad B_T = 0. \quad (24)$$

feasible action distribution of any admissible policy belongs to the local constraint, and its inuation cost dominates B_{t+1} . Induction proves that (24) is a global randomized cost lower nd. It does not prove that the local optimizer can be combined into a feasible policy or that e support prices close a global duality gap.

or three actions and one local operating inequality, the local polytope has only pure feasible ces and two-action boundary mixtures. Our new primary calculation evaluates these vertices exact rational arithmetic on directed whole-cell bounds. Unlike the earlier diagnostic with ro global floor, it uses the nonzero F_t at every date and is paired with a feasible policy for ame uniform-initial randomized objective. It improves that lower endpoint in 23 of the 42 igurations. The prior adaptive local-LP theorem and all of its experimental records remain e retained supplement; their target is still the local relaxation, not (4).

5.3. Whole-domain randomized deployments

simple backward construction supplies feasible upper policies without changing the atomic el. Given a feasible next-date value J_{t+1} , write $q_0 = T_t^{a_0} J_{t+1}$, $q_* = \max_a T_t^a J_{t+1}$, and $V_t - \varepsilon$. If $q_0 \geq b$, retain the installed action. Otherwise mix it with a lowest-index maximizing on with probability

$$p_t^{\text{change}}(x) = \frac{b(x) - q_0(x)}{q_*(x) - q_0(x)}. \quad (25)$$

denominator is positive wherever this expression is used. The resulting operating value $= \max\{q_0, b\}$ and is feasible at every state. In the piecewise-affine model, J_t remains eewise affine while the mixing probability can be a ratio of affine functions. Cost is therefore osed rather than falsely represented as an exact piecewise-affine function.

ne verifier includes isolated switching points in every operating identity. Directed cost boxes grate against the three absolutely continuous initial laws. All affine transition slopes are tly positive, so a finite set of switching points and its finite-horizon preimages have zero ability under these laws. That fact justifies almost-everywhere cost enclosures; it does not fy ignoring knots in a restart operating constraint or using the same cost enclosure for a c initial law.

direct finite-state formulation minimizes $\sum_i \nu_i C_{0i}$ subject to (2)–(3), $J_{ti} \geq V_{ti} - \varepsilon$, common policy simplexes. The constraints are bilinear in policy probabilities and action values. A McCormick relaxation on each probability box provides a classical lower program. The baseline uses the same incumbent repair, contraction rule, box maintenance, node allowance, initial law, and all-restart constraints as interval Bellman search. The interval work is the stronger LP bound and an LP-proposed feasible candidate after repair. The LP status is not a certificate. For a minimization relaxation $c'z$ subject to $\ell \leq z \leq u$, every rational vector $\lambda \geq 0$ gives

$$\mathcal{L}(\lambda) = -\lambda'b + \sum_j \min\{(c + A'\lambda)_j \ell_j, (c + A'\lambda)_j u_j\} \leq \min c'z. \quad (26)$$

follows by adding $\lambda'(Az - b) \leq 0$ and minimizing the resulting affine function over the bounded nonnegative solver marginals are evaluated in (26) using exact rational arithmetic. If the LP falls back to the interval Bellman lower bound; a floating infeasibility declaration moves the policy box. A separate verifier rebuilds every relaxation and every search-tree node from the primitive model.

6. NUMERICAL EVIDENCE

Experiments keep model class, information, initial distribution, and operating allowance constant. The supplement reports every configuration; no failed or budget-stopped case is removed. The computation report records proof-object sizes, memory, wall time, the exact environment, and the boundary between independent arithmetic checks and shared-code validation.

6.1. Prospectively fixed common-Markov problems

In generating eight new environments, we committed their seeds, joint parameter changes, solver hashes, node cap 4,096, wall-clock cap 60 seconds per solver and case, and absolute allowance 10^{-3} . The freeze is `commit 13fd3ee`; it follows development seed 390001 and precedes all eight evaluated seeds. Both solvers see the same known model. The two-state models represent low and high condition; four-state models use a pair of condition indicators and interaction terms in their transition probabilities. Horizons are 3, 4, 8, and 12; discount rates are 0.8 or 0.95; allowances are 0.01 or 0.05; operating prices, transition persistence, transition effects, and initial masses vary jointly. The exact formulas and resulting rational objects are in the supplement and proof objects.

The supplement contains all eight pairs, including unfinished searches. A positive absolute-gap certificate is not necessarily a small relative-gap certificate when the optimum is small. We therefore compare both. The linear relaxation often improves the lower endpoint substantially, but costs more per node. The four-state longer-horizon cases expose the combinatorial burden: the remaining interval is not hidden by a success count. The comparison does not establish that interval search dominates direct global programming. Instead it identifies the value and cost of linear relaxation under matched information and caps.

6.2. The unchanged continuous-state development cohort

The original maintenance model has $x \in [0, 1]$, three actions, two shocks with probabilities $2/5$, $\beta = 19/20$, stage reward $x - c_a$ with $c = (0, 3/20, 9/20)$, terminal payoff $x/2$,

	States	T	Method	Lower	Upper	Gap/upper (%)	Seconds
54	2	3	I	2.8414	2.8425	0.03	0.99
	2	3	M	2.8414	2.8425	0.03	1.77
	2	4	I	0.0000	0.0000	0.00	0.00
	2	4	M	0.0000	0.0000	0.00	0.38
	2	8	I	7.2534	9.0046	19.45	17.19
	2	8	M	8.5290	9.0046	5.28	60.41
	2	12	I	7.1419	7.5553	5.47	16.32
	2	12	M	7.4446	7.5553	1.46	60.40
	4	3	I	2.6157	2.6217	0.23	10.82
	4	3	M	2.6206	2.6217	0.04	35.99
	4	4	I	0.0233	0.0244	4.00	0.47
	4	4	M	0.0235	0.0244	3.13	2.60
	4	8	I	4.6631	7.8481	40.58	60.04
	4	8	M	6.1990	7.8481	21.01	60.41
	4	12	I	4.8936	6.6439	26.34	60.04
	4	12	M	5.8606	6.6439	11.79	60.43

interval Bellman search. M: verified McCormick search. Bounds concern the identical common randomized Markov
 tive for each pair. Reported times include model construction and the operating solve; the search's 60-second cap
 cked between iterations, not by interrupting a proof-producing node. A zero upper cost is an exact zero-cost case.

Full stop reasons, node counts, and memory appear in the supplement.

revision weight $1 + x$. Its two affine successors for actions 0, 1, and 2 respectively are

$$/4, 3x/4 + 1/20), \quad (17x/20 + 1/10, 17x/20 + 7/50), \quad (x/10 + 4/5, x/10 + 17/20).$$

seven installed rules, horizons 4, 8, and 12, and allowances 0.01, 0.05 are unchanged. Neural
 spline rules are fixed computational incumbents, not estimated economic policies. Exact
 ating values are inexpensive in this model and their computation is charged. We do not use
 cohort to claim performance when operating approximation itself is difficult.

he inherited deterministic comparison has 12 zero-cost and 12 positive-cost integrated
 alities. Ten of the positive-cost equalities occur at horizon 4, two at horizon 8, and none at
 zon 12. The maximum deterministic relative gaps at each horizon are reported alongside
 lute gaps in the supplement. In the horizon-8 neural31003 case at $\varepsilon = 0.05$, the two initial
 functions are identical pointwise, but the arrays differ at later restart dates. Thus initial
 ality does not imply equality at every restart. The new diagnostic records the full rational
 greement locus.

he lotteries in (25) certify strict improvement over *every* feasible deterministic policy in
 f the 42 uniform-initial configurations (Table II). The horizon-12 defer case at allowance
 , for example, has a randomized interval $[15.3172, 15.4670]$ and a positive deterministic
 ration. This addresses the economically relevant policy-class difference directly, without
 tituting a fixed-restart history-conditioned frontier. The 12 zero-cost cases are already
 mal by feasibility and nonnegativity and are not counted as nontrivial search successes. All
 ositive-cost randomized intervals retain a reported nonzero width; the narrowest and widest
 ive intervals are distinguishable in the full table.

changing the initial density to $2x$ yields 28 strict deterministic separations; changing it to
 $-x$ yields 27. These are sensitivity results on the same developed environments, not
 pendent observations. The connected local program improves 23 uniform lower endpoints,

Cases	Zero-cost exact	Strict gain over K_D	Improved lower	Maximum gap/upper (%)
14	4	10	8	14.61
14	4	8	7	32.29
14	4	9	8	34.23

means that a verified randomized upper cost is below the deterministic lower bound for the same initial law constraints. It does not mean that the randomized optimum is attained by that candidate. Improved lower compares selected local-support bound with the prior randomized support lower bound. All 42 uniform-initial rows and both nonuniform sensitivity tables are retained.

TABLE III
SELECTED NONLINEAR COMMON-MARKOV INTERVALS

T	Mesh per axis	Lower	Upper	Gap/upper (%)	Independent check
8	64	0.9438	3.9348	76.01	full
16	128	4.1908	7.6644	45.32	not independent
32	64	6.5389	16.7884	61.05	full
32	128	8.1264	12.5226	35.11	full R40
32	256	8.8718	11.8492	25.13	not independent

on initial law on the square and operating allowance 1/2. Inherited full independent scalar checks cover (8, 64) and (32, 64). R40 additionally replays the complete (32, 128) object, including the candidate upper all-restart operating bound. The (32, 256) result retains directed construction and structural checks, not the independent label of adjacent rows.

g the longer-horizon cases; it thus affects the primary global interval rather than a selected local diagnostic. It still does not establish global strong duality or close every

6.3. Nonlinear two-state bounds

retained nonlinear model has $(x, y) \in [0, 1]^2$, four actions, bilinear transitions, two shocks, $(x + y)/2 - c_a$, terminal payoff $(x + y)/4$, and revision weight $1 + (x + y)/2$. Its exact primitive specification is reproduced in the supplement. Whole-box operating witnesses support supersolutions use outward arithmetic. The new lower construction uses 11 global prices and five nonnegative local multipliers, including zero, at every date. It is paired exactly the retained feasible upper policies. We run no new upper-portfolio competition not compare unequal proposal success fractions as evidence of optimality.

old zero lower endpoint was not forced by the definition of revision cost: the operating constraints can force departures from the installed rule. Nonzero support supersolutions now at restriction. At $T = 32$, $N = 256$, the interval is $[8.8718, 11.8492]$, with relative width 25%. This is a genuine improvement over $[0, 11.8492]$, but it is not a near-exact optimum one-dimensional scalability result. Both endpoints and the all-restart operating inequality independently rebuilt at $T = 32$, $N = 64$, and the R40 audit extends this to the complete $N = 128$ object. Memory for the retained support arrays grows as $8N^2(T + 1)(11 + 4)$ before temporary arrays and interpreter overhead; the machine record reports the actual and process high-water mark.

allowance 1/2 belongs to the inherited exploratory nonlinear design. It is not made able to 0.01 simply by normalizing state variables: the operating scale and remaining

configurations. None is paired with a policy certified only at $1/2$. Without a newly feasible policy those tighter-tolerance rows remain lower-bound-only results. A failed sufficient certificate is labeled uncertified, not infeasible. A box crossing an installed-rule threshold charge the upper revision weight conservatively; it does not assert that every realized state at box changes action.

7. VERIFICATION, ECONOMIC INTERPRETATION, AND LIMITS

A certificate has three distinct burdens. Its mathematical reduction must preserve the target policy class. Its finite arithmetic must check the asserted inequalities with the correct direction. Its publication must correspond to the checked objects. Hashes address the third burden, not the first two. Independent arithmetic addresses part of the second burden and can expose a self-contract error, but is not a proof that every economic assumption is appropriate. The finite-state verifier uses SymPy rationals and imports neither search implementation nor the optimizer. It rebuilds the common Markov equations, primitive contracts, every policy-split, every necessary contraction, the LP residual bounds, and the global leaf minimum. The continuous-state verifier independently checks the new lotteries, whole-cell costs, support resolutions, and local primal vertices, using the prior independent primitive parser rather than the constructor's piecewise-affine operations. The nonlinear verifier uses symbolic rational arithmetic geometry and arbitrary-precision scalar integer induction instead of the constructor's authorized box operations. It checks two complete representative objects, not merely selected ones. The verification report specifies precisely which objects passed; it never substitutes a pass-run equality or a mutation test for an inequality proof. The synthesis preserves all prior scientific material. Deterministic bounds, history-conditioned lotteries, the two-period analytic randomization example, adaptive local-LP analysis, 66 economic sensitivities, failed runs, and the stopped-control derivations remain available in full. The supplement attaches the prior article and technical supplement verbatim, with a content hash. The older 349-page historical annex remains unchanged as an archival object rather than a requisite for reading this article. The stopped consumption–portfolio problem is not relabeled: its inherited full-domain bound of approximately 7.1818 does not meet its 0.01 target. A Bellman comparison supports witness propagation but supplies neither missing action coverage nor a diffusion-discretization error bound (Kushner and Dupuis, 2001). The economic conclusion established by the new primary calculation is precise. When random action assignment is admissible and costs attach to realized departures, common Markov policies can strictly reduce the minimum implementation cost relative to every deterministic rule. This conclusion holds on the certified configurations and initial laws. It does not estimate how frequently organizations permit such lotteries, how costly they are to administer, the monetary welfare effect of a real deployment. Fixed randomization overhead, parameter uncertainty, and empirical calibration would require additional primitives and certificates. The ϵ -perturbation holdout study tests a fixed algorithm beyond its development seed, but is not a statistical generalization theorem.

7.1. Model uncertainty and the economic margin for randomization

The certificates can be translated into a robustness statement without treating designed empirical primitives as estimated data. Fix a nominal model, initial law, and policy class. Suppose a nonempty set of alternative models has verified uniform bounds

$$|J_t^{p,M} - J_t^{p,0}| \leq a, \quad |V_t^M - V_t^0| \leq a, \quad |C_0^{p,M} - C_0^{p,0}| \leq b \quad (27)$$

POSITION 2—Portable operating and cost intervals: *Let $\varepsilon > 2a$. A nominal policy at allowance $\varepsilon - 2a$ is feasible at allowance ε in every alternative model. If L_+ is a lower bound at allowance $\varepsilon + 2a$ and U_- is the nominal cost of such a tightened policy, then every model-specific common-policy optimum belongs to*

$$[\max\{0, L_+ - b\}, U_- + b].$$

The interval encloses the optimum that minimizes worst-case implementation cost over all policies feasible in all the alternative models. In the latter assertion the upper endpoint uses a common policy, not a different policy for each model.

PROOF: For the proposed policy, $J^{p,M} \geq J^{p,0} - a \geq V^0 - \varepsilon + a \geq V^M - \varepsilon$. Conversely, if a policy is feasible in a model M satisfies $J^{p,0} \geq J^{p,M} - a \geq V^M - \varepsilon - a \geq V^0 - \varepsilon - 2a$. Its cost is at least L_+ and its cost in M at least $L_+ - b$. Take infima for the model-specific optimum. For the robust problem, the first implication supplies one policy for all models and it applies to every robustly feasible policy in each model. Taking the worst-case cost yields the bounds. *Q.E.D.*

Exact randomized separation also has an economically interpretable administration-cost certificate. Let U_R be the certified cost of a feasible common lottery and L_D a lower bound for a feasible deterministic policy, for the same operating constraint and initial law. Suppose the operating payoffs and dynamics are unchanged by administering the lottery. A nonnegative extra administration charge at date t bounded by h_t increases its expected discounted cost by at most $\sum_{t=0}^{T-1} \beta^t h_t$. If

$$U_R + H < L_D, \quad (28)$$

the lottery remains cheaper than every deterministic feasible policy even after this additional charge. This conclusion does not require administration at every visited state; charging the charge everywhere is conservative. Deterministic rules are assumed not to pay the randomization-charge.

Given a common per-date ceiling h , the certified break-even threshold is

$$h_* = (L_D - U_R)_+ / \sum_{t=0}^{T-1} \beta^t. \quad (29)$$

Equality at h_* preserves the strict ranking. Equality at h_* yields no strict conclusion. For model uncertainty, use a deterministic nominal lower bound at $\varepsilon + 2a$ and a nominal lottery feasible at $\varepsilon - 2a$; then replace the numerator by $(L_{D,+} - U_{R,-} - 2b)_+$. The current development reports $a = b = 0$: they calculate economic margins under known designed primitives, not under uncertainty radii.

T	Installed rule	ε	Det. lower	Lottery upper	h_* lower bound
4	defer	0.01	4.93668	4.73011	0.055684
5	defer	0.05	4.93206	3.89012	0.280857
4	neural31001	0.01	0.11117	0.08928	0.005902
8	defer	0.01	10.81072	10.63794	0.025667
8	defer	0.05	10.75071	9.85440	0.133151
8	neural31001	0.01	0.46864	0.40232	0.009853
12	defer	0.01	15.61281	15.46691	0.015872
12	defer	0.05	15.47129	14.72413	0.081278
12	neural31001	0.01	3.49393	3.36015	0.014553

form initial law, unchanged development economy. Endpoints and the threshold are rounded outward. A positive threshold means that every smaller constant per-date overhead preserves a strict cost advantage over every feasible deterministic policy. All 126 rule/horizon/tolerance/initial-law margins are retained in the machine-readable results, including zeros.

8. CONCLUSION

local operating requirement becomes a global revision method only when one can control continuation compatibility and the cost of restoring feasibility. The backward repair provides that control for positive tolerance and strict discounting. Combined with complete policy search, it produces a convergent certificate for the actual common randomized Markov objective. Combined with a verified reset approximation, it yields an explicit end-to-end error budget for an unrestricted measurable-policy problem. These results retain the optimization method rather than replacing it by a necessary-action or finite-price relaxation. The computations show both the benefit and the burden of this construction. Whole-domain queries resolve a substantive deterministic-versus-randomized comparison on the original development cohort. Connected support bounds improve the same primary objective, and zero nonlinear lower bounds replace feasibility-only intervals. A matched direct global-programming comparator is often stronger per explored box. Longer horizons and four-state perturbations retain material gaps under fixed budgets. The stopping theorem explains complete refinement can close them; the reported execution does not claim to have done so. The remaining numerical challenge is to exploit economic structure strongly enough that complete certification becomes practical at larger state and horizon dimensions without enlarging its policy class or silently weakening its operating constraints.

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This supplement supplies proof details, exact model contracts, complete result tables, and the distinction between independent verification and inherited evidence. The complete R38 article and supplement are appended without modification. The main article is self-contained and does not require the archival annex.

1. OBJECTS, ASSUMPTIONS, AND VERIFICATION BOUNDARIES

The economic object throughout the new theory is one randomized Markov policy used at states and dates. Each inequality $J_t^p(x) \geq V_t(x) - \varepsilon$ concerns that same policy. Its cost is graded under a named initial law. None of the new convergence results treats a collection of ε -optimal policies as a solution under a dispersed initial law.

TABLE I: Assumptions and certificate scope

Result	Target and assumptions	Computation and check
Repair and modulus	Borel state space, finite actions, bounded rewards and costs, $0 < \beta < 1$, $\varepsilon > 0$; common randomized Markov policy; any initial law	Analytic proof. Borel measurability and the model bounds are assumptions, not facts inferred from a hash.
Fitness-driven repair	Checked $L \leq V \leq U$, $U_T = g$, and positive residual margin $(1 - \beta)\varepsilon - d$	Analytic bracket plus 48 exact multi-action test cases. A failed margin check is not economic infeasibility.
Model perturbation	Uniform error bounds over all common policies and every alternative model; $\varepsilon > 2a$	One tightened policy is portable across models. Error radii are assumptions, not estimated confidence sets here.
Administration large	Same operating model and initial law, feasible lottery upper, valid deterministic lower, bounded non-negative overhead	All 126 exact known-model margins retained. Strict superiority requires overhead strictly below its threshold.
Finite complete search	Finite states; same all-restart constraints and initial masses; compact policy product	Rational evaluation and verified box search. Full independent SymPy checks for 16 executed search trees.
Continuous transfer	Certified reset-kernel approximation in operator total variation; initial mixture identity; $\varepsilon > 2a_h$	Analytic end-to-end theorem. No claim that atomic benchmark kernels satisfy a vanishing reset error.

Target and assumptions		Computation and checks
Lottery	Unchanged one-state, atomic affine-shock model; all-state operating feasibility; uniform $2x$ or $2(1-x)$ initial density	Exact operating identities including isolated points; directed cost boxes; 42 independent lottery and local-bound checks.
Deterministic policies	Borel finite-action necessary correspondence; deterministic policies only	R38 proof objects and independent verification remain retained. The deterministic lower bound is not a randomized lower bound.
Linear support	Two-state bilinear dynamics; uniform square; $\varepsilon = 1/2$ for paired upper policies	Directed vector arithmetic on all 20 meshes; independent scalar checks for two inherited objects and a new complete (32, 128) replay. Four $\varepsilon = 1/20$ rows are lower-only.
Local adaptation	A continuous one-step relaxation, not the common-policy global optimum	Prior exact whole-cell certificates retained; new connected lower recursion independently checked.
Diffusion	Continuous actions, boundary settlement and diffusion approximation still require coverage	Archival derivation, not a solved numerical target or external validation.

2. DETAILS OF THE REPAIR AND COMPLETE-SEARCH ARGUMENTS

2.1. Policy perturbation and terminal rewards

total variation $\text{TV}(p, q) = \frac{1}{2} \sum_a |p_a - q_a|$ for action distributions. If an action-dependent V takes values in an interval of length D , replacing p by q changes its expectation by at most $D \cdot \text{TV}(p, q)$. With h stages remaining, a stage cost at distance s can be affected by the current action a and by s earlier transitions. Thus the cost perturbation constant is $K \sum_{s=0}^{h-1} (s+1)\beta^s$. Stage rewards contribute $2R \sum_{s=0}^{h-1} (s+1)\beta^s$, and the terminal reward contributes $2Gh\beta^h$. A maximum over all remaining horizons is necessary: $h\beta^h$ need not be increasing in h . Taking a horizon- T terminal term would not provide a uniform all-restart bound. The main result uses this maximum explicitly.

2.2. Common-continuation repair and approximate minimizers

The original policy be feasible at allowance $\varepsilon + \delta$. At each backward step, all next-date values have already been fixed. The old current action probabilities evaluated against that continuation give $q_p \geq V_t - \varepsilon - \delta$. The best current action gives $q_* \geq V_t - \beta\varepsilon$. If $\alpha = \frac{V_t - \varepsilon - q_p}{q_* - q_p}$, the required mixture share is

$$\alpha = \frac{V_t - \varepsilon - q_p}{q_* - q_p} \leq \frac{\delta}{(1 - \beta)\varepsilon + \delta}.$$

This occurs only on the Borel set with a positive numerator and denominator. The first maximum over a fixed action ordering is Borel because the action set is finite. Operating value weakly

simultaneously. A coin selected once at time zero and remembered would not prove this result; construction does not use such a coin.

For a general Borel model an optimizer need not exist. Apply the repair to a policy whose cost is within $\epsilon > 0$ of the relaxed infimum. The repaired cost is at most that relaxed infimum plus $\xi + D_C \rho_\epsilon(\delta)$. Letting $\xi \downarrow 0$ proves the value modulus without assuming attainment. In the finite-state model, the product of policy simplexes is compact, the finite-horizon Bellman equations are polynomials in its coordinates, and the feasible set is closed and nonempty. There the minimum is attained. The pure-action finite-state problem is a finite enumeration; the randomized repair does not preserve deterministic policies and is not presented as a deterministic repair.

2.3. Why necessary contractions preserve completeness

Extend the already contracted next-date probability intervals. Their rectangular operating Bellman version has one maximizing Markov policy that attains its upper value at all states, by backward induction. The inequality used to contract a current interval is therefore necessary for every feasible policy remaining in the original box. Contracting a current coordinate leaves next-date values unchanged. Its retained endpoint set, when nonempty, still permits an operating value at least $V_t - \epsilon$. Proceeding backward establishes this fact simultaneously at all dates of the contracted box. Capping an operating upper value by V_t does not reduce a true rectangular infimum, since an unconstrained operating optimum already dominates every box-restricted policy.

The implementation also computes lower operating enclosures and may discard a box when its lower enclosure exceeds its upper enclosure. That is an inconsistency test, not a failure of the current sufficient candidate. Lower operating clipping at $V_t - \epsilon$ uses a property of every feasible policy. Such additional necessary tests cannot remove a feasible policy. The separate cost lower version uses the retained probability rectangle without imposing incompatible, restart-specific continuation choices. Ordinary cost Bellman minimization attains this rectangular lower value at every state.

Let w be the largest retained interval width. The midpoint differs by at most w in total variation from the rectangular operating maximizer and the rectangular cost minimizer. Hence the operating deficit is at most $D_J w$, and its cost is at most the rectangular lower cost plus w . Repair adds at most $D_C \rho_\epsilon(D_J w)$. This proves the stated box error, including the effect of contractions.

For the finite stopping count, put $a = \beta^{T-1} w_*$. Any physical width exceeding w_* has weighted width exceeding or equal to a . Each weighted coordinate starts at at most one and, whenever selected, is at least halved; contractions can only reduce it further. Along any path, after at most $\lceil \log_2(1/a) \rceil$ halvings per coordinate, no weighted width can remain above a . Because the largest weighted coordinate is selected, no more than nT times that number of splits are needed before all physical widths are at most w_* . A positive-gap search never needs to refine a node whose repaired midpoint is already within the requested gap of its lower bound. There are only finitely many possible nodes above this depth. The count can be enormous, and does not imply that the fixed-budget experiments exhaust it.

2.4. Rational policy lattices and a deterministic benchmark

For m actions, rounding down the first $m - 1$ probabilities and assigning the remainder to the last action gives a total-variation error at most $(m - 1)/M$. The relaxed feasibility test at

us $D_C\theta_M$ is a lower bound. Repair of its minimizing policy gives a feasible upper bound. The theorem bounds the gap by $D_C[\theta_M + \rho_\varepsilon(D_J\theta_M)]$. Lattice levels need not be nested: the maximum of all lower endpoints and the minimum of all feasible upper endpoints are nested certificates. The executed algorithm is a repair of the minimizing policy. This repair lattice was run.

3. CONTINUOUS-STATE TRANSFER WITHOUT A HIDDEN POLICY SIEVE

3.1. Averaging identity

Reset model, after a transition into E_j , the conditional within-cell law is exactly μ_j , independently of the predecessor state and action. For an arbitrary Borel policy p , define $p_t = \int_{E_i} p_t(a | x) \mu_i(dx)$. At the terminal date the payoff is cellwise constant. If the next-date values coincide with those of the finite model, integrating the reset Bellman equation over μ_i gives the finite Bellman equation with action probabilities $\bar{p}_t$. Backward induction proves the identity for operating values and costs. It uses the same averaged policy at each start, not a separately optimized policy for each i .

Form primitive comparison bounds $|J^p - J_h^p|$ by a_t and $|C^p - C_h^p|$ by b_t for the very same policy. The comparison for optimal operating values is obtained directly from the contraction and locality of finite-horizon Bellman operators. An original feasible policy, when averaged, satisfies the finite constraints at allowance $\varepsilon + 2a_h$. A finite policy feasible at $\varepsilon - 2a_h$ lifted to the original model and satisfies its original constraints. The initial-mixture identity integrates the cost exactly in the reset model; b_h controls the remaining original-model error. Combining both maps with the repair modulus produces the global error budget in the

3.2. Verification obligations for an application

The theorem needs a bound for every bounded Borel test function, not merely weak convergence for smooth functions. A verified uniform density approximation can provide such a bound through an L_1 estimate. For a compact domain with a Lipschitz density in both source and destination coordinates, piecewise-constant density integration gives a computable linear error modulus, with cell volume and domain diameter included in the constant. Discontinuous transition rules require an aligned partition or a separately bounded cost error. The approximation must attain the declared initial-law mixture. These conditions are sufficient, not automatically satisfied by a chosen mesh.

Affine and bilinear benchmark kernels contain atoms. Replacing their images by uniform measures does not furnish a vanishing operator-total-variation error. Their direct whole-cell certificates do not depend on this transfer theorem. We retain their unresolved structural errors and do not present a mesh sequence as a proof of convergence for those atomic models. Stochastic diffusion would additionally need continuous-action coverage, boundary settlement, discretization, and diffusion/quadrature error bounds before the same end-to-end conclusion could be claimed.

4. EXACT MODEL CONTRACTS AND PROSPECTIVE EVALUATION

4.1. Finite-state joint perturbations

The protocol commits seeds 391101–391108 after development seed 390001, with the case studies and horizons in Table II. The two-condition state rewards are $(1/5, 4/5)$. The

on $a \in \{0, 1\}$ has operating reward $s_i - ac$, terminal reward $s_i/2$, and revision cost $a(1 + s_i)$ against the installed action zero. The seed fixes $c \in \{11, \dots, 20\}/100$ and transition jitters; the complete generated rational matrices are inside every proof object.

For two states, the probability of the high-condition destination is

$$P^a(i, \text{high}) = \{22\mathbf{1}_{i=0} + 65\mathbf{1}_{i=1} + j_{ai} + 40a\}/120, \quad j_{ai} \in \{0, \dots, 8\}.$$

For four states, writing the source as $(i_x, i_y) \in \{0, 1\}^2$, the two destination indicators are conditionally independent given the source and action, with probabilities

$$p_x = (18 + 39i_x + 7i_xi_y + j_x + 37a)/120, \quad p_y = (20 + 36i_y + 5i_xi_y + j_y + 37a)/120,$$

where the seed gives $j_x, j_y \in \{0, \dots, 6\}$. Their product probabilities specify the four-state row. The seeds use uniform initial masses; even seeds use $\nu_i = (i + 1)/\sum_{j=1}^n j$. This is a designed i-perturbation suite, not a calibrated economic panel or a random sample of environments. The protocol at commit 13fd3ee0eec78db5fa6aed34e63819c5ed546209 fixes the solver hashes, all eight cases, 4,096 nodes, 60 seconds per solver/case, and absolute stopping tolerance $1/1000$. No solver coefficient was changed after these eight environments were generated. A subsequent CI execution is a replay of the frozen experiment, not a second touched holdout. Wall-clock stopping can change the explored node count across machines. The first local execution summary is retained separately; tables identify the published execution results manifest.

TABLE II: Complete finite-state work and termination records

Seed	Method	Absolute gap	Nodes	Stop	MiB RSS	Seconds	Gap met
1101	I	0.000993	1057	gap	19.8	0.99	yes
1101	M	0.000981	241	gap	77.4	1.77	yes
1102	I	0.000000	1	budget	19.0	0.00	yes
1102	M	0.000000	1	budget	76.0	0.38	yes
1103	I	1.751049	4095	budget	32.6	17.19	no
1103	M	0.475442	4077	time_cap	115.7	60.41	no
1104	I	0.413280	4095	budget	36.2	16.32	no
1104	M	0.110602	3035	time_cap	117.2	60.40	no
1105	I	0.005913	4095	budget	30.2	10.82	no
1105	M	0.000999	2985	gap	97.8	35.99	yes
1106	I	0.000974	159	gap	19.0	0.47	yes
1106	M	0.000763	157	gap	77.8	2.60	yes
1107	I	3.184929	3815	time_cap	46.3	60.04	no
1107	M	1.649036	1529	time_cap	105.6	60.41	no
1108	I	1.750167	2353	time_cap	41.9	60.04	no
1108	M	0.783226	957	time_cap	102.1	60.43	no

I and M denote interval Bellman and verified McCormick search. The memory figure is the maximum peak resident memory; it includes interpreter and imported libraries, not only the search space. Seconds include model construction and the operating solve. The search cap is checked after every proof-producing iteration, so a final iteration can exceed 60 seconds. Serialization

4.2. The nonlinear primitive contract

ate lies in $[0, 1]^2$, $\beta = 19/20$, and shock $z \in \{0, 1\}$ has probabilities $3/5, 2/5$. For each ent, the transition is $(s \text{ own} + bxy + c + dz)/100$. The coefficient pairs for actions are

a	x component (s, b, c, d)	y component (s, b, c, d)
0	(65, 5, 0, 3)	(70, 3, 0, 2)
1	(15, 3, 70, 2)	(70, 3, 0, 2)
2	(65, 5, 0, 3)	(15, 4, 70, 2)
3	(10, 2, 80, 2)	(10, 2, 80, 2).

g costs are $(0, 9/50, 9/50, 8/25)$, reward is $(x + y)/2 - c_a$, and terminal reward is 4. The installed action is 3 when both coordinates are below $1/4$; otherwise it is 1 when 0, 2 when $y < 9/20$, and 0 otherwise. Revision cost is $(1 + (x + y)/2)\mathbf{1}\{a \neq a^0(x, y)\}$. dependent checker separately partitions threshold-crossing boxes and verifies all transition by rational corner geometry. This tests the paper-to-code model contract as well as ic direction.

5. COMPLETE CONTINUOUS-STATE RESULTS

bles below have identical all-restart constraints on both sides. Lower and upper decimals ded outward, and the gap column is rounded upward. Relative gaps are descriptive percentages calculated from the exact rational interval. “Gain” means the lottery upper ictly below the deterministic lower cost for that same initial density. It does not mean lottery is a globally optimal randomized rule. Zero upper cost is exact by nonnegativity; ive randomized row is relabeled exact.

5.1. Uniform initial density

Installed rule	ε	Lower	Upper	Gap	Gap/upper (%)	Gain
refer	0.01	4.61741	4.73011	0.11269	2.38	yes
refer	0.05	3.66404	3.89012	0.22608	5.81	yes
neural31001	0.01	0.08510	0.08928	0.00418	4.68	yes
neural31001	0.05	0.02622	0.03016	0.00394	13.04	yes
neural31002	0.01	0.12047	0.12517	0.00469	3.75	yes
neural31002	0.05	0.04796	0.05403	0.00607	11.22	yes
neural31003	0.01	0.09478	0.09909	0.00430	4.34	yes
neural31003	0.05	0.03178	0.03723	0.00544	14.61	yes
occupancy_stress	0.01	3.90691	4.02072	0.11381	2.83	yes
occupancy_stress	0.05	3.02022	3.24805	0.22783	7.01	yes
pline129	0.01	0.00000	0.00000	0.00000	0.00	no
pline129	0.05	0.00000	0.00000	0.00000	0.00	no
pline33	0.01	0.00000	0.00000	0.00000	0.00	no

Installed rule	ε	Lower	Upper	Gap	Gap/upper (%)	Gain
spline33	0.05	0.00000	0.00000	0.00000	0.00	no
defer	0.01	10.50355	10.63794	0.13439	1.26	yes
defer	0.05	9.44911	9.85440	0.40528	4.11	yes
neural31001	0.01	0.34344	0.40232	0.05887	14.63	yes
neural31001	0.05	0.10408	0.14046	0.03638	25.89	no
neural31002	0.01	0.22548	0.27772	0.05223	18.81	yes
neural31002	0.05	0.04029	0.05952	0.01922	32.29	no
neural31003	0.01	0.55282	0.65391	0.10108	15.46	yes
neural31003	0.05	0.08634	0.10044	0.01410	14.03	yes
occupancy_stress	0.01	9.17362	9.36573	0.19210	2.05	yes
occupancy_stress	0.05	7.98380	8.56599	0.58218	6.80	yes
spline129	0.01	0.00000	0.00000	0.00000	0.00	no
spline129	0.05	0.00000	0.00000	0.00000	0.00	no
spline33	0.01	0.00000	0.00000	0.00000	0.00	no
spline33	0.05	0.00000	0.00000	0.00000	0.00	no
defer	0.01	15.31722	15.46691	0.14969	0.97	yes
defer	0.05	14.14203	14.72413	0.58209	3.95	yes
neural31001	0.01	2.77095	3.36015	0.58920	17.53	yes
neural31001	0.05	1.58477	2.02354	0.43876	21.68	yes
neural31002	0.01	2.05173	2.38505	0.33332	13.98	yes
neural31002	0.05	1.19573	1.45515	0.25942	17.83	yes
neural31003	0.01	0.77248	1.01028	0.23780	23.54	yes
neural31003	0.05	0.09346	0.14211	0.04865	34.23	no
occupancy_stress	0.01	13.42003	13.65506	0.23503	1.72	yes
occupancy_stress	0.05	12.15841	12.87752	0.71911	5.58	yes
spline129	0.01	0.00000	0.00000	0.00000	0.00	no
spline129	0.05	0.00000	0.00000	0.00000	0.00	no
spline33	0.01	0.00000	0.00000	0.00000	0.00	no
spline33	0.05	0.00000	0.00000	0.00000	0.00	no

5.2. Initial density $2x$

Installed rule	ε	Lower	Upper	Gap	Gap/upper (%)	Gain
defer	0.01	4.80430	4.91856	0.11426	2.32	yes
defer	0.05	3.85831	4.08355	0.22523	5.52	yes
neural31001	0.01	0.12619	0.13251	0.00632	4.77	yes
neural31001	0.05	0.03825	0.04405	0.00580	13.16	yes
neural31002	0.01	0.17588	0.18292	0.00704	3.85	yes
neural31002	0.05	0.06877	0.07759	0.00882	11.35	yes
neural31003	0.01	0.13995	0.14645	0.00650	4.43	yes
neural31003	0.05	0.04616	0.05411	0.00795	14.68	yes
occupancy_stress	0.01	3.63981	3.75184	0.11203	2.99	yes
occupancy_stress	0.05	2.80900	3.02583	0.21682	7.17	yes
spline129	0.01	0.00000	0.00000	0.00000	0.00	no

Installed rule	ε	Lower	Upper	Gap	Gap/upper (%)	Gain
pline129	0.05	0.00000	0.00000	0.00000	0.00	no
pline33	0.01	0.00000	0.00000	0.00000	0.00	no
pline33	0.05	0.00000	0.00000	0.00000	0.00	no
efer	0.01	10.69052	10.82469	0.13417	1.24	yes
efer	0.05	9.68566	10.04214	0.35647	3.55	yes
eural31001	0.01	0.30165	0.34589	0.04424	12.79	yes
eural31001	0.05	0.11746	0.14930	0.03183	21.32	no
eural31002	0.01	0.11812	0.15183	0.03371	22.20	yes
eural31002	0.05	0.01393	0.02361	0.00967	40.94	no
eural31003	0.01	0.59210	0.68149	0.08939	13.12	yes
eural31003	0.05	0.10049	0.11544	0.01494	12.94	yes
ccupancy_stress	0.01	8.90083	9.08695	0.18611	2.05	yes
ccupancy_stress	0.05	7.75429	8.29459	0.54029	6.51	yes
pline129	0.01	0.00000	0.00000	0.00000	0.00	no
pline129	0.05	0.00000	0.00000	0.00000	0.00	no
pline33	0.01	0.00000	0.00000	0.00000	0.00	no
pline33	0.05	0.00000	0.00000	0.00000	0.00	no
efer	0.01	15.50452	15.65425	0.14972	0.96	yes
efer	0.05	14.36785	14.91248	0.54462	3.65	yes
eural31001	0.01	3.00897	3.57101	0.56203	15.74	yes
eural31001	0.05	1.75968	2.30639	0.54670	23.70	yes
eural31002	0.01	2.29566	2.59770	0.30203	11.63	yes
eural31002	0.05	1.37651	1.66452	0.28800	17.30	yes
eural31003	0.01	0.70594	0.93114	0.22519	24.18	yes
eural31003	0.05	0.09731	0.14941	0.05209	34.86	yes
ccupancy_stress	0.01	13.14135	13.37437	0.23301	1.74	yes
ccupancy_stress	0.05	11.89664	12.60188	0.70523	5.60	yes
pline129	0.01	0.00000	0.00000	0.00000	0.00	no
pline129	0.05	0.00000	0.00000	0.00000	0.00	no
pline33	0.01	0.00000	0.00000	0.00000	0.00	no
pline33	0.05	0.00000	0.00000	0.00000	0.00	no

5.3. Initial density $2(1 - x)$

Installed rule	ε	Lower	Upper	Gap	Gap/upper (%)	Gain
efer	0.01	4.43053	4.54166	0.11113	2.45	yes
efer	0.05	3.46976	3.69670	0.22693	6.14	yes
eural31001	0.01	0.04401	0.04605	0.00204	4.42	yes
eural31001	0.05	0.01419	0.01627	0.00207	12.72	yes
eural31002	0.01	0.06507	0.06742	0.00235	3.47	yes
eural31002	0.05	0.02714	0.03047	0.00332	10.88	yes
eural31003	0.01	0.04961	0.05173	0.00211	4.07	yes
eural31003	0.05	0.01741	0.02036	0.00294	14.44	yes
ccupancy_stress	0.01	4.17401	4.28960	0.11559	2.69	yes

instance	horizon	lower	upper	gap	upper (%)	exact
occupancy_stress	0.05	3.23143	3.47027	0.23883	6.88	yes
spline129	0.01	0.00000	0.00000	0.00000	0.00	no
spline129	0.05	0.00000	0.00000	0.00000	0.00	no
spline33	0.01	0.00000	0.00000	0.00000	0.00	no
spline33	0.05	0.00000	0.00000	0.00000	0.00	no
defer	0.01	10.31657	10.45120	0.13462	1.29	yes
defer	0.05	9.21257	9.66666	0.45409	4.70	yes
neural31001	0.01	0.38523	0.45874	0.07350	16.02	yes
neural31001	0.05	0.09070	0.13163	0.04092	31.08	no
neural31002	0.01	0.33285	0.40361	0.07075	17.53	yes
neural31002	0.05	0.06082	0.09542	0.03460	36.26	no
neural31003	0.01	0.51355	0.62632	0.11277	18.00	yes
neural31003	0.05	0.07220	0.08545	0.01325	15.50	yes
occupancy_stress	0.01	9.44731	9.64451	0.19720	2.04	yes
occupancy_stress	0.05	8.21366	8.83739	0.62372	7.06	yes
spline129	0.01	0.00000	0.00000	0.00000	0.00	no
spline129	0.05	0.00000	0.00000	0.00000	0.00	no
spline33	0.01	0.00000	0.00000	0.00000	0.00	no
spline33	0.05	0.00000	0.00000	0.00000	0.00	no
defer	0.01	15.12991	15.27957	0.14965	0.98	yes
defer	0.05	13.91621	14.53578	0.61956	4.26	yes
neural31001	0.01	2.53293	3.14930	0.61637	19.57	yes
neural31001	0.05	1.40986	1.74069	0.33082	19.00	yes
neural31002	0.01	1.80779	2.17240	0.36460	16.78	yes
neural31002	0.05	1.01494	1.24578	0.23084	18.53	yes
neural31003	0.01	0.83901	1.08942	0.25041	22.98	yes
neural31003	0.05	0.08961	0.13482	0.04521	33.53	no
occupancy_stress	0.01	13.69954	13.93576	0.23621	1.69	yes
occupancy_stress	0.05	12.42017	13.15316	0.73298	5.57	yes
spline129	0.01	0.00000	0.00000	0.00000	0.00	no
spline129	0.05	0.00000	0.00000	0.00000	0.00	no
spline33	0.01	0.00000	0.00000	0.00000	0.00	no
spline33	0.05	0.00000	0.00000	0.00000	0.00	no

6. DETERMINISTIC GAPS AND THE EXCEPTIONAL RESTART LOCUS

The inherited deterministic headline decomposes into 12 zero-cost and 12 positive-cost equalities. Of the positive-cost equalities, ten occur at horizon 4, two at horizon 8, and none at horizon 12. The table reports every inherited absolute and relative interval and the outer vector’s maximum operating deficit. A deficit above ε is an actual violation established by exact one-dimensional representation; it is distinct from a failed sufficient nonlinear boxificate.

The installed policy has zero *revision* cost by definition. Therefore a revision gap divided by installed policy’s revision cost is undefined for every row. We do not report that ratio as ∞ , and do not substitute its operating reward as a denominator with different units. The valid normalization is gap divided by a positive feasible upper revision cost; a zero upper cost is rarely labeled exact.

State	Gap	Supplier (%)	Donor	Excess	
defer	0.01	0.000000	0.00	0.01000	0.00000
defer	0.05	0.000000	0.00	0.05000	0.00000
neural31001	0.01	0.000000	0.00	0.01000	0.00000
neural31001	0.05	0.000000	0.00	0.05000	0.00000
neural31002	0.01	0.000000	0.00	0.01000	0.00000
neural31002	0.05	0.000000	0.00	0.05000	0.00000
neural31003	0.01	0.000000	0.00	0.01000	0.00000
neural31003	0.05	0.000000	0.00	0.05000	0.00000
occupancy_stress	0.01	0.000000	0.00	0.01000	0.00000
occupancy_stress	0.05	0.000000	0.00	0.05000	0.00000
spline129	0.01	0.000000	–	0.00001	0.00000
spline129	0.05	0.000000	–	0.00001	0.00000
spline33	0.01	0.000000	–	0.00001	0.00000
spline33	0.05	0.000000	–	0.00001	0.00000
defer	0.01	0.001622	0.01	0.01170	0.00170
defer	0.05	0.019446	0.18	0.09072	0.04072
neural31001	0.01	0.010990	2.29	0.01170	0.00170
neural31001	0.05	0.103474	45.39	0.09072	0.04072
neural31002	0.01	0.010990	3.16	0.01170	0.00170
neural31002	0.05	0.069936	61.21	0.07754	0.02754
neural31003	0.01	0.006434	0.74	0.01170	0.00170
neural31003	0.05	0.000000	0.00	0.07693	0.02693
occupancy_stress	0.01	0.000000	0.00	0.01000	0.00000
occupancy_stress	0.05	0.002381	0.03	0.05763	0.00763
spline129	0.01	0.000000	–	0.00042	0.00000
spline129	0.05	0.000000	–	0.00042	0.00000
spline33	0.01	0.000000	–	0.00576	0.00000
spline33	0.05	0.000000	–	0.00576	0.00000
defer	0.01	0.001359	0.01	0.01170	0.00170
defer	0.05	0.066102	0.43	0.12738	0.07738
neural31001	0.01	0.014120	0.40	0.01170	0.00170
neural31001	0.05	0.177733	6.89	0.09072	0.04072
neural31002	0.01	0.009537	0.37	0.01170	0.00170
neural31002	0.05	0.075788	4.12	0.07897	0.02897
neural31003	0.01	0.006662	0.54	0.01170	0.00170
neural31003	0.05	0.248472	63.88	0.11215	0.06215
occupancy_stress	0.01	0.003444	0.03	0.01194	0.00194
occupancy_stress	0.05	0.060221	0.45	0.06919	0.01919
spline129	0.01	0.000000	–	0.00042	0.00000
spline129	0.05	0.000000	–	0.00042	0.00000
spline33	0.01	0.000000	–	0.00576	0.00000
spline33	0.05	0.000000	–	0.00576	0.00000

6.1. Exact disagreement for horizon 8, neural31003, allowance 1/20

Exact check reveals a stronger initial equality than the earlier verbal description suggested: as represented functions, including isolated points. The failure of the all-restart equality

tion. They are distinct restarted problems. The nonnegative Bellman cost-gap decomposition initial equality imply zero expected later gap under the selected policy and the original norm law; positive-gap intervals therefore have zero probability along that policy from this al law. They remain relevant when the controller is restarted at their states. The following ions come directly from the independent piecewise-affine parser. No decimal equality test used.

1.0.1. *Restart date 1.* Open interval with endpoints 0 and 7195348168316/157454804455. The difference has slope 55233/2000000 and intercept 289161/500000. At 0, the separately represented difference is 289161/500000.

1.0.2. *Restart date 2.* Open interval with endpoints 11274755890307953/14592000000000 and 4598974049913359/5910038365284645. The difference has slope 1031/2000 and intercept 373/1000.

Open interval with endpoints 36597632066686/46374770160891 and 4487129074/5675886884445. The difference has slope 3061039/4000000 and intercept 9987/2000000.

Open interval with endpoints 4487129063774/5675886884445 and 633416277301/787273254022275. The difference has slope 969/2000 and intercept 627/1000. At 11274755890307953/14592000000000000, the separately represented difference is 22509905322907499543/29184000000000000000.

At 36597632066686/46374770160891, the separately represented difference is 26876978886030897/46374770160891000000.

At 4487129063774/5675886884445, the separately represented difference is 2958610911/2928734202500.

1.0.3. *Restart date 3.* Open interval with endpoints 1288720115486/1669378425 and 181509165032032/231550957065375. The difference has slope 1 and intercept 1.

At 1288720115486/1669378495425, the separately represented difference is 2958610911/1669378495425.

7. NONLINEAR BOUNDS AND WORK

The support-only lower endpoint is shown beside its connected restart improvement. All upper bounds are the same retained feasible candidates; thus the improvement is not confounded by a larger proposal portfolio. A missing upper endpoint means that no upper candidate at mesh was certified by the inherited sufficient test. It does not imply true infeasibility. No comparison of unmatched success fractions is used.

T	N	Support lower	Restart lower	Upper	Gap/upper (%)	Array MiB
4	16	0.0000	0.0000	2.4975	100.00	0.1
4	32	0.0005	0.0005	2.1219	99.97	0.6
4	64	0.0118	0.0118	1.7191	99.31	2.3
4	128	0.0332	0.0334	1.3503	97.53	9.4
4	256	0.0554	0.0556	0.9992	94.43	37.5

	Upper-lower	Lower-lower	Upper	Upper-upper (1/2)	Upper-upper (1/20)
16	0.0000	0.0000	—	—	0.3
32	0.2521	0.2630	5.1263	94.87	1.1
64	0.9214	0.9438	3.9348	76.01	4.2
128	1.2956	1.3108	3.6152	63.74	16.9
256	1.4773	1.4895	3.0418	51.03	67.5
16	0.0168	0.0169	—	—	0.5
32	1.7800	1.8264	—	—	2.0
64	3.1646	3.1898	8.7556	63.57	8.0
128	4.1609	4.1908	7.6644	45.32	31.9
256	4.6932	4.7174	7.3118	35.48	127.5
16	0.4443	0.5535	—	—	1.0
32	3.4755	3.5226	—	—	3.9
64	6.4932	6.5389	16.7884	61.05	15.5
128	8.0961	8.1264	12.5226	35.11	61.9
256	8.8476	8.8718	11.8492	25.13	247.5

ray payload excludes temporary continuation ranges, the interpreter, and compression peak process memory is also saved. On a threshold-crossing box, the cost upper re charges the largest possible realized departure cost, while the lower enclosure can ero if the installed action is possible there. This conservatism concerns enclosure of the nted policy, not a claim that the policy changes at every point of that box. The nonzero s address the earlier lower-bound weakness without removing this upper-enclosure atism.

7.1. *Operating allowance 1/20*

four rows test a tolerance comparable to the primary development cohort. They are y lower-bound-only diagnostics. A feasible policy at allowance 1/2 cannot be used as r policy at allowance 1/20.

<i>T</i>	<i>N</i>	ε	Lower	Witness width	Status
4	64	0.05	1.55879	0.10794	uncertified upper
8	128	0.05	4.24475	0.08548	uncertified upper
16	128	0.05	7.76989	0.13408	uncertified upper
32	256	0.05	12.45297	0.10222	uncertified upper

8. LOCAL-LP ERROR VERSUS WORK AND DEGENERACIES

herited adaptive calculation is retained in full. Its finest tolerances required substantially lls and work. The table and two generated figures display all 12 executions, not only e rows. They concern a continuous local relaxation with zero global support floor, and silently combined with the new nonzero-floor global recursion as if their errors were

Witness	ϵ	Local tolerance	Calls	Local interval width	Seconds
constructed	0.01	1/100	3159	0.018550	11.36
constructed	0.01	1/200	6275	0.009275	32.39
constructed	0.01	1/400	12529	0.004638	106.30
exact	0.01	1/100	3156	0.018550	8.91
exact	0.01	1/200	6272	0.009275	25.40
exact	0.01	1/400	12526	0.004638	82.27
constructed	0.05	1/100	171	0.018550	0.51
constructed	0.05	1/200	309	0.009275	1.03
constructed	0.05	1/400	588	0.004638	2.35
exact	0.05	1/100	168	0.018550	0.40
exact	0.05	1/200	306	0.009275	0.84
exact	0.05	1/400	585	0.004638	1.96

The new local calculation uses a fixed represented partition, not an adaptive nondegeneracy termination assumption. For one operating inequality, it enumerates feasible pure actions and action mixtures straddling the boundary. When a pure action lies exactly on the boundary already included. When two operating values coincide, no division by their difference is formed; either their face is infeasible or its linear cost minimum is attained at an included vertex. Thus rational coincidences do not create an unreported termination assumption for new calculation. The old logs did not instrument an occurrence count for every degeneracy in their adaptive proof construction. We do not infer such a count from successful termination. These old claims remain subject to their explicitly conditional theorem.

9. INDEPENDENT CHECKS, PUBLICATION CHECKS, AND PRESERVATION

The 16 finite proof trees are checked using SymPy rationals without importing either solver, repair constructor, or the LP package. The verifier reconstructs the economic stage contract, non-policy Bellman equalities, operating feasibility, box coverage, contractions, all accrued nonnegative multiplier residuals, and the minimum over terminal leaves. Nine distinct mutations of policy class, probabilities, terminal value, initial law, revision cost, lower endpoint, multiplier sign, or root coverage are rejected. These mutation tests supplement rather than prove the verifier's all-input correctness.

The 42 continuous-state lottery objects and their 42 connected local lower objects receive independent operating, cost, support, and vertex checks. The separate randomized and deterministic inherited endpoints retain their provenance. Most final lower endpoints are either the newly checked local integral or the newly checked exact support integral. A retained R34 randomized endpoint may dominate those at a particular row; such an inherited component is not converted to a fresh R39 independent proof merely by taking a maximum. The publication audit records source components. Likewise deterministic strict separation uses the retained R38 deterministic certificates, whose own independent checks remain part of the archive. The independent label for new lotteries and local bounds does not expand automatically to all historical components. The two nonlinear independent checks cover the whole $(T, N) = (8, 64)$ and $(32, 64)$ objects, including policy cost and operating feasibility. They use rational corner geometry and scalar arbitrary-precision integer recursions, rather than vectorized construction arithmetic. Other linear meshes, except the new R40 $(32, 128)$ replay, have directed construction and structural checks, not the same independent status. These labels are distinct from hash identity, repeated mutation, and publication-integrity checks.

checked lower envelopes and positive affine pullbacks against independent SymPy evaluation and open-cell representatives, and rejects 400 malformed partitions. The retained states its seed and comparison count. This tests isolated values, threshold crossings, continuities beyond the nine hand-specified parameters and technical probability of false acceptance.

10. ADDITIONAL WITNESS, ROBUSTNESS, AND ARITHMETIC AUDITS

The main article proves a witness-driven feasibility repair and a lattice error budget that checked operating witnesses rather than an exact operating optimum. Its repair slack $(\beta)\varepsilon - d > 0$, where d bounds the upper witness's Bellman residual. Approximation of operating value and approximation of the policy remain separately visible. The new face perturbation proposition preserves one policy across every model in its uncertainty. Bank-even administration charges are derived for all 126 primary initial-law configurations, giving zero margins.

The new rational repair constructor does not compute or accept the exact optimum. In 48 test models, a separate exact operating oracle constructs TEST witnesses; it is not input to the constructor. Signed stage and terminal rewards, two or three actions, two or three states, and zones 2 through 5 are covered. SymPy matrix evaluation checks every operating and cost against the Fraction constructor. All 48 deliberately nonpositive witness-slack cases are marked as uncertified, not declared infeasible. These tests supplement the analytic theorem.

The needed piecewise-affine audit checks 200 discontinuous pairs, including separately tested knot values, with 5756 independent value comparisons and 400 malformed-partition cases. An additional 120 local programs are checked by an independently assembled SymPy basis calculation; 59 contain coincident operating values and 64 have a pure action on the operating boundary. These are measured degeneracy counts for the new audit, not actively invented counts for the historical adaptive solver.

The replays all 16 finite proof trees, three declared full primary lottery/local objects, and the new nonlinear $(T, N) = (32, 128)$ object using the frozen independent verifiers. The latter replaces prior scalar verification to a finer mesh. The full inherited 42-primary-object verifier is retained by exact hash linkage; the three new primary replays are not represented as executions. None of these tests is a statistical false-acceptance guarantee.

10.1. *Preserved scientific content*

The appended R38 article and supplement contain the complete deterministic proof, fixed-history-conditioned frontier, two-period randomized example, adaptive local-LP theorem, economic sensitivity analysis, nonlinear feasibility construction, and killed-operator/stopped-derivations. They remain historical documents with their original claims and limitations; the article and this supplement specify the current conclusions. The complete older 349-STORY_R38.pdf, prior reviews, failed runs, and source hashes remain unchanged on GitHub. No historical scientific content is deleted. Preservation is not presented as additional evidence that an unresolved economic optimum has been solved.

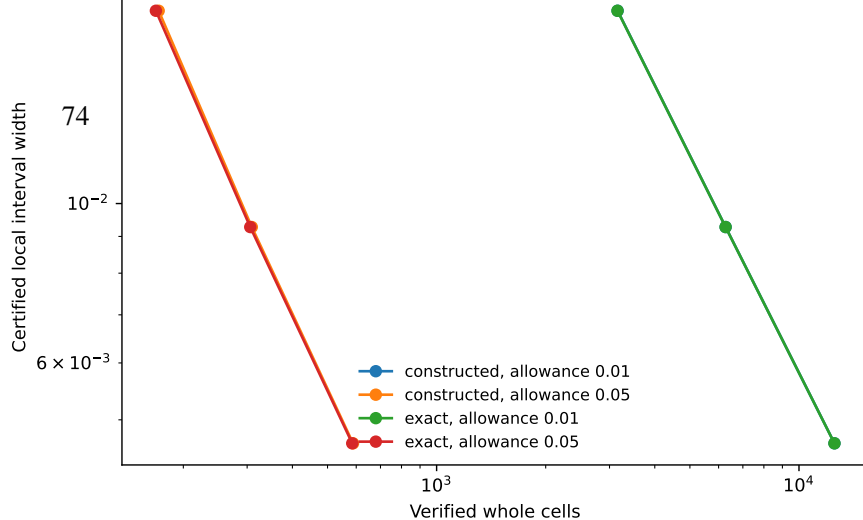


FIGURE 1.—Certified local interval width against whole cells. Every inherited local execution is displayed; the selected R39 global lower bounds solve a different recursive lower problem.

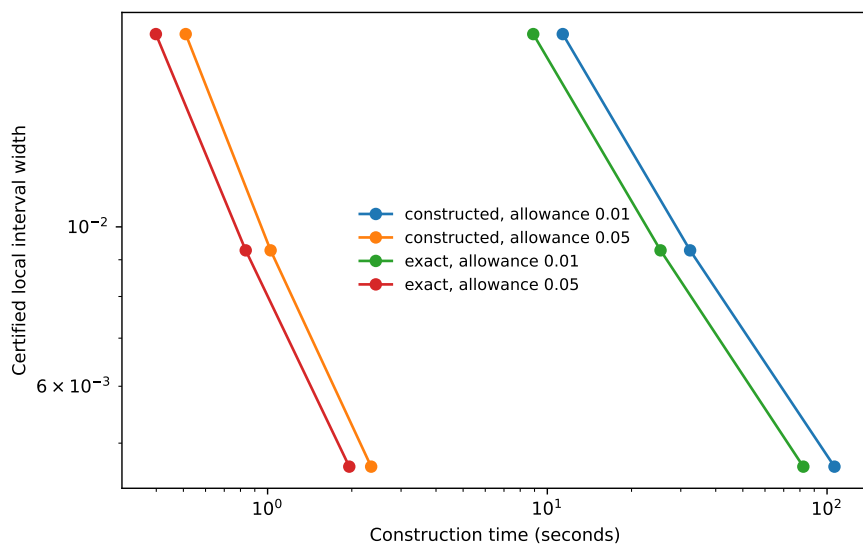


FIGURE 2.—Certified local interval width against construction seconds. The steep work increase is retained rather than hidden by an aggregate success count.

Following pages are the original article, not a newly edited substitute. Its initial cost-discussion is supplemented by the exact restart diagnostic above.

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 stic policies and cost lower bounds that can be matched by independently
 yments. In a continuous-state maintenance economy, exact arithmetic certifies
 ribution deterministic optima among 42 configurations, including 12 with
 mentation cost. An exact continuation-frontier recursion and a solved two-
 ized Markov example separate deterministic from randomized objectives.
 ed lower bounds, adaptive primal–dual cell certificates replace a finite local
 d and provide a quantitative approximation guarantee. A nonlinear two-state
 s directed enclosures beyond piecewise-affine closure, while 66 sensitivity
 measure endogenous intervention exposure. Separate rational verification,
 uns, full computational costs, and nonzero residual intervals accompany the
 ed-operator comparison connects the method to stopped economic control.

S: Dynamic programming, Policy revision, Verified computation, Constrained
 ol, Neural proposals.

1. INTRODUCTION

on rule can be inexpensive to operate and expensive to replace. A new rule
 ng performance, but its implementation cost depends on which decisions
 ates the revised rule subsequently visits. Maintenance, organizational
 oyed decision systems share this distinction. A rule that minimizes today’s
 eed not minimize the discounted cost of future changes. The economic
 a constrained control problem, not a pointwise classification problem.

cts certificates for that problem. Its operating requirement is uniform over
 tes. Its implementation objective is an expectation under a specified initial
 eployed policy’s own transition law. These are different scopes: a uniform
 nt does not turn an initial-distribution optimum into a cost optimum from
 ntain that distinction in the theory and in every numerical declaration of

on is a deterministic outer-class construction using two-sided operating
 deterministic action must be compatible with the performance target even
 imal operating continuation. Replacing that unavailable continuation by a
 ss, and optimal current value by a lower witness, preserves the necessity
 ng implementation cost over the resulting outer class gives a valid lower
 ntly evaluated feasible deployment gives an upper bound. Equality solves
 mentation objective, not merely an inner approximation. Exact operating
 g supplies a reference specialization, and its full cost is charged.

bution is a policy-class-specific constrained reference calculation. A
 recursion solves finite-shock, finite-horizon problems from a fixed restart
 ay depend on history. Its construction clips operating feasibility at each
 nly at the initial node. A two-period version of the maintenance economy
 ic randomized Markov solution for the uniform initial distribution. The

at sacrificing the lower-bound direction. The resulting error bound
 us local restart relaxation. It does not assert global strong duality for
 control problem. Separate experiments measure witness error, global
 multiplier error, and feasible upper-search-improvement
 ve material numerical consequences. Of 42 unchanged primary con-
 tly matching deterministic lower and upper initial costs; 12 of these
 All 14 horizon-four cases close. Of the 24 integrated equalities, 23
 complete cost-to-go functions. The remaining 18 deterministic cases
 nonzero intervals. The largest interval is below 0.249 in normalized
 operating tolerance is relaxed to obtain these statements.

amine the mechanism and the representation. Across 66 designed
 namic revision is strictly less costly than pointwise revision in 22
 exactly equal cost in 44. A separate bilinear two-state economy uses
 osures, horizons through 32, and meshes through 256 per dimension.
 certificates and an information-matched classical grid scalarization
 sts of a known-model numerical method, not empirical calibration or
 ee scalability.

1.1. *Relation to existing methods*

onstrained backward induction, convexification, and weak duality are
 a, 1999, Judd, 1998). We do not rename them as new principles. Our
 ictive combination of operating witnesses, policy-class-correct outer
 ble optimality test, and a numerical implementation that closes positive
 s. The frontier computation provides a direct constrained reference for
 eed objective, rather than comparing only with unconstrained operating

programming studies approximation and sampling errors under spec-
 res (Munos and Szepesvári, 2008, Scherrer, 2013). Safe policy im-
 anges in response to uncertainty about a transition model (Laroche,
 Combes, 2019). Here the model is known, approximation proposes a
 parate deterministic verification layer checks the resulting rule on the
 ral and spline proposals are retained on identical terms. We establish
 vantage nor superiority of an installed neural rule merely because a
 asible.

exploits economic and approximation structure (Judd, 1998, Brumm
 Information-relaxation methods provide a different route to bounds
 2010, Brown and Smith, 2014). Both approaches must charge bound
 mensional reference calculation exploits exact affine structure; the
 ent replaces that structure with directed box arithmetic. Verified com-
 lytic reduction as well as arithmetic error (Rump, 2010). Accordingly,
 rification is specified by its actual code boundary, rather than inferred
 cond execution path.

exist because the action set is finite. For a deterministic Markov policy π , 0, and

$$J_t^\pi = T_t^{\pi_t} J_{t+1}^\pi, \quad C_t^\pi = k_t(\cdot, \pi_t) + \beta P_t^{\pi_t} C_{t+1}^\pi. \quad (2)$$

the π^0 and a nonnegative revision weight w_t , our principal specification is $\neq \pi_t^0(x)\}$.

distribution ν and a tolerance $\varepsilon \geq 0$, write

$$\int_{(\varepsilon)} C_0^\pi d\nu, \quad \mathcal{F}_D(\varepsilon) = \{\pi : V_t(x) - J_t^\pi(x) \leq \varepsilon \text{ for every } (t, x)\}. \quad (3)$$

notes deterministic Markov policies. A randomized Markov kernel $p_t(\cdot |$ on-specific equations in (2); replacing the feasible class gives K_R . An action is charged for the action actually drawn, so its conditional expected (x, a) .

requires more notation. Let $h_t = (x_0, a_0, x_1, \dots, a_{t-1}, x_t)$ be an admissible history augmented by any remembered past randomization. Continuation rewards (h_t) are defined before the date- t action draw. The history-conditioned

$$C_t) - J_t^\pi(h_t) \leq \varepsilon \quad \text{at every admissible decision history,} \quad (4)$$

on kernels specified at off-path histories. We do not impose the same conditioning on the current action draw but before receiving its payoff. Such a could exclude precisely the mixtures studied below. Deterministic history- and randomized history-dependent policies are kept separate. A pointwise on all these histories can be integrated against any initial distribution; an at one fixed initial state cannot simply be integrated to claim a Markov on continuation choices.

tion relative to the installed rule, $J_t^\pi \geq J_t^{\pi^0}$, is checked separately. It is (3). In the primary experiment every selected deployment satisfies both deployment with a matched lower bound also solves the problem with the on restriction.

Maintenance primitives and the unchanged primary cohort

measures condition. Actions 0, 1, 2 mean defer, maintain, and replace. The are

$$r(x, a) = x - c_a, \quad (c_0, c_1, c_2) = (0, 3/20, 9/20), \quad g(x) = x/2. \quad (5)$$

1]. The primary revision weight is $1 + x$, and ν is uniform. Because
current operating cost, its justification comes from continuation value.
ons also make future implementation exposure endogenous.
unchanged: horizons 4, 8, 12, tolerances $1/100$, $1/20$, and seven frozen
horizon. These are defer, an occupancy-stress rule, two spline rules,
he neural seeds are 31001, 31002, and 31003; the spline grids have
exact serialized R34 incumbents, not refitted substitutes, are used.
separate values at partition knots are part of the new proof objects. A
work evaluation is not interchangeable with this exact compiled policy

STRUCTURE, WITNESSES AND FEASIBLE DEPLOYMENT

L_t, U_t , suppose

$$L_t \leq V_t \leq U_t, \quad L_T = U_T = g. \quad (7)$$

be constructed and checked, not supplied as a costless oracle. In the
expectation under positive affine maps, addition of affine rewards, and
nconv piecewise-affine structure. The exact representation stores rational
on each open interval, and separate values at every knot.

ive upper encoding: *Let F be continuous, convex, rational piecewise
interpolant on a rational mesh lies above F . Rounding the starting
slope upward to a common grid $2^{-q}\mathbb{Z}$ produces a convex function G
where H is the chord interpolant. If F is M -Lipschitz and mesh width
 $-F \leq Mh + 2^{1-q}$. The actual maximum defect is determined exactly
not sets.*

ders the chord slopes; upward rounding on a common lattice preserves
error is below 2^{-q} , so integration over a total length of one adds less
error. Convexity puts every chord above F , and the endpoint Lipschitz
ord error by Mh . The difference of the two represented functions is
interval, so its extrema occur at endpoint limits and stored points. *Q.E.D.*

construct $F_t = T_t U_{t+1}$ and an encoding satisfying $0 \leq U_t - F_t \leq d_t$.
 $+ \beta b_{t+1}$, set $L_t = U_t - b_t$. Backward comparison proves (7). In the
e bound propagates as $M_t \leq 1 + \beta(17/20)M_{t+1}$ before the explicit
s finite mesh and precision choices exist for every positive prescribed
ion checks the actual defects rather than relying only on a priori

efficient class. It need not solve the unrestricted feasible problem.

The cost of a deployment is always evaluated using that deployment's transition choice minimizing $k_t(x, a)$ inside $\mathcal{A}_t(x)$ discards the term $\beta P_t^a C_{t+1}$; retains it. Their comparison is meaningful only when both use the same policy's own occupancy. Section 7.3 imposes both requirements. In enrichment, a backward generator carries an actually constructed continuation; admits only actions satisfying $V_t - T_t^a J_{t+1} \leq \varepsilon$, and selects among them the one with the highest score. This generator uses exact V only in the reference study. Its search parameters, not estimated economic prices. An action failing the test is a failed candidate. The selected full policy is reevaluated before use, and attached to the generator itself.

DETERMINISTIC OUTER BOUNDS AND OPTIMALITY CERTIFICATES

based necessary set and its cost recursion by

$$\mathcal{A}_t(x) = \{a : L_t(x) - T_t^a U_{t+1}(x) \leq \varepsilon\}, \quad (9)$$

$$B_t^{L,U}(x) = \min_{a \in \mathcal{N}_t(L,U;x)} \{k_t(x, a) + \beta P_t^a B_{t+1}^{L,U}(x)\}. \quad (10)$$

because an operating-optimal action belongs to it. Unlike (8), this is an action feasible deterministic action must pass the test, but passing the test does not guarantee that the untruncated future loss remains feasible.

Define deterministic outer bound: Under (7), every feasible deterministic policy satisfies $B_t^{L,U}(x) \leq C_t^\pi(x)$ at every restart. The same inequality holds for every feasible deterministic policy satisfying (4), with $x = x_t$ and its history-conditioned feasible Markov policy $\hat{\pi}$ has

$$\int B_0^{L,U} d\nu = \int C_0^{\hat{\pi}} d\nu, \quad (11)$$

and also the infimum over the stated deterministic history-dependent class. The equality $B_t^{L,U} = C_t^{\hat{\pi}}$ additionally establishes cost optimality from the equality alone does not establish that stronger assertion.

feasible deterministic decision history with current action a , operating comparison

$$-T_t^a U_{t+1}(x) \leq V_t(x) - T_t^a V_{t+1}(x) \leq V_t(x) - J_t^\pi(h_t) \leq \varepsilon.$$

The policy uses optimal continuation as an upper bound on the policy's actual cost successor history. Hence $a \in \mathcal{N}_t$. Assuming the cost inequality at the time of P_t^a gives

$$B_t^{L,U}(x) \leq k_t(x, a) + \beta P_t^a B_{t+1}^{L,U}(x) \leq C_t^\pi(h_t).$$

comparisons, not a floating tolerance for equality. More generally, a policy at most τ certifies τ -optimality for its named deterministic objective. This criterion whenever attained; it is not a claim that every available

fining witnesses: *If $L^n \uparrow V$ and $U^n \downarrow V$, then the sets $\mathcal{N}(L^n, U^n)$ increase pointwise to $B^{V,V}$ pointwise. For a finite margin separating excluded actions from the threshold ensures sets once the witness errors are smaller than that margin. Neither with the constrained optimum unless a feasible upper policy closes*

left-hand side increases with n . An action whose exact gap exceeds ε is always included. A finite margin separates excluded actions from the threshold ensures sets once the witness errors are smaller than that margin. Neither with the constrained optimum unless a feasible upper policy closes the losses accumulate, which explains the final qualification. *Q.E.D.*

compute both $B^{L,U}$ and $B^{V,V}$. The former demonstrates that safe actions do not require an exact operating solution. The latter is a fully charged reference bound. Their difference is measured, not suppressed. When the selector itself is infeasible, it remains a valid lower-bound witness but

OPTIMIZATION AND DIRECT CONSTRAINED REFERENCES

cannot be applied to randomized policies. A mixture can satisfy the constraints by assigning positive probability to an action that fails the deterministic constraints. We retain the unrestricted support/restart bounds for the randomized reference calculation.

5.1. A continuation-frontier recursion

..., m , let $\mathcal{K}_t(x)$ be the achievable pairs of conditional operating value from a decision history ending at (t, x) , with all future decision histories independent on the installed rule is suppressed. Put $\mathcal{K}_T(x) = \{(g(x), 0)\}$.

$$(j_j, c_j) \in \mathcal{K}_{t+1}(f_{aj}(x)), \quad j = 1, \dots, m\}.$$

over frontier is needed: higher cost at the same or lower operating value
 82 the minimum second coordinate of (12) is the exact history-
 conditioned implementation optimum.

the constrained frontier: *For a finite horizon, finite actions, and finite shocks, the history-conditioned class. Rational primitives and rational state space polytopes. Primal convex combinations and supporting hyperplanes. The recursion need not compute the optimum when different histories reaching the same physical state may occur.*

terminal date the assertion is immediate. Conditional on a current action and policy must choose a continuation pair in the corresponding successor set of these pairs is exactly the action-specific set in braces. Mixing current convex hull. The half-space enforces current feasibility; successor sets enforce subsequent decision-history restriction. Conversely, choose a convex combination of its current lottery, and use the selected feasible successor continuation pair and shock. This constructs every pair in the recursion. Finite Minkowski convex hulls, and half-space intersections preserve rational polytopes. All candidate vertices and primal combinations attain the retained supporting slopes certify entire edges. The construction remembers which continuation is assigned to each history and imposes no common Markov continuation across histories. *Q.E.D.*

grow exponentially with horizon. Memoization of identical (t, x) frontier construction, but it does not impose equality of the selected continuation sets. For continuous shocks, additional validated set approximation would not infer a high-dimensional complexity rate from a finite tree. The experiment solves 70 fixed-restart horizon-four problems: the 14 primary configurations are 2, 3/4, 1. These objectives are Dirac initial distributions, not the primary ones. In 24 cases the randomized history-conditioned value is strictly below the bound at the same restart. Each full frontier has a separately implemented verification. The experiment is a direct constrained comparator for its exact initial state, not a numerical quadrature of K_R in the primary study.

2. A solved randomized Markov continuum example

on yields an exact comparison under the *same uniform initial objective* functions. Keep all maintenance primitives and install defer at both dates. At the earlier date maximizes operating value and has zero implementation cost. Thus it

$$K_R = \int_0^1 (1+x) \left(1 - \frac{\varepsilon}{\Delta(x)}\right) dx < \frac{3}{2}. \quad (13)$$

policy at every initial state; its last base continuation does not have any mixture with a smaller change probability would lose more than ε at x . Equation (13) therefore solves the randomized Markov problem, any relaxation.

$l(x) = ax + b$, the integral of $(1+x)/(ax+b)$ equals $(r-l)/a + (al+b)\ln(r-l)/(al+b)$ when $a \neq 0$, with the evident constant-denominator formula. The logarithm enclosures yield costs approximately 1.3720075152810114. The enclosure widths are below 10^{-20} . At $x=0$, $\Delta = 2323/4000$. The probability at tolerance $1/100$ is 2283/2323. This counterexample rules out any policy for the model. It supplements rather than replaces the unchanged

PROVED LOWER BOUNDS AND ADAPTIVE LOCAL CERTIFICATES

For a given support price λ , let $W_T^\lambda = \lambda g$ and

$$W_t^\lambda = \max_a \{ \lambda r_t(\cdot, a) - k_t(\cdot, a) + \beta P_t^a W_{t+1}^\lambda \}. \quad (14)$$

A policy or history-conditioned policy satisfies $W_t^\lambda \geq \lambda J_t^\pi - C_t^\pi$, hence

$$W_t^\lambda = \max \{ 0, \sup_{\lambda \in \Lambda} [\lambda(L_t(x) - \varepsilon) - W_t^\lambda(x)] \} \leq C_t^\pi(h_t). \quad (15)$$

One can proceed without claiming that its supremum exhausts global duality. The lower bound is valid after every successor decision history. Set $q_a = L_t(x) - \varepsilon + \beta P_t^a R_{t+1}(x)$, and $b = L_t(x) - \varepsilon$. Feasibility implies $\sum_a p_a q_a \geq b$. The maximization is

$$\min_{p \geq 0, \sum p_a = 1, \sum p_a q_a \geq b} \sum_a p_a z_a = \sup_{\mu \geq 0} \min_a \{ z_a + \mu(b - q_a) \}. \quad (16)$$

(16) prices a complete unrestricted continuation. The local multiplier μ is the one-step inequality in (16). They are not interchangeable.

Local restart propagation: *If F_t and R_{t+1} are valid cost lower bounds for a feasible class, then $R_t = \max\{F_t, \Phi_t(R_{t+1})\}$ is also valid. For three classes, the local optimum is attained either by a feasible pure action or by a mixture of two. The dual optimum is attained at zero or at a nonnegative intersection of two lines and equal slopes handled explicitly.*

re q_a, z_a, b are affine, a two-action primal cost is

$$P_{ij}(x) = \frac{4(q_j - b)z_i + (b - q_i)z_j}{q_j - q_i}, \quad q_i \leq b \leq q_j. \quad (17)$$

quadratic and its denominator affine. A fixed dual price gives $D_\mu(x) = \dots$, a piecewise-affine lower bound. We first split at all input knots, where the denominator sign changes. A cell is accepted only if the primal is feasible and $D_\mu \leq \rho/2$ throughout it. Multiplication by the nonnegative denominator reduces the comparison to a quadratic sign test. A quadratic's maximum on a closed interval is at an endpoint or its interior stationary point. Zero denominators at cell endpoints are handled by limits; stored isolated knots are solved separately by the exact point LP.

Refined local approximation: Suppose the preceding cell tests cover the entire domain, and the dual coefficient and isolated point value of the dual lower function decrease by at most 2^{1-q} on $[0, 1]$. Let $e = \rho/2 + 2^{1-q}$ and assume $2^{1-q} \leq \rho/2$. Refinement with this lower encoding, and the same exact floor F_t , produces

$$0 \leq R_t - \widehat{R}_t \leq e \sum_{s=0}^{T-t-1} \beta^s. \quad (18)$$

Refinement with a strictly feasible action margin on each compact cell, refinement with a positive ρ . At degeneracies, finite termination is asserted only when the refinement and cell tests admit a finite cover. A refinement cap causes an upper bound on the passing certificate.

On $[0, 1]$, decreasing slope and intercept by less than 2^{-q} decreases an affine function by at most 2^{-q} . Stored point values have the same direction. The cell test bounds the local primal optimum before rounding. Changing a continuation cost by at most βd changes the action cost by at most βd , so minimization over the unchanged feasible set is β -nonexpansive. The maximum with the same floor is nonexpansive. This gives (18). Under a strict feasibility margin, feasible primal vertices and dual values have bounded local moduli on a finite subdivision; the midpoint primal/dual error tends to zero as cell widths shrink. Compactness supplies a finite cover. At any point, such a modulus need not follow from strict interior feasibility alone, but a one-sided condition and explicit stopping test are required. *Q.E.D.*

The algorithm encountered rational serialization growth. That unsuccessful experiment used $q = 40$ and $\rho = 1/100, 1/200, 1/400$, and dyadic integration enclosures. It solves the two horizon-four neural network problems, constructs witnesses and exact V , for 12 complete certificates. The floor

ut this diagnostic. The experiment measures approximation to this
 n, not an additive decomposition of a different nonzero-support restart
 global strong duality.

EMPIRICAL EVIDENCE AND COMPUTATIONAL ACCOUNTING

erated from executable sources and exact or outward-directed proof
 36 protocol assertions are not used as observations. The primary model,
 tions, and its operating tolerances are unchanged. The computation
 ds, source hashes, execution records, arithmetic libraries, and the
 ndidate ledger.

Primary deterministic optima and residual intervals

t lower recursion and independently feasible deployments agree in 24
 including 12 strictly positive costs. Ten positive-cost equalities occur
 horizon eight. The horizon-eight neural31003 case with $\varepsilon = 1/20$ has
 ut equality of all cost-to-go functions. The other 23 exact cases have
 y. This distinction is tested, not inferred from rounded table entries.
 sary selectors are feasible, and their full cost functions match those
 deployment. This supplies a certificate-level explanation for all 14
 urations the outer relaxation itself has a feasible minimizing selector.
 every horizon-four maintenance specification has this property. At
 sary selector violates the operating target in 19 cases, by as much as
 Eighteen of those retain a positive initial cost interval; the remaining
 ly equality just described. Accumulated operating loss, not failure of
 agnosed obstruction to treating those outer selectors as deployments.
 ent passes the all-state operating constraint and preserves the installed
 didate failures and all 18 residual deterministic intervals are printed in
 ed exactly in the result ledger. The lower and upper fractions, policy
 s, frozen-input hash, policy class, objective, and exactness scope are
 2 cases. Decimal displays do not determine equality.

7.2. Controlled gap diagnostics

re compared under identical installed rules: constructed L with eight
 th the same eight prices, and exact V with fifteen prices. The eight
 56, 1024, 4096; refinement adds $1/2, 2, 8, 32, 128, 512, 2048$. This re-
 ted support solves across 21 horizon–incumbent pairs; both operating

identical initial distributions and installed rules within its own policy-class scope. Gains
e. Global prices expand from eight to fifteen; the local LP study has its own zero floor.

TABLE III

CONTINUOUS LOCAL RESTART RELAXATION: EXACT-VALUE DIAGNOSTIC

ρ	Eight-price lower	Adaptive lower	Adaptive upper
1/100	0.076186	0.088078	0.106627
1/200	0.076186	0.088107	0.097382
1/400	0.076186	0.088121	0.092759
1/100	0.020289	0.025620	0.044169
1/200	0.020289	0.025649	0.034924
1/400	0.020289	0.025665	0.030302

n neural31001, zero support floor, 40-bit directed coefficients. The endpoints enclose the
i, not the global constrained optimum.

same support functions. Pointwise monotonicity of both refinements is

ution improves 30 support floors, with maximum integrated gain below
rice refinement improves 28 floors, with maximum gain approximately
e deterministic ladder, exact V raises 30 safe outer bounds, with maximum
⁵. Upper-portfolio enrichment lowers 30 inherited deployment costs, by as
ly 0.9399. These measurements show why exact-value availability is not
ource of the residual deterministic gaps on this cohort.

t summed across policy classes. Global support floors are valid for ran-
tioned policies; the necessary-action floor is deterministic. The local LP
oor and uses only the two declared horizon-four neural31001 cases. For
mary cases, the exact deterministic interval, necessary-selector violation,
refinement, and upper-search improvement are reported as separate coordi-
n that the structural gap of the continuous global randomized problem has
ated.

l tolerance narrows a certified interval around the continuous local relax-
omparison with the eight-price local recursion is computed on identical
ero support floor. The theorem explains the narrowing without assigning it
ty.

3. *Economic sensitivities and endogenous exposure*

y uses horizon eight, three natural installed rules, two separately declared
.20, and eleven one-at-a-time primitive specifications. The baseline con-
below 0.3 and maintains below 0.6. The preventive rule replaces below

0.75. The calendar rule intervenes every fourth date, replacing below
wise, and defers at other dates. These are normalized scenarios, not
m firms.

tions change discounting to 0.90 or 0.99, the revision weight to 1 or
o $2x$ or $2(1-x)$, maintenance cost to 0.10 or 0.20, or defer persistence
named primitive changes in each scenario. Both dynamic and pointwise
efficient action sets; their costs are evaluated under their own transitions
specific initial density. All 66 configurations pass their operating checks.
ive implementation savings and 44 exact zero effects; the maximum
0.08701. The complete scenario table retains the zero effects. The
sensitivity of intervention exposure across the declared economic rules,
or a dollar welfare calibration. Computational penalties used elsewhere
se economic revision weights.

7.4. A nonlinear two-state economy

) $\in [0, 1]^2$, and actions defer, repair the first component, repair the
verhaul both components. Rewards are $(x+y)/2 - c_a$, with costs
nal payoff $(x+y)/4$, and revision weight $1 + (x+y)/2$. Discounting
remain 0.95 and (0.6, 0.4). Each coordinate transition has the form
, where $\xi \in \{0, 1\}$ and z is that coordinate. The action-specific tuples

	x -coordinate	y -coordinate
0	(65, 5, 0, 3)	(70, 3, 0, 2)
1	(15, 3, 70, 2)	(70, 3, 0, 2)
2	(65, 5, 0, 3)	(15, 4, 70, 2)
3	(10, 2, 80, 2)	(10, 2, 80, 2).

ne on the square and preserve it, but bilinearity destroys the one-
e used by the primary solver.

$N = 16, 32, 64, 128, 256$ cells per coordinate, and horizons are
arithmetic and backward floors/ceilings are performed with signed
code encloses each full transition image, enumerates every intersected
s an intermediate absolute-value bound below 2^{62} . The grid policy is
half-open box; closed transition enclosures cover ties conservatively.
ely evaluated fixed policies, appear in the supplement.

e is $1/2$ in this separate economy. It does not replace $1/100$ or $1/20$
o restart-based generators use penalties zero and four. A classical

ed/attempted. The separate operating tolerance is $1/2$. Restart and classical methods use the compiled-rule pilot uses meshes through 128. The two restart penalties are separate outputs, pieces of one comparator.

TABLE VI

ARGED PRIMARY CONSTRUCTION WORK AND REPRESENTATION

Operating DP (s)	Median revision (s)	Max. lower pieces	Max. bits
0.0046	0.0973	11	143
0.0141	0.3478	90	186
0.1198	2.3529	430	242

executed once per horizon. Revision time includes both exact-value and safe-witness outer upper portfolio. Independent verification, support diagnostics, and supplementary studies are charged overhead. Timings are generated from this publication execution.

grid solver constructs a scalarization portfolio at prices 4, 16, 64, 256; its is only a proposal and receives the same directed whole-box evaluation. e, and verification work is charged. There are 26 certified restart-generator 40 and 15 certified classical portfolios among 20. A fresh R38 audit of the installed-rule box policy fails in all 16 pilot mesh–horizon configurations at compiled-rule pilot is not a reconstruction of the unsupported R35/R36 m. The two algorithms’ candidate counts differ, so their success counts are The full ledger reports each matched mesh, horizon, tolerance, runtime, ss width, fallback count, and failure. These results demonstrate executed onstruction, not uniform dominance over the classical comparator.

7.5. Time, representation, and verification

ives are shared only across configurations with the same horizon. Frozen portized historical inputs; no neural training is described as newly executed. values, outer recursions, feasible candidate searches, support refinements, aptive LP construction, nonlinear grids, and their verification are charged action avoids comparing warm certificate work with cold neural training. verifier imports none of the Fraction/PW construction, envelope, compo- outines. It rebuilds partitions using actual primitive transition pullbacks, threshold crossings, open-cell endpoint limits, isolated values, Bellman ities, all-restart feasibility, terminal conditions, and exact initial integrals. ets pass. Twenty declared corruption categories are rejected, including g, shock probabilities, action labels, parser failures, objective scope, point tness claims. procedure independently checks the 70 fixed-restart frontier objects using didate pairs, feasible convex combinations, and supporting inequalities.

ion–portfolio program is retained with its original target. On $(u, x) \in$
horizon one,

$$Z_s^u, \quad (19)$$

$$p_s)x_s - c_s]ds + 0.2p_s x_s dZ_s^x, \quad d\langle Z^u, Z^x \rangle_s = -0.25 ds. \quad (20)$$

$[-0.5, 0.8] \times [-0.2, 0.2] \times [-0.5, 0.8]$. Running utility is $c^{1-u}/(1-u) -$
 $G = -0.02(u-2)^2 + 0.1 \log x$, and exit settlement is $G - 8(1-s)$,
the full current-state target at $k=2$ remains $\sup_{t,u,x} (V_2 - J_2^\pi) \leq 0.01$.
operator-theoretic rather than a claim that maintenance certificates solve
it, Δ a time step, and a a control held over the step. Define the killed

$$\begin{aligned} & \int_0^{\tau \wedge \Delta} e^{-rs} \ell(Z_s, a) ds + e^{-r\tau} g(t + \tau, Z_\tau) \mathbf{1}\{\tau \leq \Delta\} \\ & + e^{-r\Delta} w(Z_\Delta) \mathbf{1}\{\tau > \Delta\} \end{aligned} \quad (21)$$

is positive and has mass at most $e^{-r\Delta}$. Bellman comparison, witness
necessary-action proof therefore extend verbatim to this subprobability
tion mass replacing β and exit settlements included in the one-step
for error ξ_t enters the backward witness budget additively, and thus
corresponding discounted sum of ξ_t to the operating certificate.

l only with valid killed-transition and action-coverage enclosures. A
cannot supply a lower bound for the original continuous-control revision
s may improve it. Uniform action interpolation, time discretization,
d diffusion approximation errors must all be covered. Markov-chain
ne route to such an additional analysis (Kushner and Dupuis, 2001); it
aintenance experiment.

oundary result remains intact. At restart $(0, 61/50, 5/4)$ with $k=2$,
ant $\theta \in [-1/5, 1/5]$, the archived derivative enclosures are positive on
al and identify $\theta = 1/5$ as the unique maximizer in that scalar class.
asure are reproduced in the supplementary preservation material. The
and 7.181834580823298 is not converted into a 0.01 certificate. The
t explains precisely which parts of the certificate method transfer and
t computations remain necessary for the unchanged diffusion objective.

show why a policy-class qualifier is mathematically indispensable, rather than a mere qualification.

LP supplies a controlled approximation to a continuous restart relaxation. Global support-price study, and the remaining deterministic intervals identify risks rather than hiding them in a single gap. Nonlinear whole-box execution economic sensitivities extend the evidence beyond the exact affine reference. An object includes the proofs, source, unsuccessful computations, and actual evidence to review these claims. The original stopped-control objective and material are preserved with an explicit map to their present location.

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POLICY CLASSES, ATTAINABILITY, AND RESTART SEMANTICS

Markov feasible class in the article uses a single measurable action at each randomized counterpart uses a probability kernel at that date and state. The constraints imposed before the current action draw. Neither class is replaced by an performance constraint. Implementation cost is averaged under the specified policy after the all-restart operating inequalities have been established.

A randomized policy may remember earlier actions, shocks, and previously drawn continuation is evaluated at a decision history before the next action draw. To ensure well-defined, a policy includes continuation kernels at every admissible histories with positive probability under the initially deployed policy. The policy remains the unconstrained Markov value $V_t(x_t)$ because the underlying required inequality is conditional at every such history.

The necessary-action proof uses the action actually selected at that history. Its continuation is V_{t+1} , regardless of the particular history. A randomized policy average inequality over current actions; individual actions in its support need a stochastic test. The two-period example is an explicit instance. Conditioning on a new draw would define a different feasible class and would remove the mixture conditioning is not used here.

The recursion keeps separate continuation choices for different histories that end at the same state. Identical state/date frontier sets can be cached because the primitive of selecting two different points on the cached frontier at different histories is infeasible. Thus memoization is not a proof that a randomized Markov policy is optimal.

With finite shocks and finite horizon, the reachable decision tree is finite. That tree can be extended to unvisited decision histories by an operating-off-tree extension has finite implementation cost and zero operating regret. The Dirac initial objective. This justifies the fixed-restart reference under the definition without claiming that independently chosen references at different dates share a common Markov continuation.

ACT FINITE REPRESENTATION AND INDEPENDENT CHECKING

A witness is the finite tuple (z_i, m_i, b_i, v_i) , equal to $m_i x + b_i$ on each open interval (v_i, v_{i+1}) . The terminal and operating witnesses are continuous in the

the test of the inequality itself. The verifier uses SymPy Rational and its own tuple parser, line selection, pullback, and primitive Bellman evaluator. It does not import the con-
 nnel. For each object it checks the ~~problem node against an application~~ canonical hash, the terminal equations, $V = TV$, the upper-witness
 recursion, necessary-action masks, lower-cost minimum and attain-
 ing and cost equations, all-restart regret, and exact initial integrals. It
 ed and functionwise equality flags.
 categories include primitive discount, transition, shock probability,
 objective scope, terminal conditions, missing dates, knot order, mal-
 legal actions, fractional actions, isolated values, affine coefficients, false
 witness defects, and provenance/hash failures. The complete category
 comes are in the independent verification ledger. The independence
 ate arithmetic/partition implementation. Both programs intentionally
 omic primitives; this is not formal proof-assistant verification or an
 interpretation errors are impossible.
 comparison also hashes canonical JSON for 187 scientific objects: 42
 -restart frontiers, 12 local LP objects, 21 global support bundles, and
 This cross-environment equality check is distinct from independent
 imary and frontier groups are covered by the independent SymPy
 are not part of these 187 canonical scientific objects.

CONSTRUCTION, SUFFICIENT ACTIONS, AND OUTER RESTRICTIONS

$V_T = g$ and suppose $0 \leq U_t - T_t U_{t+1} \leq d_t$. Induction gives $V_t \leq U_t$. If
 $_{t+1}$, then

$$V_t \leq d_t + T_t U_{t+1} - T_t V_{t+1} \leq d_t + \beta b_{t+1} = b_t.$$

lower witness. Actual realized defects can produce a tighter witness
 on allowance. The R38 safe outer calculation uses the realized defect
 proof object.

class, suppose $e_t = \eta_t + \beta e_{t+1}$ and an action satisfies $U_t - T_t^a U_{t+1} \leq \eta_t$.
 satisfies $J_{t+1} \geq U_{t+1} - e_{t+1}$, then

$$U_t - T_t^a J_{t+1} \leq \eta_t + \beta e_{t+1} = e_t.$$

nonempty sufficient action sets therefore gives an operating-feasible
 on over those sets is ordinary dynamic programming, but it solves
 In contrast, every feasible deterministic action is included in $\{a :$
 cost minimum over this latter class is a lower bound on the original

ve seven named attempts. An empty backward admissible set is explicitly is seven, not a number of independently optimized global policies. A rating certificate is not silently repaired by increasing the tolerance.

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EXACT POINT LP AND WHOLE-CELL PRIMAL-DUAL TESTS

local LP is $\min p \cdot z$ subject to $p \geq 0$, $\sum p = 1$, and $p \cdot q \geq b$. Every vertex it vector, or a mixture of two actions i, j with $q_i < b < q_j$ and weights

$$p_i = \frac{q_j - b}{q_j - q_i}, \quad p_j = \frac{b - q_i}{q_j - q_i}.$$

sible unit vector handles the endpoint. When $q_i = q_j$, the pair does not ts cheaper feasible unit vector or another crossing pair suffices. Ties are nistic order, but every tied optimal value remains valid.

s are zero and nonnegative values $(z_i - z_j)/(q_i - q_j)$ for unequal q_i, q_j . checks exact primal-dual equality at every sampled basis-selection point ted knot. These exact point checks select a certificate; they do not verify a

ell, all q, z, b are affine. Additional splits ensure that action feasibility and not change in the interior. The proposed two-action primal has nonnegative one throughout the cell; these conditions are checked at endpoint limits. N and denominator $D > 0$. For a fixed $\mu \geq 0$, write $d_a = z_a + \mu(b - q_a)$. the inequality $P - D_\mu \leq \delta$ is equivalent to

$$N - D(d_a + \delta) \leq 0 \quad \text{for every action } a.$$

s quadratic. Its maximum is calculated using both endpoints and, for a vertex when interior. A zero denominator at an endpoint is treated by the ension and a separate exact point LP at the endpoint. A denominator that rior rejects the cell.

e dual minimum can change active action within an accepted cell. All such ed, and the checker verifies that each stored affine line is the corresponding hout its subcell, not only at its midpoint. The union of accepted cells must aps or overlaps. The exact knot LP value is stored independently of the ula.

each slope, intercept, and point value is rounded downward to a denom-an affine slope/intercept rounding changes the value by at most 2^{-39} in The exact difference is checked again before propagation. For arbitrary e theorem requires enough bits to keep this rounding term below half the its are the implemented choice for the three displayed tolerances, not a

process. The quantized algorithm and the final whole-cell verifier are
 An engineering recursion-depth cap of 60 causes explicit failure. The
 theorem is not a promise that this cap suffices for arbitrary models.

FRONTIER PRIMAL AND DUAL VERIFICATION

ates the action-specific Minkowski sums of efficient successor vertices,
 er hull, and intersects it with the required operating half-space. Pruning
 another achievable pair has no smaller operating value and no larger
 lies above a supporting lower edge. Because feasibility is a lower
 a higher-value cheaper pair dominates for all preceding lower-budget

ntier vertex, the proof stores either one candidate or a convex com-
 es that attains it. Its cost-support line $c = \mu j + \eta$ is checked against
 airs. Supporting slopes are nonnegative. Adjacent vertices determine
 that line against every candidate proves the entire edge, including
 operating-budget line. At the first feasible frontier point, either the
 a zero supporting slope certifies an unconstrained cost minimum. The
 tier coordinate is also checked against the independent operating value.
 constructs all candidate action/child combinations from the primitive
 ified child frontiers. It compares canonical candidate counts/hashes and
 ty. It does not import the constructor's hull or optimization functions.
 singleton $(g, 0)$. These tests certify the serialized recursive frontier,
 agation, rather than just its reported initial number.

m-initial randomized Markov calculation uses a different analytic
 al interval the integrand is rational with an affine denominator. For a
 n argument r , range reduction writes $r = 2^m s$ with $1 \leq s < 2$, using
 With $y = (s - 1)/(s + 1)$,

$$\sum_{i=0}^{n-1} \frac{y^{2i+1}}{2i+1} + R_n, \quad 0 \leq R_n \leq \frac{2y^{2n+1}}{(2n+1)(1-y^2)}.$$

oses $\log 2$. Forty terms give the recorded rational integral enclosures.
 e coefficients determine which endpoint is used. This analytic calcula-
 count of 70 independently verified frontier objects.

NEAR WHOLE-BOX ARITHMETIC AND CERTIFICATE SCOPE

$1)/N] \times [j/N, (j+1)/N]$ for range bounding. The represented policy
 nt, with the upper outside boundary included in the final box. All
 he nonlinear experiment are nonnegative. Therefore the lower/lower
 orner enclose the image of the entire box, including the bilinear term.

range spans at most one step in each coordinate, so evaluating all four
 /upper indices covers every possible successor box. An exact upper image
 cludes the neighboring assigned box, conservatively. It is not rounded into
 cessor. 96

values, revision weights, and discounted transition expectations receive
 s and ceilings at scale 2^{34} . For an expectation, the numerator is $19(3v_1 +$
 ator is 100. Its absolute value is checked below 2^{62} before integer division.
 ed lower recursion, multipliers are at most 64 and bounded model values
 pressions within signed 64-bit capacity. This finite experiment’s overflow
 is not an arbitrary-horizon machine-arithmetic theorem.

nuous threshold rule can change inside a box. Cost bounds therefore
 ative set of *possible installed actions on the full closed box*. A candidate
 revision cost whenever it might equal the installed action; it has zero upper
 en it is the unique possible installed action. Otherwise the upper revision
 his prevents a midpoint installed-action comparison from becoming an
 st certificate.

pilot itself chooses the installed action at each box midpoint and then
 oughout the box. It is a proposal, not exactly the continuous installed rule
 boxes. Likewise, the classical solver uses midpoint rewards and bilinear
 propose a fixed box policy. Both receive the same whole-box operating and
 roof pertains to the resulting serialized box policies, not to an unrecorded
 on of their actions.

of lower and upper operating action enclosures bound the unconstrained
 deterministic necessary-action recursion uses the lower operating witness,
 on bound, and lower implementation costs. The randomized local-grid
 s a dual minimum over *all* actions. The code checks both nonnegativity and
 bound dominates this randomized bound. For a policy with $U_t - J_t^{\text{lower}} \leq$
 date, operating feasibility follows. Its implementation upper enclosure
 ble optimum from above; a failed operating certificate leaves only a bound
 vn cost.

enclosure does not prove that the candidate policy is truly infeasible.
 tly use “certificate pass” rather than treating every failure as a mathe-
 result. The unsuccessful pilot, fallback boxes, coarse-mesh failures, and
 cal outcomes are retained. Witness construction is charged once for each
 each proposal and reevaluation is charged to its method. This is a nonlinear
 alibration or a claimed advantage at arbitrary dimension.

7. ALL PRIMARY CONFIGURATIONS

rral31002	0.01	0.149457	0.149457	0.000000	I/P
rral31002	0.05	0.112614	0.112614	0.000000	I/P
rral31003	0.01	0.121140	0.121140	0.000000	I/P
rral31003	0.05	0.084296	0.084296	0.000000	I/P
cy_stress	0.01	4.206802	4.206802	0.000000	I/P
cy_stress	0.05	4.189476	4.189476	0.000000	I/P
spline129	0.01	0.000000	0.000000	0.000000	I/P
spline129	0.05	0.000000	0.000000	0.000000	I/P
spline33	0.01	0.000000	0.000000	0.000000	I/P
spline33	0.05	0.000000	0.000000	0.000000	I/P
defer	0.01	10.810725	10.812347	0.001622	–
defer	0.05	10.750714	10.770159	0.019445	–
rral31001	0.01	0.468642	0.479631	0.010989	–
rral31001	0.05	0.124513	0.227986	0.103473	–
rral31002	0.01	0.336417	0.347406	0.010989	–
rral31002	0.05	0.044313	0.114248	0.069935	–
rral31003	0.01	0.865305	0.871739	0.006434	–
rral31003	0.05	0.345015	0.345015	0.000000	I
cy_stress	0.01	9.490961	9.490961	0.000000	I/P
cy_stress	0.05	9.248564	9.250944	0.002380	–
spline129	0.01	0.000000	0.000000	0.000000	I/P
spline129	0.05	0.000000	0.000000	0.000000	I/P
spline33	0.01	0.000000	0.000000	0.000000	I/P
spline33	0.05	0.000000	0.000000	0.000000	I/P
defer	0.01	15.612820	15.614178	0.001359	–
defer	0.05	15.471299	15.537400	0.066101	–
rral31001	0.01	3.493934	3.508053	0.014119	–
rral31001	0.05	2.402580	2.580313	0.177732	–
rral31002	0.01	2.536828	2.546365	0.009537	–
rral31002	0.05	1.764791	1.840578	0.075787	–
rral31003	0.01	1.227595	1.234256	0.006661	–
rral31003	0.05	0.140483	0.388955	0.248472	–
cy_stress	0.01	13.699252	13.702695	0.003443	–
cy_stress	0.05	13.242542	13.302762	0.060220	–
spline129	0.01	0.000000	0.000000	0.000000	I/P
spline129	0.05	0.000000	0.000000	0.000000	I/P
spline33	0.01	0.000000	0.000000	0.000000	I/P
spline33	0.05	0.000000	0.000000	0.000000	I/P

initial integrated equality; P additionally denotes all-restart function and complete hashes are in results/primary_exact_values.json; no

The machine-readable exact table contains both cost endpoints as witness lower endpoint, the exact gap, both equality flags, the installed-policy hash, lower-function hash, and full proof-object hash. In each

TABLE II: All 66 economic sensitivity outcomes

Outcome	Rule	ε	Pointwise	Dynamic	Saving
98	condition	0.05	10.470918	10.470918	0.000000
	preventive	0.05	10.680893	10.679056	0.001838
	calendar	0.05	8.721989	8.721989	0.000000
	condition	0.20	10.434669	10.434669	0.000000
	preventive	0.20	10.640825	10.630435	0.010390
	calendar	0.20	8.653418	8.653418	0.000000
90	condition	0.05	8.933154	8.933154	0.000000
90	preventive	0.05	9.131155	9.131155	0.000000
90	calendar	0.05	7.458884	7.458884	0.000000
90	condition	0.20	8.906508	8.906508	0.000000
90	preventive	0.20	9.075142	9.074068	0.001074
90	calendar	0.20	7.387071	7.387071	0.000000
99	condition	0.05	11.907352	11.907352	0.000000
99	preventive	0.05	12.107270	12.102384	0.004886
99	calendar	0.05	9.894955	9.894955	0.000000
99	condition	0.20	11.866806	11.866806	0.000000
99	preventive	0.20	12.085757	12.068172	0.017585
99	calendar	0.20	9.842486	9.830059	0.012428
n_weight	condition	0.05	5.733254	5.733254	0.000000
n_weight	preventive	0.05	5.898295	5.893513	0.004782
n_weight	calendar	0.05	4.826153	4.826153	0.000000
n_weight	condition	0.20	5.733254	5.733254	0.000000
n_weight	preventive	0.20	5.896158	5.880418	0.015740
n_weight	calendar	0.20	4.803912	4.803912	0.000000
ion_weight	condition	0.05	15.208582	15.208582	0.000000
ion_weight	preventive	0.05	15.463492	15.462576	0.000916
ion_weight	calendar	0.05	12.617824	12.617824	0.000000
ion_weight	condition	0.20	15.136084	15.136084	0.000000
ion_weight	preventive	0.20	15.385492	15.379338	0.006154
ion_weight	calendar	0.20	12.502923	12.502923	0.000000
ion_density	condition	0.05	10.897170	10.897170	0.000000
ion_density	preventive	0.05	11.004643	11.003208	0.001435
ion_density	calendar	0.05	8.787898	8.787898	0.000000
ion_density	condition	0.20	10.870687	10.870687	0.000000
ion_density	preventive	0.20	10.973545	10.964602	0.008943
ion_density	calendar	0.20	8.703869	8.703869	0.000000
ion_density	condition	0.05	10.044666	10.044666	0.000000
ion_density	preventive	0.05	10.357144	10.354904	0.002240
ion_density	calendar	0.05	8.656080	8.656080	0.000000
ion_density	condition	0.20	9.998652	9.998652	0.000000
ion_density	preventive	0.20	10.308105	10.296267	0.011838
ion_density	calendar	0.20	8.602967	8.602967	0.000000

0	calendar	0.20	8.626888	8.626888	0.000000
0	condition	0.05	10.512273	10.512273	0.000000
0	preventive	0.05	10.696023	10.696023	0.000000
0	calendar	0.05	8.822815	8.772793	0.050022
0	condition	0.20	10.068605	10.055233	0.013372
0	preventive	0.20	10.577155	10.577155	0.000000
0	calendar	0.20	8.257042	8.170031	0.087011
ce_065	condition	0.05	10.469741	10.469741	0.000000
ce_065	preventive	0.05	10.679481	10.678541	0.000940
ce_065	calendar	0.05	8.721618	8.721618	0.000000
ce_065	condition	0.20	10.434168	10.434168	0.000000
ce_065	preventive	0.20	10.640244	10.630954	0.009290
ce_065	calendar	0.20	8.653691	8.653691	0.000000
ce_085	condition	0.05	8.309909	8.309909	0.000000
ce_085	preventive	0.05	9.281022	9.281022	0.000000
ce_085	calendar	0.05	6.696913	6.696913	0.000000
ce_085	condition	0.20	7.808322	7.792568	0.015754
ce_085	preventive	0.20	8.734932	8.692720	0.042212
ce_085	calendar	0.20	6.142199	6.131385	0.010814

se passed the same-class operating check for both policies. All values
l implementation costs; exact rational values, changed primitives, and
tivity.json and its proof objects.

ALL FIXED-RESTART CONSTRAINED REFERENCES

III: All exact fixed-restart history-conditioned references

ε	x_0	Randomized	Det. lower	Nodes	Vertices
0.01	0	4.149907	4.365233	253	314
0.01	1/4	4.447063	4.658161	257	318
0.01	1/2	4.743143	4.951089	253	314
0.01	3/4	5.032875	5.244017	256	316
0.01	1	5.422816	5.615933	212	264
0.05	0	3.287636	4.365233	253	314
0.05	1/4	3.600854	4.658161	257	318
0.05	1/2	3.909072	4.951089	253	314
0.05	3/4	4.188360	5.244017	256	316
0.05	1	4.647604	5.615933	212	264
0.01	0	0.000000	0.000000	253	253
0.01	1/4	0.000000	0.000000	257	258
0.01	1/2	0.000000	0.000000	253	254

001	0.05	3/4	0.335335	1.750000	256	260
001	0.05	1	0.000000	0.000000	212	212
002	0.01	0	0.000000	0.000000	253	253
002	0.01	1/4	0.000000	0.000000	257	257
002	0.01	1/2	0.000000	0.000000	253	253
002	0.01	3/4	1.467067	1.750000	256	258
002	0.01	1	0.000000	0.000000	212	212
002	0.05	0	0.000000	0.000000	253	253
002	0.05	1/4	0.000000	0.000000	257	257
002	0.05	1/2	0.000000	0.000000	253	253
002	0.05	3/4	0.335335	1.750000	256	258
002	0.05	1	0.000000	0.000000	212	212
003	0.01	0	0.000000	0.000000	253	253
003	0.01	1/4	0.000000	0.000000	257	258
003	0.01	1/2	0.000000	0.000000	253	254
003	0.01	3/4	1.467067	1.750000	256	261
003	0.01	1	0.000000	0.000000	212	213
003	0.05	0	0.000000	0.000000	253	253
003	0.05	1/4	0.000000	0.000000	257	258
003	0.05	1/2	0.000000	0.000000	253	254
003	0.05	3/4	0.335335	1.750000	256	261
003	0.05	1	0.000000	0.000000	212	213
y_stress	0.01	0	4.149907	4.365233	253	312
y_stress	0.01	1/4	4.447063	4.658161	257	316
y_stress	0.01	1/2	3.774393	3.954007	253	303
y_stress	0.01	3/4	4.036576	4.234366	256	302
y_stress	0.01	1	0.000000	0.000000	212	237
y_stress	0.05	0	3.287636	4.365233	253	312
y_stress	0.05	1/4	3.600854	4.658161	257	316
y_stress	0.05	1/2	3.051087	3.954007	253	303
y_stress	0.05	3/4	3.252671	4.234366	256	302
y_stress	0.05	1	0.000000	0.000000	212	239
	0.01	0	0.000000	0.000000	253	253
	0.01	1/4	0.000000	0.000000	257	257
	0.01	1/2	0.000000	0.000000	253	253
	0.01	3/4	0.000000	0.000000	256	256
	0.01	1	0.000000	0.000000	212	212
	0.05	0	0.000000	0.000000	253	253
	0.05	1/4	0.000000	0.000000	257	257
	0.05	1/2	0.000000	0.000000	253	253
	0.05	3/4	0.000000	0.000000	256	256
	0.05	1	0.000000	0.000000	212	212
	0.01	0	0.000000	0.000000	253	253
	0.01	1/4	0.000000	0.000000	257	257
	0.01	1/2	0.000000	0.000000	253	253

0.05	3/4	0.000000	0.000000	256	256
0.05	1	0.000000	0.000000	212	212

ALL NONLINEAR CANDIDATES AND FAILURES

IV: All nonlinear candidates, including failed certificates

Method	Pass	Regret bound	Global LB	Policy UB	Seconds
R0	yes	0.4977	0.0000	2.6751	0.011
R4	yes	0.4977	0.0000	2.4974	0.010
P	no	1.1074	0.0000	1.6213	0.007
C	yes	0.4151	0.0000	3.0150	0.024
R0	no	0.5157	0.0000	2.1002	0.012
R4	yes	0.4996	0.0000	2.1219	0.012
P	no	0.9009	0.0000	0.6582	0.012
C	yes	0.3095	0.0000	2.6452	0.038
R0	yes	0.5000	0.0000	1.7191	0.030
R4	yes	0.4996	0.0000	1.9953	0.031
P	no	0.8199	0.0000	0.3489	0.028
C	yes	0.2182	0.0000	2.6183	0.083
R0	yes	0.5000	0.0000	1.3502	0.103
R4	yes	0.5000	0.0000	1.5348	0.105
P	no	0.7748	0.0000	0.1618	0.100
C	yes	0.1651	0.0000	2.5997	0.279
R0	yes	0.5000	0.0000	0.9991	0.369
R4	yes	0.5000	0.0000	1.1250	0.377
C	yes	0.1539	0.0000	2.5915	1.049
R0	no	0.7217	0.0000	7.8440	0.014
R4	no	0.7207	0.0000	7.5944	0.014
P	no	1.9218	0.0000	3.5345	0.014
C	no	1.8675	0.0000	3.2901	0.048
R0	no	0.5306	0.0000	5.3481	0.024
R4	yes	0.4999	0.0000	5.1263	0.024
P	no	1.4938	0.0000	1.0518	0.023
C	yes	0.4683	0.0000	5.2667	0.075
R0	yes	0.5000	0.0000	4.5263	0.060
R4	yes	0.4998	0.0000	3.9348	0.060
P	no	1.3586	0.0000	0.5982	0.057

R0	yes	0.5000	0.0000	3.0417	0.720
R4	yes	0.5000	0.0000	3.0816	0.722
C	yes	0.1591	0.0000	5.1651	2.051
R0	no	1.1505	0.0000	12.7079	0.028
R4	no	1.1499	0.0000	12.5424	0.028
P	no	3.0806	0.0000	6.0818	0.027
C	no	3.0440	0.0000	5.9726	0.095
R0	no	0.6609	0.0000	13.1739	0.047
R4	no	0.6250	0.0000	12.3093	0.046
P	no	2.3354	0.0000	1.6657	0.045
C	no	2.3030	0.0000	1.8973	0.148
R0	yes	0.5000	0.0000	8.7753	0.124
R4	yes	0.5000	0.0000	8.7555	0.119
P	no	2.0828	0.0000	1.1280	0.107
C	yes	0.4225	0.0000	9.0306	0.348
R0	yes	0.5000	0.0000	7.9683	0.400
R4	yes	0.5000	0.0000	7.6643	0.393
P	no	1.9345	0.0000	0.2580	0.372
C	yes	0.2597	0.0000	8.9885	1.118
R0	yes	0.5000	0.0000	7.3118	1.436
R4	yes	0.5000	0.0000	7.4019	1.430
C	yes	0.1931	0.0000	8.9669	4.103
R0	no	1.6264	0.0000	17.9085	0.055
R4	no	1.6261	0.0000	17.8356	0.056
P	no	4.3523	0.0000	8.9351	0.054
C	no	4.2975	0.0000	8.8716	0.188
R0	no	0.8575	0.0000	24.2374	0.091
R4	no	0.8417	0.0000	23.5439	0.094
P	no	3.2834	0.0000	2.3702	0.087
C	no	3.2552	0.0000	2.6072	0.293
R0	no	0.5056	0.0000	16.2448	0.242
R4	no	0.5013	0.0000	14.0445	0.238
P	no	2.8959	0.0000	1.7564	0.211
C	yes	0.4440	0.0000	16.7883	0.684
R0	yes	0.5000	0.0000	12.5711	0.797
R4	yes	0.5000	0.0000	12.5225	0.832
P	no	2.6814	0.0000	0.3477	0.705
C	yes	0.3209	0.0000	13.1873	2.276
R0	yes	0.5000	0.0000	11.8492	2.867
R4	yes	0.5000	0.0000	12.1034	2.876
C	yes	0.2305	0.0000	13.1578	8.256

art generators; C is the classical four-price portfolio; P is the midpoint-
e pilot. Global LB is the deterministic outer-class lower bound. Policy UB

ε	ρ	Grid LB	Adaptive LB	Adaptive UB	Cells
0.01	1/100	0.076185	0.088078	0.106627	3159
0.01	1/200	0.076185	0.088107	0.097381	6272
0.01	1/400	0.076185	0.088121	0.092758	12529
0.01	1/100	0.076186	0.088078	0.106627	3156
0.01	1/200	0.076186	0.088107	0.097382	6272
0.01	1/400	0.076186	0.088121	0.092759	12526
0.05	1/100	0.020289	0.025619	0.044169	171
0.05	1/200	0.020289	0.025649	0.034924	309
0.05	1/400	0.020289	0.025665	0.030302	588
0.05	1/100	0.020289	0.025620	0.044169	168
0.05	1/200	0.020289	0.025649	0.034924	306
0.05	1/400	0.020289	0.025665	0.030302	585

Each cell passes exact feasibility and quadratic gap tests; stored isolated LP solutions. The floor is zero in all rows.

OPERATORS: ERROR PROPAGATION AND ACTION COVERAGE

Step as $T_t^a v = h_t^a + Q_t^a v$, where Q_t^a is a positive subprobability kernel on $\mathcal{V}_t$. Monotonicity and the estimate $\|T_t^a v - T_t^a w\|_\infty \leq \alpha_t \|v - w\|_\infty$ validated approximation obeys $\|\tilde{T}_t^a v - T_t^a v\|_\infty \leq \xi_t$ for every action a by the construction. If upper encoding adds defect d_t , then the backward

$$b_t = d_t + \xi_t + \alpha_t b_{t+1}.$$

Step by ξ_t and taking the corresponding lower envelope supplies safe lower-kernel argument proves the deterministic necessary-action lower bound recursion with compatible stopping and nonnegative costs. Continuity is essential. A discrete subset gives a lower operating value but not an optimum. It also creates an overly narrow implementation policy class. To approximate a continuous-control target, one must enclose all omitted actions, for continuity regularity and interval covers. State interpolation, killed-transition coverage, and time-discretization error must remain separately identifiable and not nonlinear maintenance calculation validates a distinct finite-action approximation constants.

RETAINED SCALAR STOPPED-CONTROL OPTIMUM

Example (4) with $k = 2$, $c = 3/4$, $p = 0$, and constant $\theta \in [-1/5, 1/5]$, the optimal payoff is strictly increasing in θ . The derivative lower bounds

the primitives, controls, and restart are those stated immediately above, $\theta = 3/4$, and $x(0) = 5/4$,

$$x(t) = \frac{75}{2} - \frac{145}{4}e^{t/50}. \quad (1)$$

$\leq (1 - 1/50)^{-1} = 50/49$, hence $x(t) \geq 25/49 > 1/2$. It is decreasing, so no wealth boundary is reached. Let y denote preference. Direct unchanged settlement gives

$$\begin{aligned} &= \frac{(3/4)^{1-y}}{1-y} - \theta^2 + 8 - .04\theta(y-2) - .00005 + .002 - \frac{.075}{x(t)} \\ &\quad + .0008(y-2)^2 - .004\log x(t) + .32(1-t), \end{aligned} \quad (2)$$

$$= -2\theta - .04(y-2). \quad (3)$$

but $z = y - 1 \in [1/5, 9/5]$ and $L = \log(4/3) < 1/3$. The utility is $-e^{Lz}/z$, on this interval because $Lz < 1$. Its lower bound is $-5e^{1/15} \geq -75/14$. $\log x(t) \geq 1/2$, and $\log x(t) \leq \log(5/4) < 1/4$. Dropping nonnegative terms

$$-\frac{1}{20000} - \frac{3}{20} - \frac{1}{1000} - \frac{4}{625} - \frac{1}{25} - \frac{75}{14} = \frac{342357}{140000} > 2. \quad (4)$$

is $e^{Lz}(1 - Lz)/z^2 \geq 10/81$. Thus

$$\partial_y h_\theta \geq \frac{10}{81} - \frac{4}{3125} - \frac{1}{125} = \frac{28901}{253125} > 0. \quad (5)$$

antly from $b = 14/5$ to larger y . The extension remains bounded, nondecreasing. Its parameter derivative retains the bounds in (3). Conservative ≤ 20 and $|\partial_\theta h| \leq 2$ hold for this extension.

13.2. A coupling lower bound for the derivative

σW_t , killed at $a = 6/5$, where $u_0 = 61/50$ and $\sigma = 1/20$. If drift increases, h is higher at every date and lower-boundary survival events are nested. For $h \geq 2$, paths surviving under both drifts have a nonnegative difference. Newly surviving paths contribute at least two. Taking a positive difference

$$\partial_\theta \mathbb{E}_\theta[h(t, Y_t) \mathbf{1}\{t < \tau_a\}] \geq 2\partial_\theta \Pr_\theta(t < \tau_a). \quad (6)$$

of h_θ on θ must be added, not included in this frozen-integrand comparison. It is at least $-4/125$ for $\theta \in [-1/5, 0]$ and at least $-54/125$ for $\theta \in [0, 1/5]$, after

ave. Indeed, with $r = \varphi/\Phi$, its second derivative is $-r(z)\{z+r(z)\} <$
 tary Gaussian tail inequality gives $r(z) > -z$. Equation (7) is therefore
 imum on each of the two half intervals is at least the smaller of its

normal arguments at $\theta = 0$ and $\theta = -1/5$ are bounded below by -2
 The latter follows by maximizing $4\sqrt{t} + (2/5)/\sqrt{t}$ on that compact
 $14/5$. The rational inequalities

$$\begin{aligned} & < 25, \quad \Phi(-2) > 1/50, \quad \Phi(-14/5) > 1/500, \\ & \Phi(-4/5) > 1/5, \quad \Phi(6/5) > 22/25. \end{aligned} \quad (8)$$

the entire negative drift half interval.
 endpoint arguments at $\theta = 0$ and $\theta = 1/5$ are at least $-4/5$ and $6/5$.
 on the positive drift half interval. Since $e^{-.04t} \geq 24/25$, the two lower
 of the lower-killed occupation expression are

$$2\frac{4}{25}\frac{24}{25}\frac{8}{25} - \frac{4}{125} = .066304, \quad (9)$$

$$2\frac{3}{4}\frac{24}{25}\frac{352}{625} - \frac{54}{125} = .379008. \quad (10)$$

appears in this argument. The intervals cover every admissible constant
 point zero.

3. Distant-boundary error and rational checks

distant but not discarded. Under all admissible drifts, a path hitting it
 excursion of at least $(b - u_0 - 1/5)/\sigma$. Reflection and a Gaussian tail

$$2\exp\left[-\frac{(b - u_0 - 1/5)^2}{2\sigma^2}\right] = 2e^{-380.88} < 2^{-379}. \quad (11)$$

ference supported on that event, the drift likelihood score has second
 auchy–Schwarz bounds the derivative discrepancy between the actual
 integral and the extended lower-killed one by

$$\sqrt{p_b} + 2p_b < 4002^{-189} + 22^{-379} < 1/1000. \quad (12)$$

in the preceding half-line bounds gives precisely $8163/125000$ and
 increasing continuous function on the compact drift interval has the

with the direction of the bound preserved. The code records the resulting bounds and requires them to exceed all displayed thresholds. The current theorem at $\theta = 1/5$ is inherited from the unchanged validated R30 killed-kernel hash, Git blob hash, and the fact that it is not recomputed are recorded in `-09-24-r34/results/boundary.json`. The current theorem is an endpoint evaluation to the optimum of the entire constant feedback optimality or satisfy the original all-domain current-state target.

14. PRESERVATION MAP AND HISTORICAL EVIDENCE

source, result, or review is removed. The branch starts at the latest R36 and R38. The article, supplement, response, and computational report from page-for-page in `HISTORY_R38.pdf`, followed by the earlier recursive original stopped consumption–portfolio specification, its full-domain target, quadrature record, the 47-coordinate derivative/residual bridge discussion, rollback derivations, inventory comparisons, and operating-mode studies are archived pages.

protocol files remain unchanged in the repository. Their unmaterialized are not treated as verified observations. R38 supplies a newly executed explicit provenance. In particular, the fresh compiled-rule nonlinear pilot historical zero-penalty run. The complete prior unfavorable neural and retained; none is converted into a new neural acceleration claim.

a historical annex, not additional experiments for the R38 cohort. Its former revision status are read at their original dates. The current main article and which numerical objectives have new certificates and which original targets bounds. This organization shortens the main narrative without deleting The main article prints the references cited in its current theorem and n. The complete inherited bibliography is retained in `history/refer` in the unchanged R34 source, and in the historical PDF annex; uncited s have not been removed from the scientific archive.